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Bone and Tendon Graft Substitutes and Adjuncts

Number: 0411

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Last Review
07/10/2025
Effective: 05/04/2000
Next Review: 04/23/2026

Review History

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses bone and tendon graft substitutes and adjuncts.

I. **Medical Necessity**

A. *INFUSE Bone Graft (Bone Morphogenic Protein-2)*

Aetna considers the INFUSE Bone Graft medically necessary for lumbar spinal fusion procedures in skeletally mature persons who meet the following criteria:

1. The member meets medical necessity criteria for lumbar spinal fusion in CPB 0743 - Spinal Surgery: Laminectomy and Fusion (../700_799/0743.html); *and*
2. INFUSE Bone Graft is to be implanted via an anterior (ALIF) or lateral (OLIF, DLIF, XLIF or LLIF) approach.

The INFUSE Bone Graft is considered medically necessary for treating skeletally mature persons with acute, open tibial shaft fractures that have been stabilized with intramedullary nail fixation after appropriate wound management, when INFUSE Bone Graft is applied within 14 days after the initial fracture.

Aetna considers the INFUSE Bone Graft experimental, investigational, or unproven for all other indications, including its use in ankle fusions and cervical fusions, because its effectiveness for indications other than the ones listed above has not been established.

Note: The INFUSE Bone Graft is also known as bone morphogenic, or morphogenetic protein-2, BMP-2.

B. *Allograft for Spinal Fusion and Osteochondral Defects*

Aetna considers cadaveric allograft and demineralized bone matrix medically necessary for spinal fusions and for filling osteochondral defects (bone void fillers). Allograft materials that are 100% bone are considered medically necessary for these indications regardless of the shape of the implant.

C. *Polymethylmethacrylate (PMMA) or Calcium Sulfate (e.g., Osteoset Resorbable Mini-Bead) Antibiotic Beads*

Aetna considers PMMA antibiotic beads or calcium sulfate antibiotic beads (e.g., Osteoset Resorbable Mini-Bead) medically necessary for use in conjunction with intravenous

antibiotics in the treatment of chronic osteomyelitis.

D. *Mesenchymal Stem Cell Therapy/Bone Marrow Aspirate/Progenitor Cells*

Aetna considers bone marrow injections medically necessary in the treatment of bone cysts (unicameral/simple). Aetna considers the use of bone marrow aspirate experimental, investigational, or unproven for all other orthopedic applications including nonunion fracture, repair or regeneration of musculoskeletal tissue, osteoarthritis, and as an adjunct to spinal fusion because there is insufficient evidence to support its use for these indications.

E. *Hydroxyapatite Bone Substitute*

Aetna considers hydroxyapatite bone substitute (e.g., OtoMimix) medically necessary for middle ear surgery.

F. *Beta Tri-Calcium Phosphate (B-TCP)-Based Bone Graft Extenders and Substitutes Products*

Aetna considers beta tri-calcium phosphate (b-TCP)-based bone graft extenders and substitutes medically necessary for spinal fusions that meet criteria in CPB 0743 - Spinal Surgery: Laminectomy and Fusion (../700_799/0743.html).

II. **Experimental, Investigational, or Unproven**

Aetna considers the following procedures experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

A. Autologous platelet concentrates, alone or in combination with bone graft materials for maxillary sinus augmentation;
B. Autologous platelet-rich plasma, gels and growth factors (e.g., Procuren), alone or in conjunction with bone grafting materials for acute primary and secondary closure wounds, chronic wound healing, augmentation procedures (e.g., for dental

implants and for the floor of the maxillary sinus), soft tissue injuries, and all indications;

C. Beta-tricalcium phosphate combined with native bone proteins for ankle and hindfoot arthrodesis;
D. Bone micro-indentation testing for evaluation of bone quality;
E. Injection of a calcium-based biomaterial (AGN1) to prevent and treat osteoporotic hip fractures (i.e., OSSURE Local Osteo-Enhancement Procedure [LOEP]);
F. Low-crystallinity apatite for bone reconstruction;
G. Mesenchymal stem cell therapy (e.g., AlloStem, Osteocel, Osteocel Plus, Ovation, Regenexx, and Trinity Evolution) and/or the use of progenitor cells for all orthopedic applications including repair or regeneration of musculoskeletal tissue, osteochondritis dissecans, spinal fusion, and bone nonunions;
H. PerOssal (ceramic bone substitute) for the treatment of osteomyelitis and nonunions;
I. Platelet-rich fibrin for lateral window sinus lifts;
J. Subchondroplasty for the treatment of bone osteoarthritis and all other indications;
K. Surgical mesh composed of porcine intestinal submucosa for rotator cuff repair surgery, repair of anorectal fistula, and for other indications;
L. The following interventions:

- Acellular human dermal allograft (e.g., Alloderm and Arthrex allograft) for nasal septal repair
- Adipose-derived stromal vascular fraction cells
- Anterior cruciate ligament-derived stem cells for ligament tissue engineering
- Anti-microbial bone graft substitutes for the treatment of osteomyelitis
- Autologous stem cells for use after screw removal in orthopedic surgery
- Avive tendon wrap
- Bioglass 45S5 and other ceramic products (except for beta tri-calcium phosphate (b-TCP) for spine fusions)
- Ceracell Ortho Foam for use in cervical instrumentation
- ChronOS beta tri-calcium phosphate bone graft substitute for indications other than spinal fusion

- Collagen-based bio-inductive implants (e.g., Regeneten) for repair of rotator cuff tears
- Cook anal fistula plug
- DeNovo NT natural tissue (allogeneic minced cartilage) graft
- EmCell (fetal stem cell therapy)
- Genetically corrected autologous epidermal grafts for the treatment of recessive dystrophic epidermolysis bullosa
- Gore anal fistula plug
- Gracilis cadaveric graft for hallux valgus repair
- Human growth factors (e.g., fibroblast growth factor, insulin-like growth factor) to enhance bone healing
- i-Factor peptide enhanced bone graft
- Kartogenin-treated autologous tendon graft
- Knee Creation nanocrystalline calcium phosphate bone substitute
- Ligament and Joint Regeneration and Neuvo-generation Medicine (LaJRaN)
- Nacre (mother-of-pearl)
- Nucleus pulposus allograft (e.g., VIA Disc NP) (see CPB 0602 - Intradiscal Procedures (../600_699/0602.html))
- Surgisis collagen plug for the treatment of anal fistulas
- Tendon Wrap Tendon Protector
- Tooth-bone graft
- Xenograft implantation into articular surfaces.

III. Related CMS Coverage Guidance

This Clinical Policy Bulletin (CPB) supplements but does not replace, modify, or supersede existing Medicare Regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs). The supplemental medical necessity criteria in this CPB further define those indications for services that are proven safe and effective where those indications are not fully established in applicable NCDs and LCDs. These supplemental medical necessity criteria are based upon evidence-based guidelines and clinical studies in the peer-reviewed published medical literature. The background section of this CPB includes an explanation of the rationale that supports adoption of the medical necessity criteria and a summary of evidence that was

considered during the development of the CPB; the reference section includes a list of the sources of such evidence. While there is a possible risk of reduced or delayed care with any coverage criteria, Aetna believes that the benefits of these criteria – ensuring patients receive services that are appropriate, safe, and effective – substantially outweigh any clinical harms.

Code of Federal Regulations (CFR): 42 CFR 417; 42 CFR 422; 42 CFR 423.

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual; CMS IOM Publication 100-03 Medicare National Coverage Determination Manual.

Medicare Coverage Determinations: Centers for Medicare & Medicaid Services (CMS), Medicare Coverage Database [Internet]. Baltimore, MD: CMS; updated periodically. Available at: Medicare Coverage Center (https://www.cms.gov/medicare/coverage/center?redirect=/center/coverage.asp). Accessed November 7, 2023.

IV. **Related Policies**

- CPB 0016 - Back Pain - Invasive Procedures (../1_99/0016.html)
- CPB 0244 - Skin and Soft Tissue Substitutes (../200_299/0244.html)
- CPB 0364 - Allograft Transplants of the Extremities (../300_399/0364.html)
- CPB 0743 - Spinal Surgery: Laminectomy and Fusion (../700_799/0743.html)
- CPB 0784 - Blood and Adipose Product Injections for Selected Indications (../700_799/0784.html)

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Bone and Tendon Graft Substitutes and Adjuncts:	
CPT codes not covered for indications listed in the CPB:	
0565T	Autologous cellular implant derived from adipose tissue for the treatment of osteoarthritis of the knees; tissue harvesting and cellular implant creation
0566T	injection of cellular implant into knee joint including ultrasound guidance, unilateral
0841T	Digitization of glass microscope slides for pathology consultation during surgery; first tissue block, with frozen section(s), single specimen (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
20690 - 20694	Uniplane and multiplane fixation systems
20900	Bone graft, any donor area; minor or small (e.g., dowel or button)
20902	major or large
20955	Bone graft with microvascular anastomosis; fibula
20962	other than fibula, iliac crest, or metatarsal
20974	Electrical stimulation to aid bone healing, noninvasive (nonoperative)
20975	invasive (operative)
20979	Low intensity ultrasound stimulation to aid bone healing, noninvasive (nonoperative)
22548 - 22819	Arthrodesis, spine [spinal fusion]
22853	Insertion of interbody biomechanical device(s) (eg, synthetic cage, mesh) with integral anterior instrumentation for device anchoring (eg, screws, flanges), when performed, to intervertebral disc space in conjunction with interbody arthrodesis, each interspace (List separately in addition to code for primary procedure)

Code	Code Description
22854	Insertion of intervertebral biomechanical device(s) (eg, synthetic cage, mesh) with integral anterior instrumentation for device anchoring (eg, screws, flanges), when performed, to vertebral corpectomy(ies) (vertebral body resection, partial or complete) defect, in conjunction with interbody arthrodesis, each contiguous defect (List separately in addition to code for primary procedure)
22859	Insertion of intervertebral biomechanical device(s) (eg, synthetic cage, mesh, methylmethacrylate) to intervertebral disc space or vertebral body defect without interbody arthrodesis, each contiguous defect (List separately in addition to code for primary procedure)
27301 - 27499	Femur (thigh region) and knee joint surgery
29065 - 29085	Application cast; upper extremity
29305 - 29355	Lower extremity casts
77072	Bone age studies
HCPCS codes covered if selection criteria are met:	
Opteform DBM, Bio DBM, Bioset, Propel DBM, Allosource femoral head bone graft, Optium allograft, Beta tri-calcium phosphate (b-TCP), Prime DBM HD, Kore Fiber, InterGro DBM, OsteoAMP, OsteoAMP Select and OsteoSelect, fiberFUSE demineralized bone matrix (DBM) allograft – no specific code	

Code	Code Description
HCPCS codes not covered for indications listed in the CPB :	
AlloStem; Arthrex biopaste (BioCartilage); ChronOS bone graft substitute, Cortiva Acellular Dermal Matrix, Acell biologic graft, BioD Restore, EmCell, kartogenin-treated autologous tendon graft, nacre (mother-of-pearl), Osteofuse, Rybone, SureFuse, SXBarrier (Surgilogix), Vivex Via graft (Amendia), Trufuse,Bioglass (45S5), Genetically corrected autologous epideral grafts, I-factor peptide enhanced bone graft, OsteoVive and synthetic bone graft, Intergro DBM Fibers, Vivigen, Osteomatrix or Arthrex Quickset (calcium phosphate cement), Avive tendon wrap, CartiMax allograft, Cerament bone void filler, collagen-based bio-inductive implants, Celling Biosciences Solum IV allograft, Equivabone Graft, Healos Bone Graft Replacement and Healos Sponge, Ceracell Ortho Foam, SurGenTec ViMax viable fiber allograft and Tactoset injectable bone substitute, AttraX Putty, Neocore, Ossifuse (boats, strips, and putty), Wright Ignite, Marlex mesh, Low-crystallinity apatite for bone reconstruction - no specific code	
C1763	Connective tissue, non-human (includes synthetic)
Other HCPCS codes related to the CPB:	
E0747	Osteogenesis stimulator, electrical, noninvasive, other than spinal applications
E0749	Osteogenesis stimulator, electrical, surgically implanted
Q4001 - Q4048	Casting supplies
ICD-10 codes not covered for indications listed in the CPB:	
M80.051A - M80.059S	Age-related osteoporosis with current pathological fracture, femur
M80.851A - M80.859S	Other osteoporosis with current pathological fracture, femur
Osteogenic Protein-1 (OP-1):	
Other CPT codes related to the CPB:	
22548 - 22819	Arthrodesis, spine [spinal fusion]
ICD-10 codes covered if selection criteria are met:	
Numerous options	Subsequent encounter for fracture with nonunion, clavicle [Codes not listed due to expanded specificity]
Numerous options	Subsequent encounter for fracture with nonunion, humerus [Codes not listed due to expanded specificity]

Code	Code Description
Numerous options	Subsequent encounter for fracture with nonunion, ulna and radius [Codes not listed due to expanded specificity]
Numerous options	Subsequent encounter for fracture with nonunion, metacarpal bone(s) [Codes not listed due to expanded specificity]
Numerous options	Subsequent encounter for fracture with nonunion, femur [Codes not listed due to expanded specificity]
Numerous options	Subsequent encounter for fracture with nonunion, tibia and fibula [Codes not listed due to expanded specificity]
Numerous options	Subsequent encounter for fracture with nonunion, metatarsal bone(s) [Codes not listed due to expanded specificity]
Numerous options	Subluxation and dislocation of vertebrae [Codes not listed due to expanded specificity]
C41.2	Malignant neoplasm of vertebral column
C41.4	Malignant neoplasm of pelvic bones, sacrum and coccyx
C70.1	Malignant neoplasm of spinal meninges
C79.31	Secondary malignant neoplasm of brain
C79.32	Secondary malignant neoplasm of cerebral meninges
C79.49	Secondary malignant neoplasm of other parts of nervous system
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.8	Benign neoplasm of pelvic bones, sacrum and coccyx
D32.1	Benign neoplasm of spinal meninges
D33.4	Benign neoplasm of spinal cord
D42.0 - D42.9	Neoplasm of uncertain behavior of meninges
D43.0 - D43.2	Neoplasm of uncertain behavior of brain
D43.4	Neoplasm of uncertain behavior of spinal cord
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage
G06.1	Intraspinal abscess and granuloma

Code	Code Description
M40.00 - M40.37, M40.50 - M40.57	Kyphosis and lordosis
M41.00 - M41.35, M41.80 - M41.9	Scoliosis
M43.00 - M43.19	Spondylolysis and spondylolisthesis
M46.20 - M46.28	Osteomyelitis of vertebra
M46.30 - M46.39	Infection of intervertebral disc (pyogenic)
M48.061 - M48.07	Spinal stenosis, lumbar and lumbosacral region
M48.50xA - M48.58xS	Collapsed vertebra, not elsewhere classified
M80.08XA - M80.08XS	Age-related osteoporosis with current pathological fracture, vertebra(e)
M84.48xA – M84.48xS, M84.58xA, - M84.58xS, M84.68xA – M84.68xS	Pathological fracture [vertebrae]
M86.18, M86.28, M86.68	Acute, subacute and other chronic osteomyelitis [spinal]
M89.68	Osteopathy after poliomyelitis, other site [spinal]
M90.88	Osteopathy in diseases classified elsewhere, other site [spinal]
M96.0	Pseudarthrosis after fusion or arthrodesis
M96.2 - M96.3	Postradiation and postlaminectomy kyphosis

Code	Code Description
M96.5	Postradiation scoliosis
M99.10 - M99.15	Subluxation complex (vertebral)
Q76.2	Congenital spondylolisthesis
S12.000A - S12.9xxS [S14.101A - S14.159S also required]	Fracture of cervical vertebra with cervical spinal cord injury
S22.000A - S22.089S [S24.101A - S24.159S also required]	Fracture of thoracic vertebra with thoracic spinal cord injury
S31.000A - S31.001S	Open wound of lower back and pelvis
S32.000A - S32.2xxS [S34.101A - S34.139S, S34.3xxA - S34.3xxA - S34.3xxS also required]	Fracture of lumbar vertebra, sacrum and coccyx with injury of lumbar and sacral spinal cord and cauda equina
S33.100A - S33.141S	Subluxation and dislocation of lumbar vertebra
Z98.1	Arthrodesis status [nonunion of prior fusion]
ICD-10 codes not covered for indications listed in the CPB:	
M43.20 – M43.28	Spinal fusion
O09.00 - O09.93	Supervision of high risk pregnancy

Code	Code Description
O10.011 - O21.9, O23.00 - O26.43, O26.611 - O26.839, O26.86, O26.891 - O26.93, O29.011 - O29.93, O31.00x0 - O31.03x9, O35.7xx0 - O35.7xx9, O36.80x0 - O36.80x9, O36.8210 - O36.8299, O44.00 - O60.23x9, O67.0 - O67.9, O86.11, O86.13 - O86.29, O90.5 - O90.81, O98.011 - O99.03, O99.280 - O99.325, O99.340 - O99.345, O99.511 - O99.835, O9A.111 - O9A.53	Maternal disorders predominantly related to pregnancy and childbirth
Q81.2	Epidermolysis bullosa dystrophica

Code	Code Description
Z34.00 - Z34.93	Encounter for supervision of normal pregnancy
Z39.0 - Z39.2	Encounter for maternal postpartum care and examination
Z85.00 - Z85.71, Z85.79 - Z85.9	Personal history of malignant neoplasm
InFuse Bone Graft (Bone Morphogenic Protein-2):	
InFuse Bone graft [covered for lumbar spine fusion; not covered for cervical fusion, ankle fusion] - no specific code:	
Other CPT codes related to the CPB:	
22548 – 22586	Arthrodesis, anterior or anterolateral approach technique
22590 – 22634	Arthrodesis, posterior, posterolateral or lateral transverse process technique
ICD-10 codes covered if selection criteria are met:	
Numerous options	Open fracture of shaft of tibia [for skeletally mature persons stabilized with intramedullary nail fixation after appropriate wound management and applied within 14 days after the initial fracture] [Codes not listed due to expanded specificity]
M43.25 - M43.27	Fusion of lumbar spine
M51.360 - M51.379	Other lumbar and lumbosacral intervertebral disc degeneration
Pro Osteon Hydroxyapatite Bone Graft Substitute:	
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
Numerous options	Nonunion of fracture [Codes not listed due to expanded specificity]
Numerous options	Subluxation and dislocation of vertebrae [Codes not listed due to expanded specificity]
C40.00 - C40.02	Malignant neoplasm of scapula and long bones of upper limb
C40.20 - C40.22	Malignant neoplasm of long bones of lower limb
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow

Code	Code Description
D16.00 - D16.02	Benign neoplasm of scapula and long bones of upper limb
D16.20 - D16.22	Benign neoplasm of long bones of lower limb
K60.30 - K60.329, K60.50 - K60.529	Anal and anorectal fistula
M40.00 - M41.9	Kyphosis, lordosis and scoliosis
M43.00 - M43.19, M43.8X1 - M43.9	Other deforming dorsopathies
M50.00 - M50.03	Cervical disc disorder with myelopathy
M50.30 - M50.33	Other cervical disc degeneration
M51.04 - M51.06	Thoracic, thoracolumbar and lumbosacral intervertebral disc disorders with myelopathy
M51.34 - M51.379	Other thoracic, thoracolumbar and lumbosacral intervertebral disc degeneration
M51.9	Unspecified thoracic, thoracolumbar and lumbosacral intervertebral disc disorder
M85.00 - M85.09	Fibrous dysplasia (monostotic)
M85.40 - M85.69	Solitary bone cyst, aneurysmal bone cyst and other cyst of bone
M96.1 - M96.5	Postprocedural complications and disorders of musculoskeletal system, not elsewhere classified
M99.10 - M99.15	Subluxation complex (vertebral)
Q67.5	Congenital deformity of spine

Code	Code Description
Q76.2	Congenital spondylolisthesis
Q76.3	Congenital scoliosis due to congenital bony malformation
Q76.411 - Q76.49	Other congenital malformations of spine, not associated with scoliosis
S12.000A - S12.9xxS [S14.101A - S14.159S also required]	Fracture of cervical vertebra with cervical spinal cord injury
S22.000A - S22.089S [S24.101A - S24.159S also required]	Fracture of thoracic vertebra with thoracic spinal cord injury
S32.000A - S32.2xxS [S34.101A - S34.139S, S34.3xxA- S34.3xxS also required]	Fracture of lumbar vertebra, sacrum and coccyx with injury of lumbar and sacral spinal cord and cauda equina
S42.451A, S42.452A, S42.453A, S42.454A, S42.455A, S42.456A, S42.461A, S42.462A, S42.463A, S42.464A, S42.465A, S42.466A	Closed fracture of lateral and medial condyle of humerus

Code	Code Description
S49.101A, S49.102A, S49.109A, S49.111A, S49.112A, S49.119A, S49.121A, S49.122A, S49.129A, S49.131A, S49.132A, S49.139A, S49.141A, S49.142A, S49.149A, S49.191A, S49.192A, S49.199A	Closed physeal fracture of lower end of humerus
S52.601A, S52.602A, S52.609A, S52.611A, S52.612A, S52.613A, S52.614A, S52.615A, S52.616A, S52.691A, S52.692A, S52.699A	Closed fracture of lower end of ulna

Code	Code Description
S59.001A, S59.002A, S59.009A, S59.011A, S59.012A, S59.019A, S59.021A, S59.022A, S59.029A, S59.031A, S59.032A, S59.039A, S59.041A, S59.042A, S59.049A, S59.091A, S59.092A, S59.099A	Closed physeal fracture of lower end of ulna
S72.021A - S72.021C, S72.022A - S72.022C, S72.023A - S72.023C, S72.024A - S72.024C, S72.025A - S72.025C, S72.026A - S72.026C	Fracture of epiphysis (separation) (upper) of femur
S72.441A, S72.442A, S72.443A, S72.444A, S72.445A, S72.446A	Closed fracture of lower epiphysis (separation) of femur

Code	Code Description
S79.001A, S79.002A, S79.009A, S79.011A, S79.012A, S79.019A, S79.091A, S79.092A, S79.099A	Closed physeal fracture of upper end of femur
S79.101A, S79.102A, S79.109A, S79.111A, S79.112A, S79.119A, S79.121A, S79.122A, S79.129A, S79.131A, S79.132A, S79.139A, S79.141A, S79.142A, S79.149A, S79.191A, S79.192A, S79.199A	Closed physeal fracture of lower end of femur
Platelet-Rich Plasma:	
CPT codes not covered for indications listed in the CPB:	
0232T	Injection(s), platelet rich plasma, any tissue, including image guidance, harvesting and preparation when performed [Platelet-rich fibrin for lateral window sinus lifts]
0481T	Injection(s), autologous white blood cell concentrate (autologous protein solution), any site, including image guidance, harvesting and preparation, when performed

Code	Code Description
Other CPT codes related to the CPB:	
12001 - 12021	Repair - simple
12031 - 12057	Repair - intermediate
HCPCS codes covered if selection criteria are met :	
P9020	Platelet rich plasma, each unit
HCPCS codes not covered for indications listed in the CPB:	
S9055	Procuren or other growth factor preparation to promote wound healing
Other HCPCS codes related to the CPB:	
C1602	Orthopedic/device/drug matrix/absorbable bone void filler, antimicrobial-eluting (implantable)
C9359	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Putty, Integra OS Osteoconductive Scaffold Putty), per 0.5 cc
C9362	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Strip), per 0.5 cc
D6010	Surgical placement of implant body: endosteal implant
D6040	Implant-supported porcelain-fused-to-metal crown
D7951	Sinus augmentation via lateral open approach
D7952	Sinus augmentation via vertical approach
P9022	Red blood cells, washed, each unit
ICD-10 codes covered if selection criteria are met :	
D47.3	Essential (hemorrhagic) thrombocythemia
D69.41 - D69.6	Other primary and secondary thrombocytopenia
Beta-tricalcium phosphate:	
CPT codes not covered for indications listed in the CPB:	
Beta-tricalcium phosphate, Native bone proteins - No specific code	
Other CPT codes related to the CPB:	
27870	Arthrodesis, ankle, open
27871	Arthrodesis, tibiofibular joint, proximal or distal
28705	Arthrodesis; pantalar
28715	Arthrodesis; triple

Code	Code Description
28725	Arthrodesis; subtalar
29899	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; with ankle arthrodesis
Porcine Intestinal Submucous Surgical Mesh:	
CPT codes not covered for indications listed in the CPB:	
46707	Repair of anorectal fistula with plug (e.g., porcine small intestine submucosa [SIS])
Nucleus pulposus allograft:	
CPT codes not covered for indications listed in the CPB:	
0627T	Percutaneous injection of allogeneic cellular and/or tissue-based product, intervertebral disc, unilateral or bilateral injection, with fluoroscopic guidance, lumbar; first level
0628T	Percutaneous injection of allogeneic cellular and/or tissue-based product, intervertebral disc, unilateral or bilateral injection, with fluoroscopic guidance, lumbar; each additional level
0629T	Percutaneous injection of allogeneic cellular and/or tissue-based product, intervertebral disc, unilateral or bilateral injection, with CT guidance, lumbar; first level
0630T	Percutaneous injection of allogeneic cellular and/or tissue-based product, intervertebral disc, unilateral or bilateral injection, with CT guidance, lumbar; each additional level
ICD-10 codes not covered for indications listed in the CPB:	
M54.50 - M54.59	Low back pain
Allograft for Spinal Fusion:	
CPT codes covered if selection criteria are met:	
20930	Allograft for spine surgery only; morselized
20931	Allograft for spine surgery only; structural
HCPCS codes covered if selection criteria are met:	
C1762	Connective tissue, human (includes fascia lata)
ICD-10 codes covered if selection criteria are met:	
M43.20 - M43.28	Fusion of spine

Code	Code Description
Bone Void Fillers for Nonunions:	
HCPCS codes covered if selection criteria are met:	
NuVasive Propel DBM putty- no specific code:	
C9359	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Putty, Integra OS Osteoconductive Scaffold Putty), per 0.5 cc
C9362	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Strip), per 0.5 cc
ICD-10 codes covered if selection criteria are met:	
M43.20 - M43.28	Fusion of spine
ICD-10 codes not covered for indications listed in the CPB:	
Numerous options	Delayed healing of fracture [Codes not listed due to expanded specificity] 7th character G [when structural integrity is required, subsequent encounter for fracture with delayed healing]
Numerous options	Malunion of fracture [Codes not listed due to expanded specificity] 7th character K [when structural integrity is required, covered as an adjunct to approved biologic or non-biologic implants]
Numerous options	Nonunion of fracture [Codes not listed due to expanded specificity] 7th character P
Polymethylmethacrylate (PMMA) Antibiotic beads:	
CPT codes covered if selection criteria are met:	
Calcium sulfate antibiotic beads (e.g., Osteoset Resorbable Mini-Bead) – no specific code:	
0707T	Injection(s), bone substitute material (eg, calcium phosphate) into subchondral bone defect (ie, bone marrow lesion, bone bruise, stress injury, microtrabecular fracture), including imaging guidance and arthroscopic assistance for joint visualization
11981	Insertion, non-biodegradable drug delivery implant
11982	Removal, non-biodegradable drug delivery implant
11983	Removal with reinsertion, non-biodegradable drug delivery implant

Code	Code Description
+20700	Manual preparation and insertion of drug-delivery device(s), deep (eg, subfascial) (List separately in addition to code for primary procedure)
+20701	Removal of drug-delivery device(s), deep (eg, subfascial) (List separately in addition to code for primary procedure)
+20702	Manual preparation and insertion of drug-delivery device(s), intramedullary (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
M86.30 – M86.69	Chronic osteomyelitis [PMMA antibiotic beads are covered when used with IV antibiotics in the treatment of chronic osteomyelitis]
Mesenchymal Stem Cell Therapy/Bone Marrow Aspirate:	
CPT codes not covered for indications listed in the CPB:	
38232	Bone marrow harvesting for transplantation; autologous
38240 - 38241	Hematopoietic progenitor cell (HPC) transplantation
Other CPT codes related to the CPB:	
20615	Aspiration and injection for treatment of bone cyst
20939	Bone marrow aspiration for bone grafting, spine surgery only, through separate skin or fascial incision (List separately in addition to code for primary procedure)
22548 - 22819	Arthrodesis, spine
HCPCS codes not covered for indications listed in the CPB:	
S2142	Cord blood-derived stem-cell transplantation, allogeneic
S2150	Bone marrow or blood-derived stem cells (peripheral or umbilical), allogeneic or autologous, harvesting, transplantation, and related complications; including: pheresis and cell preparation/storage; marrow ablative therapy; drugs, supplies, hospitalization with outpatient follow-up; medical/surgical, diagnostic, emergency, and rehabilitative services; and the number of days of pre- and post-transplant care in the global definition
ICD-10 codes covered if selection criteria are met:	
C00.0 - D49.9	Neoplasms

Code	Code Description
D50.0 - D89.9	Diseases of the Blood and Blood-forming organs and certain disorders involving the immune mechanism
M85.40 - M85.69	Cyst of bone
ICD-10 codes not covered for indications listed in the CPB:	
Numerous options	Malunion of fractures [Codes not listed due to expanded specificity]
Numerous options	Nonunion of fracture [Codes not listed due to expanded specificity]
M93.20 - M93.29	Osteochondritis dissecans
M96.0	Pseudarthrosis after fusion or arthrodesis
Z98.1	Arthrodesis status
Hydroxyapatite bone substitute (e.g., OtoMimix) for middle ear surgery:	
No specific code	
Experimental and investigational Substitutes and Adjuncts:	
HCPCS codes not covered for indications listed in the CPB:	
Q4116	Alloderm, per square centimeter
Q4173	Palingen or palingen xplus, per square centimeter
Q4174	Palingen or promatrx, 0.36 mg per 0.25 cc
Q4195 - Q4197	Puraply
Tendon Wrap Tendon Protector:	
HCPCS codes not covered for indications listed in the CPB:	
C9356	Tendon, porous matrix of cross-linked collagen and glycosaminoglycan matrix (Tenoglide Tendon Protector Sheet), per square centimeter [Tendon Wrap Tendon Protector]
Bone Void Fillers for use in spinal fusions:	
Other CPT codes related to the CPB:	
22548 - 22819	Arthrodesis, spine [spinal fusion]
HCPCS codes covered if selection criteria are met:	
C9359	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Putty, Integra OS Osteoconductive Scaffold Putty), per 0.5 cc

Code	Code Description
C9362	Porous purified collagen matrix bone void filler (Integra Mozaik Osteoconductive Scaffold Strip), per 0.5 cc
Anti-microbial bone graft substitutes :	
HCPCS codes not covered for indications listed in the CPB :	
C1602	Orthopedic/device/drug matrix/absorbable bone void filler, antimicrobial-eluting (implantable) [PerOssal (ceramic bone substitute)]
ICD-10 codes not covered for indications listed in the CPB:	
M86.00 - M86.9	Osteomyelitis
Numerous options	Subsequent encounter for fracture with nonunion, clavicle, humerus, ulna and radius, metacarpal bone(s), femur, tibia and fibula, metatarsal bone(s), [Codes not listed due to expanded specificity]
Bone micro-indentation testing:	
CPT codes not covered for indications listed in the CPB:	
0547T	Bone-material quality testing by microindentation(s) of the tibia(s), with results reported as a score
Subchondroplasty:	
CPT codes not covered for indications listed in the CPB:	
0707T	Injection(s), bone-substitute material (eg, calcium phosphate) into subchondral bone defect (ie, bone marrow lesion, bone bruise, stress injury, microtrabecular fracture), including imaging guidance and arthroscopic assistance for joint visualization
0869T	Injection(s), bone-substitute material for bone and/or soft tissue hardware fixation augmentation, including intraoperative imaging guidance, when performed
Xenograft implantation into articular surfaces:	
CPT codes not covered for indications listed in the CPB:	
0737T	Xenograft implantation into the articular surface

Background

Osteogenic proteins, also referred to as bone morphogenetic, or morphogenic proteins (BMPs), are a family of bone-matrix polypeptides isolated from a variety of mammalian species. Implantation of OPs induces a sequence of cellular events that lead to the formation of new bone. Some of the potential clinical applications of OPs are: (i) as a bone graft substitute to promote spinal fusion and to aid in the incorporation of metal implants, (ii) to improve the performance of autograft and allograft bone, and (iii) as an agent for osteochondral defects. Biologic allograft materials can include allografts made from bone, allografts containing stem cells or other materials besides bone, or a combination of both.

Recombinantly produced human osteogenic protein-1 (OP-1), also known as BMP-7, was developed by Stryker Biotech (Hopkinton, MA), a division of Stryker Corporation. The OP-1 Implant was approved by the Food and Drug Administration (FDA) as a Humanitarian Use Device (HUD). As defined in the Federal Food, Drug and Cosmetic Act (21 CFR 814.124), a HUD "is a device that is intended to benefit patients in the treatment and diagnosis of diseases or conditions that affect or is manifested in fewer than 4,000 individuals in the United States per year." The FDA developed the HUD categorization to provide an incentive for the development of devices for use in the treatment or diagnosis of diseases affecting small patient populations.

The manufacturer submitted to the FDA results from a multi-center Long Bone Treatment Study, where 10 patients with long bone nonunions having prior failed autograft were treated with OP-1 implant. Seven of the 10 patients had clinical healing (pain and function), and 2 of 10 had radiographic healing (bridging in 3 or 4 cortices).

The manufacturer also submitted the results of the multi-center Tibial Nonunion Study, where a subset of 14 patients with prior failed autograft was treated with the OP-1 Implant, and 13 patients were treated with autograft. Twelve of patients receiving the OP-1 Implant had clinical resolution (pain and function) of their nonunion, and 8 patients had radiographic healing (bridging in three views). By comparison, 12 of 13

patients receiving autograft had clinical resolution of their nonunion, and 12 of 13 had radiographic healing. The FDA concluded that, although the OP-1 implant was an effective treatment for nonunions, the implant was not as effective as autograft. Therefore, the FDA product labeling states that the OP-1 bone morphogenic protein is indicated "for use as an alternative to autograft in recalcitrant long bone nonunions *where use of autograft is unfeasible* and alternative treatments have failed" (emphasis added).

As of 2014, the manufacturer of the OP-1 implant ceased production and removed the product from the United States.

Friedlaender et al. (2001) reported on the results of a randomized, controlled, single- blind multi-center clinical trial where 122 patients with 124 tibial nonunions were assigned to either OP-1 Implant or bone autograft. The OP-1 Implant was found to be less effective than bone autograft. After 9 months of treatment, 81 % of the OP-1-treated nonunions and 85 % of patients receiving autogenous bone were judged by clinical criteria to have been treated successfully, and 75 % of OP-1 treated patients and 84 % of autograft-treated patients had healed fractures by radiographic criteria.

In a randomized study, Johnsson et al (2002) examined whether OP-1 (BMP-7) in the OP-1 Implant yields better stabilizing bony fusion than autograft bone in patients undergoing posterolateral fusion between L5 and S1. A total of 20 patients were randomized to fusion with either OP-1 Implant (n = 10) or autograft bone from the iliac crest (n = 10). The patients were instructed to keep the trunk straight for 5 months after surgery with the aid of a soft lumbar brace. At surgery 0.8-mm metallic markers were positioned in L5 and the sacrum, enabling radio-stereometric follow-up analysis during 1 year. No significant difference was observed between the radio-stereometric and radiographic results of fusion with the OP-1 Implant and fusion with autograft bone. Thus, the OP-1 Implant did not yield better stabilizing bony fusion than autograft bone.

Sandhu et al (2003) stated that OP-1 has been studied in limited pilot studies of posterolateral fusion. It is unclear whether the addition of OP-1 ensures arthrodesis in this application.

Vaccaro et al (2008) examined the safety and the clinical and radiographical efficacy of OP-1 (rhBMP-7) Putty as compared with an iliac crest bone autograft control in un-instrumented, single-level postero-lateral spinal arthrodesis. A total of 335 patients were randomized in 2:1 fashion to receive either OP-1 Putty or autograft for degenerative spondylolisthesis and symptomatic spinal stenosis. Patients were observed serially with radiographs, clinical examinations, and appropriate clinical indicators, including Oswestry Disability Index (ODI), Short-Form 36, and visual analog scale scores. Serum samples were examined at regular intervals to assess the presence of antibodies to OP-1. The primary end point, "overall success", was analyzed at 24 months. The study was extended to include additional imaging data and long-term clinical follow-up at 36+ months. At the 36+ month time point, computed tomography (CT) scans were obtained in addition to plain radiographs to evaluate the presence and location of new bone formation. Modified overall success, including improvements in ODI, absence of re-treatment, neurological success, absence of device-related serious adverse events, angulation and translation success, and new bone formation by CT scan (at 36+ months), was then calculated using the 24-month primary clinical endpoints, updated retreatment data, and CT imaging and radiographical end points. OP-1 Putty was demonstrated to be statistically equivalent to autograft with respect to the primary end point of modified overall success. The use of OP-1 Putty when compared to autograft was associated with statistically lower intra-operative blood loss and shorter operative times. Although patients in the OP-1 Putty group demonstrated an early propensity for formation of anti-OP-1 antibodies, this resolved completely in all patients with no clinical sequelae. The authors concluded that OP-1 Putty is a safe and effective alternative to autograft in the setting of un-instrumented postero-lateral spinal arthrodesis performed for degenerative spondylolisthesis and symptomatic spinal stenosis.

Bone morphogenetic protein-2 (BMP-2) was approved by the FDA as a bone graft substitute in anterior lumbar interbody fusions. It has also been used off-label in anterior cervical fusions. Smucker and colleagues (2006) examined if BMP-2 is associated with an increased incidence of clinically relevant post-operative pre-vertebral swelling problems in patients undergoing anterior cervical fusions. A total of 234 consecutive patients (aged 12 to 82 years) undergoing anterior cervical fusion with

and without BMP-2 over a 2-year period at one institution comprised the study population. The incidence of clinically relevant pre-vertebral swelling was calculated. The populations were compared and statistical significance was determined. A total of 234 patients met the study criteria, 69 of whom underwent anterior cervical spine fusions using BMP-2; 27.5 % of those patients in the BMP-2 group had a clinically significant swelling event versus only 3.6 % of patients in the non-BMP-2 group. This difference was statistically significant ($p < 0.0001$) and remained so after controlling for other significant predictors of swelling. The authors concluded that off-label use of BMP-2 in the anterior cervical spine is associated with an increased rate of clinically relevant swelling events.

In a systemic review, Mussano et al (2007) examined if BMPs are more effective in treating bone defects than traditional techniques, such as grafting autologous bone. An electronic search was made in the databases of MEDLINE, EMBASE (through MeSH and Emtree), and the Cochrane Central Register of Controlled Trials with no linguistic restrictions. Randomized controlled trials (RCTs) that compared bone regeneration achieved through BMPs versus that obtained by traditional methods entered the study. The 17 publications that met the criteria, divided into subgroups by type of bone, were tabulated by salient characteristics and evaluated through the items proposed by van Tulder et al. However, as the studies differed widely (in terms of site, sample size, dosage of active principle, carrier, clinical and radiological data recording), it was possible to carry out a meta-analysis of clinical and radiological outcome only for the subgroup that evaluated the vertebrae, where it was observed that BMPs offer a slightly but statistically significant greater efficacy than do traditional techniques. The authors concluded that the use of BMPs at the vertebrae can eliminate the need for surgery to harvest autologous bone. The only large study carried out on the other sites suggested that BMPs should be used at a concentration of 1.5 mg/ml to treat fractures of the tibia. The authors stated that further RCTs of good methodological quality are needed to clarify the effectiveness of BMPs in clinical practice.

The Pro Osteon Bone Graft Substitute (Interpore International) is a hydroxyapatite bone allograft material made from marine coral. The product was approved by the FDA in 1992 as a bone void filler for repair of metaphyseal defects and long bone cyst and tumor defects. The

product is to be used in conjunction with rigid internal fixation, as the Pro Osteon does not possess sufficient strength to support the reduction of a defect site prior to hard tissue ingrowth. External stabilization is not sufficient.

Pro Osteon coralline hydroxyapatite is not indicated for spinal fusion or fractures of the epiphyseal plate. A prospective randomized controlled clinical study directly compared coralline hydroxyapatite to iliac crest grafts in spinal fusion and found that the coralline graft “does not possess adequate structural integrity to resist axial loading and maintain disc height or segmental lordosis during cervical interbody fusion” (McConnell et al, 2003).

The INFUSE Bone Graft/LT-CAGE Lumbar Tapered Fusion Device (Medtronic Sofamor Danek) includes recombinant human bone morphogenic protein 2 (rhBMP-2) in a collagen absorbable sponge and a tapered titanium spinal cage, and has been approved for spinal fusion in persons with single-level degenerative disc disease from L4 to S1, where the patient has had at least 6 months of nonoperative treatment, and the device is to be used via an anterior approach. Studies submitted to the FDA compared the INFUSE Bone Graft to autogenous iliac crest bone graft in patients with degenerative lumbar disc disease. These studies showed clinically equivalent fusion rates between the 2 groups, with similar outcomes in terms of back pain, leg pain, disability and neurological status. The primary advantage of use of the device is that it does not require harvesting of autologous bone.

The California Technology Assessment Forum (CTAF) (Feldman, 2005) concluded that rhBMP-2 carried on a collagen sponge used in conjunction with an FDA approved device meets CTAF criteria for the treatment of patients undergoing single level anterior lumbar interbody spinal fusion for symptomatic single level degenerative disease at L4 to S1 of at least 6 months duration that has not responded to non-operative treatments. The California Technology Assessment Forum concluded that all other uses of rhBMP-2 including its use in cervical spinal fusions and for treatment of open tibial fracture do not meet CTAF criteria.

An evidence review prepared for the Ontario Ministry of Health and Long-Term Care (2004) found that "[t]he largest number of spinal fusion cases using BMP devices has been for anterior lumbar interbody fusion. Although radiologic fusion occurs at a consistently faster rate among recipients of the BMP device than among recipients of autologous bone grafts, clinical outcomes (pain and disability) appear no different. Regardless of technique, improvements in pain and disability are reported by similar proportions of participants in all the arms of all the trials."

In a study on occipito-cervical fusion using recombinant human BMP-2, Shahlaie and Kim (2008) stated that INFUSE should not be routinely used for occipito-cervical fusion. They noted that further studies are needed to determine if modified techniques such as intra-operative steroids and extended post-operative use of wound drains, can improve safety of its use in the posterior cervical region.

On July 1, 2008, the FDA issued a Public Health Notification to health care providers regarding life-threatening complications arising from off-label use of INFUSE Bone Graft in cervical spinal fusion. These complications included swelling of the neck and throat tissue that caused difficulty breathing, swallowing or speaking. People who suffered these adverse events needed respiratory support with intubation, medication or tracheotomy.

On April 30, 2018, Medtronic announced the FDA approval of INFUSE Bone Graft for use with additional spine implants made of PEEK in oblique lateral interbody fusion (OLIF 25 and OLIF 51) and anterior lumbar interbody fusion (ALIF) procedures at a single level (Medtronic, 2018).

- Use in OLIF 51 procedures with Divergence-L Interbody Fusion Device at a single level from L5-S1
- Use in OLIF 25 procedures with Pivox Oblique Lateral Spine System at a single level from L2-L5
- Use in ALIF procedures with Divergence-Linterbody Fusion Device at a single level from L2-S1

Platelet-Rich Plasma

Regeneration of guided bone is an established procedure used in implant dentistry to increase the quality and quantity of the host bone in sites of localized alveolar defects. Improvement in the osteo-inductive properties of currently available grafting materials is needed because of the lack of predictability in osseous regenerative procedures with these materials. Platelet-rich plasma (PRP), a modification of fibrin glue derived from autologous blood, is being used to deliver growth factors in high concentration to areas requiring osseous grafting. Growth factors released from the platelets include platelet-derived growth factor, transforming growth factor beta, platelet-derived epidermal growth factor, platelet-derived angiogenesis factor, insulin-like growth factor 1, and platelet factor 4. These factors signal the local mesenchymal and epithelial cells to migrate, divide, and increase collagen and matrix synthesis. PRP, as an adjunctive material with bone grafts during augmentation procedures, has been suggested to increase quality of bone regeneration and the rate of bone deposition.

In a randomized controlled study (n = 10), Kassolis and Reynolds (2005) compared bone formation after sub-antral maxillary sinus augmentation with freeze-dried bone allograft (FDBA) plus PRP versus FDBA plus resorbable membrane. The authors reported that the combination of FDBA and PRP enhanced the rate of formation of bone compared with FDBA and membrane, when used in sub-antral sinus augmentation. The investigators concluded, however, that more studies are needed to determine if such incremental enhancements in bone formation affect clinical outcome.

In a randomized controlled study, Camargo et al (2005) compared the clinical effectiveness of a combination therapy consisting of bovine porous bone mineral (BPBM), guided tissue regeneration (GTR), and PRP in the regeneration of periodontal intra-bony defects in humans. Twenty-eight paired intra-bony defects were surgically treated using a split-mouth design. Defects were treated with BPBM, GTR, and PRP (experimental), or with open-flap debridement (control). Clinical parameters evaluated included changes in attachment level, pocket depth, and defect fill as revealed by re-entry at 6 months. Pre-operative pocket depths, attachment levels, and trans-operative bone

measurements were similar for the 2 groups. Post-surgical measurements taken at 6 months revealed that both treatment modalities significantly decreased pocket depth and increased clinical attachment and defect fill compared to baseline. The differences between the experimental and control groups were 2.22 (+/- 0.39) mm on buccal and 2.12 (+/- 0.34) mm on lingual sites for pocket depth, 3.05 (+/- 0.51) mm on buccal and 2.88 (+/- 0.46) mm on lingual sites for gain in clinical attachment, and 3.46 (+/- 0.96) mm on buccal and 3.42 (+/- 0.02) mm on lingual sites for defect fill. These differences between groups were statistically significant in favor of the experimental defects. The combined therapy was also clinically more effective than open-flap debridement. The authors stated that the superiority of the experimental group could not be attributed solely to the surgical intervention and was likely a result of the BPBM/GTR/ PRP application. The authors concluded that combining BPBM, GTR, and PRP was an effective modality of regenerative treatment for intra-bony defects in patients with advanced periodontitis.

Lekovic and colleagues (2003) examined the effectiveness of PRP, BPBM and GTR used in combination as regenerative treatment for grade II molar furcation defects in humans (n = 52). These investigators concluded that the PRP/BPBM/GTR combined technique is an effective modality of regenerative treatment for mandibular grade II furcation defects. Moreover, they stated that further studies are necessary to elucidate the role played by each component of the combined therapy in achieving these results.

Recent reviews have reached contradictory findings regarding the effectiveness of PRP for bone grafting. Marx (2004) stated that PRP remains the only effective growth factor preparation available to oral and maxillofacial surgeons as well as other dental specialists for outpatient use. In contrast, Freymiller and Aghaloo (2004) stated: "Practitioners involved with bone grafting have high hopes that PRP will be proven to be of benefit in bone graft healing. However, at this early stage of investigation, the results are inconclusive. There is still much to learn regarding PRP before this adjunctive material should be considered for routine use. Unfortunately, this has not been the case because an entire industry has developed to manufacture the equipment and supplies needed for surgeons to prepare PRP in the office or operating room.

Courses are being offered throughout the United States touting the benefits of PRP. Considering the meager volume and contradictory nature of the currently available evidence, there appears to be a disproportionate use of PRP in clinical practice." These authors concluded that more research (especially well-designed, rigorous, standardized human trials) is needed before evidence-based surgeons can feel confident in recommending this procedure/material to their patients.

These conclusions are in agreement with the observations of Sanchez et al (2003) and Grageda (2004). Sanchez et al (2003) stated that “there is clearly a lack of scientific evidence to support the use of PRP in combination with bone grafts during augmentation procedures. This novel and potentially promising technique requires well-designed, controlled trials to provide evidence of effectiveness.” Grageda (2004) stated that since the introduction of PRP, several investigators have examined its effectiveness using various bone grafting materials. There have been different protocols as well as different types of clinical cases. The author concluded that “there is an urgent need not just for more, but for standardized research studies in this subject to provide evidence-based dentistry to patients. Without the standardization of these protocols, it will be extremely difficult to ascertain whether PRP enhances bone healing when it is used alone or in conjunction with bone grafting materials."

A systematic evidence review of surgical techniques for placing dental implants prepared for the Cochrane Collaboration (Coulthard et al, 2003) concluded that there is no strong evidence that the use of PRP or other variations in surgical technique described in the review for placing implants have superior success rates.

Devices to prepare PRP have been cleared by the FDA based on 510(k) premarket notification. The FDA has required that the product labeling for one such device state that “[t]he Platelet Rich Plasma prepared by this device has not been evaluated for any clinical indications” (Golding, 2004).

Recent studies also produced contradictory findings on the clinical value of PRP. While Okuda et al (2005) reported that treatment with a combination of PRP and porous hydroxyapatite (HA) compared to HA with saline led to a significantly more favorable clinical improvement in intra-bony periodontal defects (n = 70), and Sammartino et al (2005) found that PRP is effective in inducing and accelerating bone regeneration for the treatment of periodontal defects at the distal root of the mandibular second molar after surgical extraction of a mesioangular, deeply impacted mandibular third molar (n = 18), results from other studies indicated that PRP does not provide any added benefits.

In a randomized controlled study (n = 24), Huang et al (2005) examined the effects of PRP in combination with coronally advanced flap (CAF) for the treatment of gingival recession. These investigators concluded that the application of PRP in CAF root coverage procedure provides no clinically measurable enhancements on the final therapeutic outcomes of CAF in Miller's Class I recession defects. Furthermore, in a controlled clinical trial (n = 10), Monov et al (2005) found that the instillation of PRP during implant placement in the lower anterior mandible did not add additional benefit. These findings are in agreement with the observation of Raghoebard et al (2005) who noted that no beneficial effect of PRP on wound healing and bone remodeling of autologous bone grafts used for augmentation of the floor of the maxillary sinus.

In a review on the role of PRP in sinus augmentation, Boyapati and Wang (2006) stated that although the lateral wall sinus lift is a predictable clinical procedure to increase vertical bone height resulting in implant success rates comparable to that of native bone, the issue of extended healing periods remains troublesome. Clinicians and researchers have investigated several methods, including addition of growth factors and peptides, to reduce this healing time and enhance bone formation within the subantral environment. Platelet-rich plasma is an autologous blood product containing high concentrations of several growth factors and adhesive glycoproteins. The incorporation of PRP into the sinus graft has been proposed as a method to shorten healing time, enhance wound healing, and improve bone quality. These investigators noted that currently, the literature is conflicting with respect to the adjunctive use of PRP in sinus augmentation. Factors that may contribute to this variability include variable/inappropriate study design, under-powered studies,

differing platelet yields, and differing graft materials used. In addition, methods of quantifying bone regeneration and wound healing differ between studies. Currently, because of limited scientific evidence, the adjunctive use of PRP in sinus augmentation cannot be recommended. The authors stated that further prospective clinical studies are urgently needed.

In a randomized controlled trial, de Vos et al (2010) examined if a PRP injection would improve outcome in chronic mid-portion Achilles tendinopathy. A stratified, block-randomized, double-blind, placebo-controlled study at a single center of 54 randomized patients aged 18 to 70 years with chronic tendinopathy 2 to 7 cm above the Achilles tendon insertion were carried out. The trial was conducted between August 28, 2008, and January 29, 2009, with follow-up until July 16, 2009. Subjects received eccentric exercises (usual care) with either a PRP injection (PRP group) or saline injection (placebo group). Randomization was stratified by activity level. Main outcome measure was the validated Victorian Institute of Sports Assessment-Achilles (VISA-A) questionnaire, which evaluated pain score and activity level; and was completed at baseline and 6, 12, and 24 weeks. The VISA-A score ranged from 0 to 100, with higher scores corresponding with less pain and increased activity. Treatment group effects were evaluated using general linear models on the basis of intention-to-treat. After randomization into the PRP group (n = 27) or placebo group (n = 27), there was complete follow-up of all patients. The mean VISA-A score improved significantly after 24 weeks in the PRP group by 21.7 points (95 % confidence interval [CI]: 13.0 to 30.5) and in the placebo group by 20.5 points (95 % CI: 11.6 to 29.4). The increase was not significantly different between both groups (adjusted between-group difference from baseline to 24 weeks, -0.9; 95 % CI: -12.4 to 10.6). This CI did not include the pre-defined relevant difference of 12 points in favor of PRP treatment. The authors concluded that among patients with chronic Achilles tendinopathy who were treated with eccentric exercises, a PRP injection compared with a saline injection did not result in greater improvement in pain and activity.

In a decision memorandum, the Centers for Medicare & Medicaid Services (CMS, 2008) determined that the evidence is inadequate to conclude that autologous PRP for the treatment of chronic non-healing cutaneous wounds, acute surgical wounds when the autologous PRP is

applied directly to the closed incision, or dehiscent wounds improves health outcomes. Therefore, CMS determined that PRP is not reasonable and necessary for the treatment of these indications. Consequently, CMS issued a non-coverage determination for acute surgical wounds when the autologous PRP is applied directly to the closed incision and for dehiscent wounds. CMS also maintained the current non-coverage for chronic, non-healing cutaneous wounds.

In a systematic review on the safety and effectiveness of the use of autologous PRP for tissue regeneration, Martínez-Zapata et al (2009) concluded that PRP improves the gingival recession but not the clinical attachment level in chronic periodontitis. In the complete healing process of chronic skin ulcers, the results are inconclusive. There are little data regarding the safety of PRP. There are several methodological limitations and, consequently, future research should focus on strong and well-designed RCTs that evaluate the safety and effectiveness of PRP.

Guidelines from the Work Loss Data Institute (2008) on work-related disorders of the elbow state that platelet-rich plasma and autologous blood donation are under study and are not specifically recommended.

An assessment by the Institute for Clinical Effectiveness and Health Policy (IECS, 2008) concluded that, "although in vitro, PRP has demonstrated to release growth factors and to improve tendon structure, so far, there is no evidence supporting its use in human beings."

Porcine Intestinal Submucosa Surgical Mesh

The rotator cuff is comprised of four muscles (i.e., infraspinatus, subscapularis, supraspinatus and teres minor) that originate from the scapula. The tendons of these muscles form a single tendon unit, which inserts onto the greater tuberosity of the humerus. These “structures” combine to form a “cuff” over the head of the humerus. The rotator cuff helps to lift and rotate the arm as well as to stabilize the ball of the shoulder within the joint.

Tears of the rotator cuff tendons are one of the most common causes of pain, loss of motion, and disability in adults. Traditional treatments include conservative interventions (e.g., rest and limited overhead activity,

use of a sling, non-steroidal anti-inflammatory drugs, oral glucocorticoid, strengthening exercise and physical therapy, intra-articular or subacromial glucocorticosteroid injection), and surgery (arthroscopic or open). Non-surgical treatments, which may take several weeks or months, produce pain relief in approximately 50 % of patients and no improvement in strength at long-term follow-up, whereas surgical intervention results in pain relief in about 85 % of patients and a better return of strength (Ruotolo and Nottage, 2002). Following rotator cuff repair surgery, the arm is immobilized to allow the tear to heal. The length of immobilization is usually dependent on the severity of the tear. Furthermore, patients' commitment/compliance to rehabilitation is important to attain a good surgical outcome.

Recent developments in rotator cuff repair surgery include newer arthroscopic and mini-open surgical techniques. These new techniques are intended to allow for smaller, less painful incisions and faster recovery time. Many of these advances use dissolvable anchors, which hold sutures in place or hold sutures down to bone until the repair has healed and then are absorbed by the body. There is also ongoing research on orthobiologic tissue implants that is intended to enhance healing and promote growth of new tissue.

A surgical mesh composed of porcine small intestinal submucosa (Restore Orthobiologic Soft Tissue Implant, DePuy Orthopaedics, Inc., Warsaw, IN) was cleared for marketing based on a FDA 510(k) premarket notification in December 2000. The implant is manufactured from 10 layers of small intestine submucosa derived from porcine small intestine and is mainly composed of water and collagen. According to the FDA, this surgical mesh implant is intended for use in general surgical procedures for reinforcement of soft tissue where weakness exists. The device is intended to act as a resorbable scaffold that initially has sufficient strength to assist with a soft tissue repair, but then resorbs and is replaced by the patient's own tissue. In addition, the implant is intended for use in the specific application of reinforcement of the soft tissues, which are repaired by suture or suture anchors, limited to the supraspinatus, during rotator cuff surgery. According to the manufacturer, this surgical mesh implant is intended to give the surgeon a less invasive treatment when the rotator cuff tissue is of poor quality or the repair needs reinforcement.

Although the Restore orthobiologic implant has been cleared by the FDA for marketing, there is a lack of adequate evidence on the effectiveness of this implant in rotator cuff repair. Malcarney et al (2005) presented a case series of 25 patients who underwent rotator cuff repair by one surgeon using this implant to augment the repaired tendon or fill a defect. Four of 25 patients (16 %) experienced an overt inflammatory reaction at a mean of 13 days post-operatively. All patients underwent open irrigation and debridement of the rotator cuff and the implant. The authors concluded that these porcine surgical mesh implants should be used with caution and with the understanding that an early post-operative non-specific inflammatory reaction can occur that may cause breakdown of the repair. Furthermore, these investigators stated that more studies are needed to further characterize the reaction and determine which patients are susceptible.

Zheng et al (2005) stated that the small intestinal submucosa (SIS) that is used in this implant is not an acellular collagenous matrix, and contains porcine DNA. They suggested that further studies should be conducted to evaluate the clinical safety and effectiveness of SIS implant biomaterials.

The most frequent side effects encountered in soft tissue repair include infection, adhesions, sterile effusion, instability, increased stiffness post-operatively, and general risks associated with surgery and anesthesia such as neurological, cardiac, and respiratory deficit. Potential device-related risks include stretching or tearing of the device, stiffness, chronic synovitis or effusion, prolonged post-operative rehabilitation, delayed or failed incorporation of the device as well as immunological reaction. Moreover, the porcine surgical mesh implant is contraindicated in patients with massive chronic rotator cuff tears that cannot be mobilized, or where the muscle tissue has undergone substantial fatty degeneration.

Fibrin glue has been used to treat anorectal fistulas in an attempt to avoid more radical surgical intervention. Fibrin glue treatment is simple and repeatable; failure does not compromise further treatment options; and sphincter function is preserved. However, reported success rates vary widely. Suturable bioprosthetic plugs (Surgisis, Cook Surgical, Inc.) have been employed to close the primary opening of fistula tracts. Surgisis is a new 4- or 8-ply bioactive, prosthetic mesh for hernia repair derived from

porcine SIS. In a review on resorbable extra-cellular matrix grafts in urological reconstruction, Santucci and Barber (2005) noted that recent problems with inflammation following 8-ply pubo-vaginal sling use and failures after 1- and 4-ply SIS repair of Peyronie's disease underscore the need for research before wide adoption.

In a prospective cohort study, Johnson and Armstrong (2006) compared fibrin glue versus the anal fistula plug. Patients with high trans-sphincteric fistulas, or deeper, were prospectively enrolled. Patients with Crohn's disease or superficial fistulas were excluded. Age, gender, number and type of fistula tracts, and previous fistula surgeries were compared between groups. Under general anesthesia and in prone jack-knife position, the tract was irrigated with hydrogen peroxide. Fistula tracts were occluded by fibrin glue versus closure of the primary opening using a Surgisis anal fistula plug. A total of 25 patients were prospectively enrolled: 10 patients underwent fibrin glue closure, and 15 used a fistula plug. Patient's age, gender, fistula tract characteristics, and number of previous closure attempts was similar in both groups. In the fibrin glue group, 6 patients (60 %) had persistence of one or more fistulas at 3 months, compared with 2 patients (13 %) in the plug group ($p < 0.05$, Fisher exact test). The authors concluded that closure of the primary opening of a fistula tract using a suturable biologic anal fistula plug is an effective method of treating anorectal fistulas. The method seems to be more reliable than fibrin glue closure. The greater efficacy of the fistula plug may be the result of the ability to suture the plug in the primary opening, therefore, closing the primary opening more effectively. These investigators noted that further prospective, long-term studies are warranted.

A guidance document from the National Institute for Health and Clinical Excellence (NICE, 2006) found insufficient evidence to support the use of porcine intestinal submucosa plugs for repair of anorectal fistula. The NICE assessment concluded: "Current evidence suggests that there are no major safety concerns associated with the closure of anal fistula (fistula in ano) using a suturable bioprosthetic plug. However, evidence on the efficacy of the procedure is not adequate for it to be used without special arrangements for consent and for audit or research." The specialist advisors to NICE commented that there was uncertainty about recurrence rates and the long-term outcomes of this procedure.

Schwandner and Fuerst (2009) analyzed the efficacy of the Surgisis(R) AFP(TM) anal fistula plug and the Surgisis(R) mesh for the closure of complex fistulas in Crohn's disease. All patients with peri-anal Crohn's disease suffering from trans-sphincteric and recto-vaginal fistulas who underwent surgery using the Surgisis(R) anal fistula plug or the Surgisis(R) mesh were prospectively enrolled in this study. Inclusion criteria included trans-sphincteric single-tract fistulas and recto-vaginal fistulas. Surgery was performed using a standardized technique, including irrigation of the fistula tract, placement and internal fixation of the Surgisis(R) anal fistula plug, and combined trans-anal/trans-vaginal excision of recto-vaginal fistula with trans-vaginal placement of the mesh. Success was defined as closure of both internal and external (peri-anal or vaginal) openings, absence of drainage without further intervention, and absence of abscess formation. Follow-up information was obtained from clinical examination 3, 6, 9, and 12 months post-operatively. Within the observation period, a total of 16 procedures were performed. After a mean follow-up of 9 months and 1 patient lost to follow-up, the overall success rate was 75 %. For trans-sphincteric fistulas, the success rate was 77 %, whereas it was 66 % in recto-vaginal fistulas associated with Crohn's disease. All 4 patients with failure had re-operation. Rate of stoma reversal in those patients who had fecal diversion was 66 %. No deterioration of continence was documented. The authors concluded that the short-term success rates are promising; further analysis is needed to explain the definite role of this technique in comparison with traditional surgical techniques.

Safar et al (2009) analyzed the efficacy of the Cook Surgisis AFP anal fistula plug for the management of complex anal fistulas. This was a retrospective review of all patients prospectively entered into a database at the authors' institution who underwent treatment for complex anal fistulas using Cook Surgisis AFP anal fistula plug between July 2005 and July 2006. Patient's demographics, fistula etiology, and success rates were recorded. The plug was placed in accordance with the inventor's guidelines. Success was defined as closure of all external openings, absence of drainage without further intervention, and absence of abscess formation. A total of 35 patients underwent 39 plug insertions (22 men; mean age of 46 (range of 15 to 79) years). Three patients were lost to follow-up, therefore, 36 procedures to be analyzed. The fistula etiology was crypto-glandular in 31 (88.6 %) patients and Crohn's disease

associated in the other 4 (11.4 %). There were 11 smokers and 3 patients with diabetes. The mean follow-up was 126 days (standard = 69.4). The overall success rate was 5 of 36 (13.9 %). One of the 4 Crohn's disease-associated fistulas healed (25 %) and 4 of 32 (12.5 %) procedures resulted in healing of crypto-glandular fistulas. In 17 patients, further procedures were necessary as a result of failure of treatment with the plug. The reasons for failure were infection requiring drainage and seton placement in 8 patients (25.8 %), plug dislodgement in 3 (9.7 %), persistent drainage/tract and need for other procedures in 20 patients (64.5 %). The authors concluded that the success rate for Surgisis AFP anal fistula plug for the treatment of complex anal fistulas was (13.9 %), which is much lower than previously described. They stated that further analysis is needed to explain significant differences in outcomes.

Demineralized Bone Matrix

Autologous Iliac Crest Bone Grafting (ICBG) is considered the gold-standard graft choice for spinal arthrodesis; however, it is associated with donor site morbidity and a limited graft supply.

Demineralized bone matrix products are a class of commercially available grafting agents that are prepared from allograft bone. There is some evidence for the use of demineralized bone matrix products in spinal fusions as an alternative to allograft. Cammisa, et al. (2004) conducted a prospective equivalency trial of Grafton DBM and iliac crest autograft in spine fusion, with each patient serving has his own control The investigators stated that, 2hile autograft remains the preferred graft material to facilitate spine fusion, the supply is limited and harvesting produces may have undesirable clinical consequences. A total of 120 patients underwent posterolateral spine fusion with pedicle screw fixation and bone grafting. Iliac crest autograft was implanted on one side of the spine and a Grafton DBM/autograft composite was implanted on the contralateral side in the same patient. An independent, blinded reviewer evaluated anteroposterior and lateral flexion-extension radiographs. The fusion mass lateral to the instrumentation on each side was judged fused or not, and the mineralization of the graft was rated absent, mild, moderate, or extensive. The degree of correspondence in outcomes between sides was estimated by computing the percentage agreement and kappa statistic. The investigators reported that nearly 70% of patients

(81 of 120) provided complete 24-month radiographic studies. The bone graft mass was fused in 42 cases (52%) on the Grafton DBMside and in 44 cases (54%) on the autograft side. The overall percentage agreement for fusion status between sides was approximately 75% (61 of 81), indicating moderately strong statistical correspondence (kappa = 0.51, P < 0.0001). Bone mineralization ratings also were similar between treated sides. Perfect agreement was realized in almost 60% of patients (48 of 81) with moderate statistical correspondence (weighted kappa = 0.54, P < 0.0001). The authors concluded that Grafton DBM can extend a smaller quantity of autograft than is normally required to achieve a solid spinal arthrodesis. Consequently, a reduced amount of harvested autograft may be required, potentially diminishing the risk and severity of donor site complications.

Kang et al (2012) conducted a 2-year prospective, multicenter, randomized controlled clinical trial comparing the outcomes of Grafton demineralized bone matrix (DBM) Matrix with local bone with that of iliac crest bone graft (ICBG) in a single-level instrumented posterior lumbar fusion. Forty-six patients were randomly assigned (2:1) to receive Grafton DBM Matrix with local bone (30 patients) or autologous ICBG (16 patients). The mean age was 64 (females [F] = 21, males [M] = 9) in the DBM group and 65 (F = 9, M = 5) in the ICBG group. An independent radiologist evaluated plain radiographs and computed tomographic scans at 6-month, 1-year, and 2-year time points. Clinical outcomes were measured using Oswestry Disability Index (ODI) and Medical Outcomes Study 36-Item Short Form Health Survey. The investigators reported that 41 patients (DBM = 28 and ICBG = 13) completed the 2-year follow-up. Final fusion rates were 86% (Grafton Matrix) versus 92% (ICBG) (P = 1.0 not significant). The Grafton group showed slightly better improvement in ODI score than the ICBG group at the final 2-year follow-up (Grafton [16.2] and ICBG [22.7]); however, the difference was not statistically significant (P = 0.2346 at 24 mo). Grafton showed consistently higher physical function scores at 24 months; however, differences were not statistically significant (P = 0.0823). Similar improvements in the physical component summary scores were seen in both the Grafton and ICBG groups. There was a statistically significant greater mean intraoperative blood loss in the ICBG group than in the Grafton group (P < 0.0031). The investigators concluded that, at 2-year follow-up, subjects who were

randomized to Grafton Matrix and local bone achieved an 86% overall fusion rate and improvements in clinical outcomes that were comparable with those in the ICBG group.

Aghdasi et al (2013) conducted a systematic review of the evidence for demineralized bone matrix for spinal fusion. The authors found that demineralized bone matrix has been evaluated in animal models and human clinical trials of spine fusion. Results of animal studies indicate variation in performance within and among demineralized bone matrix products. The majority of human clinical trials report high fusion rates when DBM is employed as a graft extender or a graft enhancer. The authors found that few prospective randomized controlled trials have been performed comparing DBM to autologous iliac crest bone graft in spine fusion. The authors concluded that, although many animal and human studies demonstrate comparable efficacy of DBM when combined with autograft or compared to autograft alone, additional high level of evidence studies are required to clearly define the indications for its use in spine fusion surgeries and the appropriate patient population that will benefit from DBM.

Bone Void Fillers for Nonunions

Minimally invasive injectable graft (MIIG) (Wright Medical Technology, Inc., Arlington, TN) is an example of a bone void filler, and is a paste made with calcium sulphate (plaster of Paris). It is injected into osseous defects that are created surgically or as a result of trauma. The paste cures in-situ, resorbs, and then is replaced with bone during the healing process. The cured paste provides a temporary support media for bone fragments during the surgical procedure but does not provide structural support during the healing process. Injection of MIIG is usually performed in conjunction with another procedure, such as reduction of a fracture. Minimally invasive injectable graft was cleared by the FDA through the 510(k) process since it is substantially equivalent to other bone void fillers on the market.

Integra Mozaik Osteoconductive Scaffold (OS) putty (Integra LifeSciences Corp., Plainsboro, NJ) is a synthetic bone void filler manufactured from beta tri-calcium phosphate and type I bovine collagen. Combined with bone marrow aspirate, Integra Mozaik OS is

intended for use as a bone void filler of the skeletal system in the extremities, spine, and pelvis. Integra Mozaik OS putty was cleared by the FDA through the 510(k) process since it is substantially equivalent to another bone void filler on the market. According to the FDA 510(k) letter to the manufacturer, it is specifically indicated for use in the treatment of surgically treated osseous defects or osseous defects created from traumatic injury to the bone. Following placement in the body void or gap (defect), Integra Mozaik putty is resorbed and replaced with bone during the healing process.

There is insufficient evidence to support the use of MIIG, Integra Mozaik OS putty, or other bone void fillers as a treatment for delayed union or nonunions. Furthermore, a technology assessment prepared by ECRI for Agency for Healthcare Research and Quality (2005) concluded that there is no reliable evidence to support the use of calcium sulphate or other bone void fillers as treatments for delayed fracture healing.

A retrospective case series examined the use of AlloMatrix injectable putty in nonunions in multiple bone types (Wilkins and Kelly, 2003). The nonunions were also treated using standard internal/external fixation techniques. The publication did not report prior treatment or the duration of the nonunions prior to the AlloMatrix putty treatment. A technology assessment prepared by the ECRI Institute (Schoelles et al, 2005) for the Agency for Healthcare Research and Quality, commenting on this study, stated that "[w]ithout this information, interpretation of the results is difficult". The study also did not report whether all consecutively treated patients were included or if dropouts occurred during the treatment period. The reported healing rate was 30 of 35 (85 %) in an average of 3.5 months, but healing rates per bone type were not reported.

A subsequent study by Ziran and colleagues (2007) reported on an unacceptably high rate of complications with the use of Allomatrix for nonunions. A consecutive series of patients requiring bone grafting for atrophic/avascular nonunions were retrospectively studied. Patients were monitored for healing and adverse effects, which included local or systemic reactions, wound problems, infection, and any secondary surgery caused by graft complications. The investigators reported that over half of the patients (51 %) developed post-operative drainage. Of the 41 patients, 13 (32 %) had drainage that required surgical intervention

and 14 (34 %) developed a deep infection. Eleven patients with deep infections also required surgical treatment of drainage. In addition, 19 (46 %) patients did not heal and required secondary surgical intervention. The investigators reported that there were correlations between infection and a history of previously treated infection ($p < 0.007$), as well as wound drainage ($p < 0.001$). Failure of treatment correlated to the presence of a post-operative infection ($p < 0.001$). Other analyses were not performed because of the small sample size, which was because of early termination of the study. The investigators concluded that the use of Allomatrix putty as an alternative for autogenous bone graft in the treatment of nonunions resulted in an unacceptably high rate of complications. The investigators stated: "[a]lthough we recommend further study, we do not recommend the use of Allomatrix for the treatment of nonunions, especially if there is a large volumetric defect or a history of any prior contamination of the tissue bed".

Mesenchymal Stem Cell

Mesenchymal stem cells or MSCs are multipotent stem cells that can differentiate into a variety of cell types. Mesenchymal stem cells have been classically obtained from the bone marrow, and have been shown to differentiate into various cell types, including osteoblasts, chondrocytes, myocytes, adipocytes, and neuronal cells.

Helm and colleagues (2001) stated that although autologous bone remains the gold standard for stimulating bone repair and regeneration, the advent in molecular biology as well as bioengineering techniques has produced materials that exhibit potent osteogenic activities. Recombinant human osteogenic growth factors (e.g., BMP) are now produced in highly concentrated and pure forms and have been shown to be extremely potent bone-inducing agents when delivered in vivo in rats, dogs, primates, and humans. They noted that the delivery of MSCs, derived from adult bone marrow, to regions requiring bone formation is also compelling, and it has been shown to be successful in inducing osteogenesis in many pre-clinical animal studies. Finally, the identification of biological and non-biological scaffolding materials is a crucial component of future bone graft substitutes, not only as a delivery vehicle for bone growth factors and MSCs, but also as an osteo-conductive matrix to stimulate bone deposition directly.

Recently, MSCs has been studied for its use in orthopedic application (e.g., healing long bone defects, intervertebral disc repair and regeneration as well as spinal arthrodesis procedures). Acosta et al (2005) noted that although important obstacles to the survival and proliferation of MSCs within the degenerating intervertebral disc need to be overcome, the potential for this therapy to slow or reverse the degenerative process remains substantial. Leung et al (2006) stated that in the past several years, significant progress has been made in the field of stem cell regeneration of the intervertebral disc. Autogenic MSCs in animal models can arrest intervertebral disc degeneration or even partially regenerate it, and the effect is suggested to be dependent on the severity of degeneration. Mesenchymal stem cells are able to escape alloantigen recognition which is an advantage for allogenic transplantation. A number of injectable scaffolds have been described and various methods to pre-modulate MSCs' activity have been tested. They noted that more work is needed to address the use of MSCs in large animal models as well as the fate of the implanted MSCs, especially the long-term outcomes.

Mclain et al (2005) noted that successful arthrodesis in challenging clinical scenarios is facilitated when the site is augmented with autograft bone. The iliac crest has long been the preferred source of autograft material, but graft harvest is associated with frequent complications and pain. Connective tissue progenitor cells aspirated from the iliac crest and concentrated with allograft matrix and demineralized bone matrix provide a promising alternative to traditional autograft harvest. The vertebral body, an even larger reservoir of myeloproliferative cells, should provide progenitor cell concentrations similar to those of the iliac crest. In this study, a total of 21 adults (11 men and 10 women with a mean age of 59 +/- 14 years) undergoing posterior lumbar arthrodesis and pedicle screw instrumentation underwent transpedicular aspiration of connective tissue progenitor cells. Aspirates were obtained from two depths within the vertebral body and were quantified relative to matched, bilateral aspirates from the iliac crest that were obtained from the same patient at the same time. Histochemical analysis was used to determine the prevalence of vertebral progenitor cells relative to the depth of aspiration, the vertebral level, age, and gender, as compared with the iliac crest standard. The cell count, progenitor cell concentration (cells/cc marrow), and progenitor cell prevalence (cells/million cells) were calculated. Aspirates of vertebral

marrow demonstrated comparable or greater concentrations of progenitor cells compared with matched controls from the iliac crest. Progenitor cell concentrations were consistently higher than matched controls from the iliac crest ($p = 0.05$). The concentration of osteogenic progenitor cells was, on the average, 71 % higher in the vertebral aspirates than in the paired iliac crest samples ($p = 0.05$). With the numbers available, there were no significant differences relative to vertebral body level, the side aspirated, the depth of aspiration, or gender. An age-related decline in cellularity was suggested for the iliac crest aspirates. The authors concluded that the vertebral body is a suitable site for aspiration of bone marrow for graft augmentation during spinal arthrodesis. They also stated that future clinical studies will attempt to confirm the ability to obtain fusion using only this source of connective tissue progenitor cells.

Anderson and colleagues (2005) reviewed the rationale and discussed the results of cellular strategies that have been proposed or investigated for disc degeneration. These investigators noted that although substantial work remains, the future of cellular therapies for symptomatic disc degeneration appears promising. They concluded that continued research is warranted to further define the optimal cell type, scaffolds, and adjuvants that will allow successful disc repair in human patients.

Risbud and colleagues (2006) evaluated the osteogenic potential of MSCs isolated from the bone marrow of the human vertebral body (VB). Marrow samples from VB of patients undergoing lumbar spinal surgery were collected; marrow was also harvested from the iliac crest (IC). Progenitor cells were isolated and the number of colony forming unit-fibroblastic (CFU-F) determined. The osteogenic potential of the cells was characterized using biochemical and molecular biology techniques. Both the VB and IC marrow generated small, medium, and large sized CFU-F. Higher numbers of CFU-F were obtained from the VB marrow than the IC ($p < 0.05$). Progenitor cells from both anatomic sites expressed comparable levels of CD166, CD105, CD49a, and CD63. Moreover, progenitor cells from the VB exhibited an increased level of alkaline phosphatase activity. MSCs of the VB and the IC displayed similar levels of expression of Runx-2, collagen Type I, CD44, ALCAM, and osteocalcin. The level of expression of bone sialoprotein was higher in MSC from the IC than the VB. VB and IC cells mineralized their extracellular matrix to a similar extent. The authors concluded that their

findings show that CFU-F frequency is higher in the marrow of the VB than the IC. Progenitor cells isolated from both sites respond in a similar manner to an osteogenic stimulus and express common immunophenotypes. Based on these findings, these researchers proposed that progenitor cells from the lumbar vertebral marrow would be suitable candidate for osseous graft supplementation in spinal fusion procedures. They stated that studies must now be conducted using animal models to ascertain if cells of the VB are as effective as those of the IC for the fusion applications.

Minamide et al (2007) examined the ability of BMP and basic fibroblast growth factor (FGF) to enhance the effectiveness of bone marrow-derived MSCs in lumbar arthrodesis. They found that MSCs cultured with BMP-2 and basic FGF act as a substitute for autograft in lumbar arthrodesis. This technique may yield a more consistent quality of fusion bone as compared to that with autograft. They stated that these results are encouraging and warrant further studies with the suitable dose of BMP-2 and basic FGF, and may provide a rational basis for their clinical application.

AlloStem is partially demineralized allograft bone combined with adipose derived mesenchymal stem cells; it is similar to autograft bone.

Hankemeier et al (2003) noted that the optimal operative therapy for the treatment of osteochondritis dissecans tali is still controversial. Beside bone marrow-stimulating techniques like abrasion arthroplasty, drilling and microfracturing, new techniques like autologous osteochondral transplantation and autologous chondrocyte transplantation are increasingly used. This study reviewed the clinical, radiological and subjective long-term outcome of bone marrow-stimulating therapy for 45 ankles with an osteochondritis dissecans tali stage 3 or 4 according to the classification by Berndt and Harty. All ankles were treated by the removal of the dissecate and abrasion of the subchondral bone. In 67 %, an additional antegrade drilling of the defect was performed. The average maximum size of the lesion was 1.1 cm. At follow-up examination, 10.4 years (7.1 to 13.5 years) post-operatively, the average American Orthopedic Foot & Ankle Society (AOFAS)-score was 91 points (66 to 100 points). Using the score of Mazur, the outcome of 28 ankles (62 %) was rated excellent, 12 ankles (27 %) were rated good and 5 ankles (11 %)

fair or poor. Progressive osteoarthritic changes, according to the classification of van Dijk, were seen in 7 ankles (16 %). Re-operations were necessary in 8 cases (18 %). Obesity, age older than 40 years and pre-operative osteoarthritic changes had a significant negative impact on the clinical outcome. Bone marrow stimulating therapy is an inexpensive, low invasive therapy and a good therapeutic option at least for small Berndt/Harty stage 3 and 4 ODT lesions. Autologous chondrocyte transplantation and osteochondral autografts yield encouraging 2- and 4-year results, but still have to prove their superiority in long-term follow-up studies.

Kon et al (2012) stated that osteochondritis dissecans is a relatively common cause of knee pain. These researchers described the outcomes of five different surgical techniques in a series of 60 patients with osteochondritis dissecans. Sixty patients (aged 22.4 ± 7.4 years, 62 knees) with osteochondritis dissecans of a femoral condyle (45 medial and 17 lateral) were treated with osteochondral autologous transplantation, autologous chondrocyte implantation with bone graft, biomimetic nanostructured osteochondral scaffold (MaioRegen) implantation, bone-cartilage paste graft, or a "1-step" bone-marrow-derived cell transplantation technique. Pre-operative and follow-up evaluation included the International Knee Documentation Committee (IKDC) score, the EuroQol visual analog scale (EQ-VAS) score, radiographs, and magnetic resonance imaging. The global mean IKDC score improved from 40.1 ± 14.3 pre-operatively to 77.2 ± 21.3 ($p < 0.0005$) at 5.3 ± 4.7 years of follow-up, and the EQ-VAS improved from 51.7 ± 17.0 to 83.5 ± 18.3 ($p < 0.0005$). No influence of age, lesion size, duration of follow-up, or previous surgical procedures on the result was found. The only difference among the results of the surgical procedures was a trend toward better results following autologous chondrocyte implantation ($p = 0.06$). The authors concluded that all of the techniques were effective in achieving good clinical and radiographic results in patients with osteochondritis dissecans, and the effectiveness of autologous chondrocyte implantation was confirmed at a mean follow-up of 5 years. Newer techniques such as MaioRegen implantation and the "1-step" transplantation technique are based on different rationales; the first relies on the characteristics of the scaffold and the second on the

regenerative potential of mesenchymal cells. Both of these newer procedures have the advantage of being minimally invasive and requiring a single operation.

Teo et al (2013) noted that recent advances have been made in using chondrocytes and other cell-based therapy to treat cartilage defects in adults. However, it is unclear whether these advances should be extended to the adolescent and young adult-aged patients. These researchers assessed cell-based surgical therapy for patellar osteochondritis dissecans (OCD) in adolescents and young adults by (i) determining function with the International Knee Documentation Committee (IKDC) subjective and Lysholm-Gillquist scores; and (ii) evaluating activity level using the Tegner-Lysholm scale. They retrospectively reviewed 23 patients between 12 and 21 years of age (mean of 16.8 years) treated for OCD lesions involving the patella from 2001 to 2008. Twenty patients had autologous chondrocyte implantation and 3 patients had cultured bone marrow stem cell implantation. There were 19 males and 4 females. These investigators obtained pre-operative CT scans to assess patella subluxation, tilt, and congruence angle to determine choice of treatment. They obtained IKDC subjective knee evaluation scores, Tegner-Lysholm activity levels, and Lysholm-Gillquist knee scores pre-operatively and at 6, 12, and 24 months post-operatively. Mean IKDC score, Tegner-Lysholm outcomes, and Lysholm-Gillquist scale improved from 45, 2.5, and 50, respectively, at surgery to 75, 4, and 70, respectively, at 24-month follow-up. Complications include periosteal hypertrophy observed in 2 patients. The authors concluded that cell-based therapy was associated with short-term improvement in function in adolescents and young adults with patellar OCD. (Only 3 patients received cultured bone marrow stem cell implantation). Further investigation is needed to study the value of MSC therapy in orthopedic applications before it can be used in the clinical setting.

Miscellaneous Interventions

Cheng et al (2010) had previously isolated and identified stem cells from human anterior cruciate ligament (ACL). The purpose of this study was to evaluate the differences in proliferation, differentiation, and extracellular matrix (ECM) formation abilities between bone marrow stem cells (BMSCs) and ACL-derived stem cells (LSCs) from the same donors

when cultured with different growth factors, including basic fibroblast growth factor (bFGF), epidermal growth factor, and transforming growth factor-beta 1 (TGF-beta1). Ligament tissues and bone marrow aspirate were obtained from patients undergoing total knee arthroplasty and ACL reconstruction surgeries. Proliferation, colony formation, and population doubling capacity as well as multi-lineage differentiation potentials of LSCs and BMSCs were compared. Gene expression and ECM production for ligament engineering were also evaluated. It was found that BMSCs possessed better osteogenic differentiation potential than LSCs, while similar adipogenic and chondrogenic differentiation abilities were observed. Proliferation rates of both LSCs and BMSCs were enhanced by bFGF and TGF-beta1. TGF-beta1 treatment significantly increased the expression of type I collagen, type III collagen, fibronectin, and alpha-smooth muscle actin in LSCs, but TGF-beta1 only up-regulated type I collagen and tenascin-c in BMSCs. Protein quantification further confirmed the results of differential gene expression and suggested that LSCs and BMSCs increase ECM production upon TGF-beta1 treatment. In summary, in comparison with BMSCs, LSCs proliferate faster and maintain an undifferentiated state with bFGF treatment, whereas under TGF-beta1 treatment, LSCs up-regulate major tendinous gene expression and produce a robust amount of ligament ECM protein, making LSCs a potential cell source in future applications of ACL tissue engineering.

Steinert et al (2011) noted that when ruptured, the ACL of the human knee has limited regenerative potential. However, the goal of this report was to show that the cells that migrate out of the human ACL constitute a rich population of progenitor cells and these researchers hypothesized that they display mesenchymal stem cell (MSC) characteristics when compared with adherent cells derived from bone marrow or collagenase digests from ACL. They showed that ACL outgrowth cells are adherent, fibroblastic cells with a surface immunophenotype strongly positive for cluster of differentiation (CD)29, CD44, CD49c, CD73, CD90, CD97, CD105, CD146, and CD166, weakly positive for CD106 and CD14, but negative for CD11c, CD31, CD34, CD40, CD45, CD53, CD74, CD133, CD144, and CD163. Staining for STRO-1 was seen by immunohistochemistry but not flow cytometry. Under suitable culture conditions, the ACL outgrowth-derived MSCs differentiated into chondrocytes, osteoblasts, and adipocytes and showed capacity to self-

renew in an in vitro assay of ligamentogenesis. MSCs derived from collagenase digests of ACL tissue and human bone marrow were analyzed in parallel and displayed similar, but not identical, properties. In situ staining of the ACL suggests that the MSCs reside both aligned with the collagenous matrix of the ligament and adjacent to small blood vessels. The authors concluded that the cells that emigrate from damaged ACLs are MSCs and that they have the potential to provide the basis for a superior, biological repair of this ligament.

According to information from the manufacturer, BIO MatrX Structure is a highly porous, synthetic bone graft substitute that sets hard upon implantation for a complete defect fill. The manufacturer states that the resulting osteoconductive scaffold provides inter-connected porosity and high surface area to facilitate cell mediated remodeling and new bone growth. BIO MatrX Generate is a combination of osteoconductive nano-crystalline calcium phosphate and Demineralized Bone Matrix (DBM) that is tested for osteoinductive potential by lot, after sterilization, in an in-vivo athymic nude rodent muscle pouch model. The viscous putty sets hard after closure providing an osteoconductive scaffold to facilitate new bone growth. The manufacturer states that both materials are FDA-cleared to be hydrated with saline or blood; and are indicated as bone void fillers of the pelvis, extremities and the postero-lateral spine.

The use of minced cartilage techniques are in the early stages of development. According to the manufacturer, DeNovo NT was developed as a consequence of the need for expanded treatment options for the treatment of cartilage lesions. DeNovo NT (natural tissue) graft and DeNovo ET live chondral engineered tissue graft (Neocartilage) are produced by ISTO Technologies (St. Louis, MO), and exclusively distributed by Zimmer, Inc. (Warsaw, IN). DeNovo NT consists of manually minced cartilage tissue pieces obtained from juvenile allograft donor joints. The tissue fragments are mixed intra-operatively with fibrin glue before implantation. It is thought that mincing the tissue helps with cell migration. As there are no chemicals used and minimal manipulation, it is regulated as an allograft tissue rather than a biological implant. Thus, the allograft tissue does not require FDA approval for marketing. DeNovo NT is currently available in the U.S. Neocartilage uses juvenile allogeneic cartilage cells that are isolated and expanded in-vitro, similar to other ACI techniques. Neocartilage is currently being studied in human clinical

trials under an FDA-approved investigational new drug (IND) application. The FDA approved ISTO's IND application in 2006, which allowed them to pursue clinical trials of the product in humans. There are no studies evaluating the DeNovo ET tissue graft in the published medical literature.

There are few studies evaluating the DeNovo NT graft in the published medical literature. The manufacturer of DeNovo NT has initiated a post-market, multi-center, longitudinal data collection study to collect clinical outcomes of subjects implanted with DeNovo NT. Data are to be obtained either retrospectively or prospectively from patients implanted or to be implanted with DeNovo NT for the treatment of lesion in the ankle. Data to be collected include details of the operative procedure as well as subjects' pain, function, activity levels, and healthcare resource use through a 5-year post-operative follow-up period. Four U.S. sites are participating in this manufacturer-sponsored observational study with 25 subjects; the study began in 2006 and is expected to be completed in 2013.

Ky et al (2008) evaluated the effectiveness of the Surgisis (Anal Fistula Plug) in multiple patients and presented early clinical results along with notable clinical observations from their experience. This was a prospective analysis of all patients who received the Anal Fistula Plug for treatment of anorectal fistulas between April 2006 and February 2007. All tracts were irrigated with peroxide, the plug was inserted in the tract, and buried at the internal opening with 2-0 vicryl and mucosal advancement flap. Statistical analysis was performed with Fisher's exact test. A total of 45 patients were treated with the Anal Fistula Plug and 1 patient was lost to follow-up. There were 27 males and 17 females with average age of 44.1 years treated for simple (n = 24) or complex (n = 20) fistulas. Preliminary results indicated an 84 % healing rate by 3 to 8 weeks post-operatively, which progressively declined from 72.7 % at 8 weeks to 62.4 % at 12 weeks and 54.6 % at a median follow-up of 6.5 (range of 3 to 13) months. Long-term Anal Fistula Plug closure rate was significantly higher in patients with simple than complex fistulas (70.8 versus 35 %; $p < 0.02$) and with non-Crohn's disease versus Crohn's disease (66.7 versus 26.6 %; $p < 0.02$). Patients with 2 successive plug placements had significantly lower closure rates than patients who underwent placement of the plug once (12.5 versus 63.9 %; $p < 0.02$). No significant difference in closure rates were found between patients with 1 versus multiple fistula

tracts. Post-operative complications included peri-anal abscess in 5 patients (3 Crohn's disease, 2 non-Crohn's disease). The authors concluded that Anal Fistula Plug is most successful in the treatment of simple anorectal fistulas but is associated with a high failure rate in complex fistula and particularly in patients with Crohn's disease. Repeat plug placement is associated with increased failure. Given the relatively low morbidity associated with the procedure, Anal Fistula Plug should be considered as a first-line treatment for patients with simple fistulas and as an alternative in selected patients with complex fistulas. Drawbacks of this study were: (i) small sample size, (ii) short duration of follow-up, and (iii) high failure rate.

Buchberg et al (2010) compared the Cook Surgisis AFP plug and the newer Gore Bio-A plug in the management of complex anal fistulas. A retrospective chart review of patients treated with Cook and Gore fistula plugs between August 2007 and December 2009 was performed. Success was defined as closure of all external openings and absence of drainage and abscess formation. Twelve Cook patients underwent 16 plug insertions and 10 Gore patients underwent 11 plug insertions. The overall procedural success rate in the Gore group was 54.5 % (6 of 11) versus 12.5 % (2 of 16) in the Cook group. The reasons for failure were unknown in the majority of patients and plug dislodgement in 2 patients. These short-term results with the Gore fistula plug suggested a higher procedural success rate in comparison to the Cook plug. The authors concluded that patients should be cautioned regarding potentially high failure rates; however, longer follow-up and a larger patient population are needed to confirm significant differences in fistula plug efficacy.

According to the manufacturer, Ovation, a novel cellular repair matrix, is derived from placental mesenchyme. It provides the 3 essential components of periosteum -- (i) extracellular matrix, (ii) endogenous mesenchymal stem cells and (iii) a replenishing source of growth factors -- without the need for autografting. There is currently insufficient evidence to support the use of mesenchymal stem cell therapy for orthopedic applications including repair or regeneration of musculoskeletal tissue, spinal fusion, and long bone nonunions.

An UpToDate review on “Hallux valgus deformity (bunion)” (Ferrari, 2013) does not mention the use of grafting as a therapeutic option.

Goebel et al (2005) described the advantages of using Mimix hydroxyapatite (HA) bone cement in reconstructing a variety of ossicular chain abnormalities. A total of 25 cases of HA reconstruction were included in this series (ages of 23 to 74; mean of 47 years). The examples presented include (i) HA as the sole reconstructive material for incus erosion, (ii) HA for securing total or partial ossicular replacement prosthesis, (iii) incus augmentation after crimping for revision stapedotomy with incus erosion, (iv) HA in primary stapedotomy to fix the crimped prosthesis to an intact incus, and (v) other unique situations. Pre-operative and post-operative audiograms were evaluated for 4-tone pure tone average (PTA), speech reception thresholds, word recognition scores, and air-bone gaps (ABGs). Mean follow-up was 11 months (range of 2 to 22 months). The mean PTA improved from 57 dB to 37 dB, whereas the mean ABGs decreased from 33 dB to 16 dB. There were no cases of infection or extrusion. The authors concluded that hydroxyapatite bone cement is an excellent adjunct or alternative to ossiculoplasty with preformed prostheses. Easily malleable, rapidly setting, and rapidly hardening, Mimix is particularly well-suited for middle ear work. Definitive fixation with bone cements during difficult ossicular chain reconstruction may ensure a more enduring successful outcome.

Elsheikh et al (2006) analyzed the results obtained from HA bone cement repair of ossicular discontinuity between the incus and stapes during surgery of retraction pockets. A total of 62 previously untreated patients (82 ears) with retraction pockets were studied. Hydroxyapatite bone cement was used to repair defects at the incudo-stapedial connection in 82 ears with retraction pockets. The ears were divided into 2 groups: group 1 included 48 ears with a small defect in the long process of the incus; group 2 included 34 ears with a large defect in the long process of the incus. In addition, 20 control patients underwent surgery using Plastipore partial ossicular replacement prostheses. Hearing results were reported in 4 frequencies (0.5, 1, 2, and 3 kHz). Analysis of the results was performed using the paired t-test with significance level at 0.05. Main outcome measures were anatomic and audiologic results.

Significant post-operative improvement of pure-tone air conduction threshold averages and air-bone gap averages were reported in the 3 studied groups. The post-operative air-bone gap averages showed significantly better outcome in groups 1 and 2 compared with controls ($p < 0.001$), while there was no statistically significant difference between groups 1 and 2 ($p > 0.05$). The authors concluded that bone cement ossiculoplasty offered cost-effective and significant improvement in conductive hearing loss. It provided an excellent alternative to ossiculoplasty with preformed prostheses. They believed the indications for bone cement were validated by these results.

Redaelli de Zinis et al (2008) reported hearing results using a titanium ossicular replacement prosthesis during canal wall down mastoidectomy with tympanoplasty to treat cholesteatoma. Patients with cholesteatoma treated with primary or revision canal wall down mastoidectomy with tympanoplasty in a single-stage. Patients with implanted HA prostheses composed a matched control group. Medical records were reviewed for type of ossicular condition, type of prosthesis, and hearing threshold at 1-year follow-up. Results were reported as the 5-frequency average air conduction gain, bone conduction gain, and ABG. The malleus handle was present in 24 patients, and the stapes superstructure in 22 patients. Mean (SD) air conduction gain was 7.6 (14.7) dB ($p = 0.001$); it was 8.7 (12.0) dB in the group with titanium prostheses and 6.3 (17.4) dB in the group with HA prostheses ($p = 0.54$). Bone conduction gain was 1.1 (4.9) dB ($p = 0.19$). No patients experienced post-operative impairment of bone threshold greater than 5 dB. Post-operative air-bone gap was 26.5 (15.3) dB; it was 23.8 (15.7) dB in the titanium group and 29.8 (14.6) dB in the HA group ($p = 0.18$). Air-bone gap closure was 40 %; it was 46.2 % in the titanium group and 33.3 % in the HA group ($p = 0.35$). The authors concluded that titanium is a satisfactory material for use in ossicular reconstruction and is comparable to HA, although at present, no definitive conclusion about the superiority of titanium can be drawn.

Kawano and co-workers (2010) noted that many cases of tympano-sclerotic stapes fixation are accompanied by fixation or erosion of malleus and/or incus. This status of the ossicular chain is one of the reasons that ossiculoplasty for tympano-sclerotic stapes fixation is more difficult than that for oto-sclerosis. These investigators conducted a retrospective review of 7 patients who were operated on for tympano-sclerotic stapes

fixation between 2002 and 2006. All of the patients had abnormal conditions of the malleus and/or incus and underwent stapedectomy and total ossiculoplasty with HA prosthesis (Apaceram T-7 type), which has a planar-like head portion that contacts a piece of cartilage. Post-operative hearing results were assessed in all 7 patients after at least 1 year. The post-operative ABG was closed within 10 dB in 2 of 7 patients, and was less than 20 dB in 6 of 7 patients. The mean post-operative ABG was closed within 10 dB at 1 and 2 kHz and less than 20 dB at low frequencies (0.25 and 0.5 Hz). There was almost no hearing improvement at high frequencies (4 and 8 kHz). There were no patients with post-operative SNHL. The authors concluded that the present study showed that stapedectomy and total ossiculoplasty with cartilage-connecting HA prosthesis is effective and safe for stapes fixation accompanied by fixation or erosion of the malleus and/or incus.

Van Rompaey et al (2011) studied hearing outcome in revision stapedotomy cases where extensive erosion of the long process of the incus was observed in a consecutive series where a malleo-vestibular prosthesis was used versus a consecutive series where HA bone cement was used to re-build the eroded long process of the incus and integrate the prosthesis. This study examined a total of 20 revision cases of surgically treated oto-sclerosis where extensive incus erosion was observed during revision surgery. In the earlier consecutive series, 10 cases were treated with malleo-vestibular prostheses. In the later consecutive series, 10 cases were treated with HA bone cement to re-build the incus-prosthesis interface. Air-bone gap, bone-conduction thresholds, and air-conduction thresholds were evaluated pre-operatively and at 1 to 3 months. Last audiometry available also was reported (median of 12 months). Pure-tone averages were calculated according to the guidelines of the Committee on Hearing and Equilibrium for the evaluation of conductive hearing loss. Raw data were displayed in an Amsterdam Hearing Evaluation Plot. Six male patients and 14 female patients were included. Age varied from 34 to 75 years (median of 53 years). The median post-operative ABG at last follow-up audiometry was 15.6 in the malleo-vestibular prosthesis group and 13.1 dB in the HA bone cement group. No short-term or intermediate-term adverse reactions or unsuspected bone conduction deteriorations were seen. The authors concluded that HA bone cement can be successfully used to reconstruct the long process of the incus in case of extensive erosion of

the long process. Intermediate-term hearing outcome is comparable to the outcome of a series of similar cases treated with malleo-vestibular prostheses. Because the placement of a malleo-vestibular prosthesis is technically more difficult and presents a high risk to the inner ear, the authors thought HA bone cement can be a useful alternative in these difficult cases.

Somers et al (2012) compared the hearing outcome using HA bone cement to bridge the incudo-stapedial gap versus incus re-modelling for ossiculoplasty in case of incudo-stapedial discontinuity. A non-randomized retrospective study was conducted at a tertiary referral otologic center. The intervention in 24 primary cases of conductive hearing loss was subsequent middle ear inspection where incudo-stapedial discontinuity was observed. Hydroxyapatite bone cement was used in 10 consecutive cases, and incus re-modelling was performed in 14 consecutive cases. Air-bone gap, bone-conduction (BC) thresholds, and air-conduction (AC) thresholds were evaluated pre-operatively and at 3, 6 and 12 months post-operatively. No patients were lost to follow-up. Pure-tone averages were calculated according to the guidelines of the Committee on Hearing and Equilibrium for the evaluation of conductive hearing loss. The Amsterdam Hearing Evaluation Plots are presented. The postoperative ABG closure to within 20 and 10 dB at 12 months was, respectively, 80 and 40 % in the HA bone cement group and 57.1 and 28.6 % in the standard ossiculoplasty group (no statistically significant difference). However, these researchers observed a statistically significant difference in ABG gain at 6 and 12 months favoring the HA bone cement cases. No short-term or intermediate-term adverse reactions were observed. The authors concluded that HA bone cement bridging ossiculoplasty offers a better intermediate-term ABG gain than standard ossiculoplasty. This new technique is a valuable alternative to conventional ossiculoplasty and presents the practical advantage of being easier and faster.

Ayshford et al (2003) noted that nasal septal perforations present a distinct challenge to the otolaryngologist and a significant cause of symptoms to affected patients. Many surgical techniques for the repair of septal perforations have been described. Connective tissue autografts are commonly used as inter-positional grafts between the septal flaps. Recently acellular human dermal allograft has been used with success.

In this study, a total of 17 patients with symptomatic anterior nasal septal perforations that had failed conservative treatment underwent a closed endoscopic repair of their perforations with acellular human dermal allograft (Alloderm) and an anteriorly based inferior turbinate flap; 13 patients had a successful closure of the perforation, 2 patients, despite initial success, re-perforated as a result of persistent crust picking and, in 2 patients, the graft failed. The authors concluded that with appropriate patient selection and stringent post-operative care this technique offers a good surgical outcome for the closure of septal perforations. The findings of this small study need to be validated by well-designed studies.

Chhabra and Houser (2012) noted that the closure of nasal septal perforations can be challenging based on the etiology, location, and method of closure. These researchers reported on a novel method of closure for nasal septal perforations using a unilateral mucosal rotational flap and acellular dermal interposition graft. A total of 20 patients with nasal septal perforations of various etiologies underwent this novel method of repair through a closed, endonasal approach. Out of 20 patients, 17 demonstrated successful closure of their septal perforations, consistent with an 85 % success rate. Based upon size, closure rates were 89 % for small perforations (less than 1 cm), 80 % for medium perforations (1 to 2 cm), and complete closure for a single large perforation (greater than 2 cm). Of 20 patients, 19 were completely asymptomatic following surgical intervention, and of the 3 with failed repairs, only 1 patient required revision surgery for persistent symptoms. The authors concluded that nasal septal perforations may cause bothersome symptoms and present a significant reconstructive challenge. Native septal tissue is advantageous due to a rich vascular supply and proximity to the defect, while interposition grafts act as a scaffold for the migration of respiratory mucosa. The findings of this small study need to be validated by well-designed studies.

An UpToDate review on “Osteonecrosis (avascular necrosis of bone)” (Jones and Mont, 2014) states that “Bone grafting of the lesion, which has also been used to treat small- to medium-sized lesions. Outcomes for patients treated with impaction grafting have demonstrated promising results. The objective Knee Society Score after a mean follow-up of approximately four years (range of two to eight years) was 89 (range of 70 to 100), and the functional score was 81 (range of 50 to 100). None of

the patients were revised. One study reported that a graft matrix of allogeneic cancellous bone chips augmented with enriched autogenous bone marrow aspirate yielded promising results in three patients with large lesions at two years of follow-up".

Le Huec et al (1997) presented the results of a comparative study of 2 series of postero-lateral arthrodeses for scoliosis performed using COTREL DUBOUSSET instrumentation. A total of 54 consecutive patients underwent surgery for idiopathic scoliosis using the same technique -- 30 received a graft consisting of a mixture of cortico-cancellous autologous and allogenic bone frozen at -80 degrees, and 24 patients were grafted with a mixture of cortico-cancellous autologous bone and sticks of tri-calcium phosphate (TCP, Biosorb, SBM, Lourdes, France). All patients were seen at 3, 6 and 12 months, then once a year for at least 4 years with clinical and radiological evaluation at each visit. At the final follow-up visit, no radiologic signs of pseudoarthrosis were found in either group with a minimum follow-up of 4 years. The appearance of bone callus was considered satisfactory at 6 months in all cases; moreover callus seemed to be more important in the TCP series, although this assessment was subjective. Tri-calcium phosphate resorption was total after 2 years, while allograft fragments were visible on x-rays after 2 years. Minor mechanical complications occurred but did not influence the results. Loss of correction was 8 % of that initially obtained in the allograft group and 2 % in the TCP group. Loss of correction did not progress after 6 months in the TCP group and after 2 years in the allograft group. Based upon this experience, the use of synthetic bone substitutes such as TCP would appear to be a valuable alternative to allografts in postero-lateral spinal arthrodesis for idiopathic scoliosis, and it would eliminate the risk of viral contamination inherent to allograft implantation. The authors stated that there had been no previous comparative studies concerning the use of TCP versus allograft in the literature.

Kanayama et al (2006) evaluated the osteo-inductive property of OP-1 or BMP-7 and fusion rate in human instrumented postero-lateral lumbar fusion through radiographic examination, surgical exploration, and histologic assessment. A total of 19 patients with L3 to L4 or L4 to L5 degenerative spondylolisthesis underwent postero-lateral lumbar fusion using pedicle screw instrumentation. The patients were randomized to

receive either OP-1 putty (3.5 mg OP-1/g of collagen matrix per side) alone (n = 9), or local autograft with HA-TCP granules (n = 10). Fusion status was evaluated using plain radiography and CT scan. Radiographic fusion criteria included less than 5 degrees of angular motion, less than 2 mm of translation, and evidence of bridging bone in the postero-lateral lumbar area in which the graft materials were placed following decortication. After a minimum 1-year follow-up, the patients who showed radiographic evidence of fusion underwent instrumentation removal and surgical exploration of the fusion site. Biopsy specimens were taken from the fusion mass and evaluated histologically. Radiographic fusion rate was 7 of 9 OP-1 patients and 9 of 10 control patients. Based on surgical exploration of these 16 patients, new bone formation was macroscopically observed in the postero-lateral lumbar region in all cases; however, solid fusion was observed in 4 of 7 OP-1 and 7 of 9 HA-TCP/autograft patients. Histologic assessment demonstrated viable bone in 6 of 7 OP-1 patients. All the control (HA-TCP/autograft) specimens contained viable bone and fibrous tissue surrounding ceramic granules, suggesting slow incorporation of the graft material. The authors concluded that in a human postero-lateral lumbar spine trial, OP-1 reliably induced viable amounts of new bone formation, but the fusion success rate evaluated by surgical exploration was only 4 of 7.

In a pilot study, Lerner et al (2009) compared the clinical and radiographic results of ultraporous beta-TCP (b-TCP) versus autogenous iliac crest bone graft (ICBG) as graft extenders in scoliosis surgery. In the posterior correction of scoliosis, local bone resected as part of the procedure is used as the base bone graft material. Supplemental grafting from the iliac crest is considered the gold-standard in posterior spinal fusion. However, autograft is not available in unlimited quantities, and bone harvesting is a source of significant morbidity. Ultraporous b-TCP might be a substitute for ICBG in these patients and thus eliminate donor site morbidity. A total of 40 patients with adolescent idiopathic scoliosis (AIS) were randomized into 2 treatment groups and underwent corrective posterior instrumentation. In 20 patients, ICBG harvesting was performed whereas the other half received b-TCP (VITOSS) to augment the local bone graft. If thoracoplasty was performed, the resected rib bone was added in both groups. Patients were observed clinically and radiographically for a minimum of 20 months post-operatively, with a

mean follow-up of 4 years. Overall pain and pain specific to the back and donor site were assessed using a visual analog scale (VAS). As a result, both groups were comparable with respect to the age at the time of surgery, gender ratio, pre-operative deformity, and hence length of instrumentation. There was no significant difference in blood loss and operative time. In 9 patients of the b-TCP group and 8 patients of the ICBG group, thoracoplasty was performed resulting in a rib graft of on average 7.9 g in both groups. Average curve correction was 61.7 % in the b-TCP group and 61.2 % in the ICBG group at hospital discharge ($p = 0.313$) and 57.2 and 54.3 %, respectively, at follow-up ($p = 0.109$). Loss of curve correction amounted on average 2.6 degrees in the b-TCP group and 4.2 degrees in the comparison group ($p = 0.033$). In the ICBG group, 4 patients still reported donor site pain of on average 2/10 on the VAS at last follow-up. One patient in the b-TCP group was diagnosed with a pseudarthrosis at the caudal end of the instrumentation. Revision surgery demonstrated solid bone formation directly above the pseudarthrosis with no histological evidence of b-TCP in the biopsy taken. The authors concluded that the use of b-TCP instead of ICBG as extenders of local bone graft yielded equivalent results in the posterior correction of AIS. They stated that the promising early results of this pilot study supported that b-TCP appears to be an effective bone substitute in scoliosis surgery avoiding harvesting of pelvic bone and the associated morbidity.

Larsson (2010) noted that a number of different calcium phosphate compounds such as calcium phosphate cements and solid b-TCP products have been introduced during the past 10 years. The chemical composition mimics the mineral phase of bone and as a result of this likeness, the materials seem to be re-modeled as for normal bone through a cell-mediated process that involves osteoclastic activity. This is a major difference when compared with, for instance, calcium sulphate compounds that after implantation dissolve irrespective of the new bone formation rate. Calcium phosphates are highly biocompatible and in addition, they act as synthetic osteo-conductive scaffolds after implantation in bone. When placed adjacent to bone, osteoid is formed directly on the surface of the calcium phosphate with no soft tissue interposed. Re-modeling is slow and incomplete, but by adding more and larger pores, like in ultraporous b-TCP, complete or nearly complete resorption can be achieved. The indications explored so far include filling

of metaphyseal fracture voids or bone cysts, a volume expander in conjunction with inductive products, and as a carrier for various growth factors and antibiotics. The authors concluded that calcium phosphate compounds (e.g., calcium phosphate cement and b-TCP) will most certainly be part of the future armamentarium when dealing with fracture treatment.

Larsson and Hannink (2011) stated that more than a decade has passed since the first injectable bone substitutes were introduced for use in orthopedic trauma, and over recent years the number of commercial products has increased dramatically. Despite the fact that these bone substitutes have been on the market for many years, knowledge among potential users on how and when they might be useful is still fairly limited. Most injectable bone substitutes belong to one of two major groups: by far the largest group contains products based on various calcium phosphate (CP) mixtures, whilst the smaller group consists of calcium sulphate (CS) compounds. Following mixing, the CP or CS paste can be injected into--for instance--a fracture space for augmentation as an alternative to bone graft, or around a screw for augmentation if the bone is weak. Within minutes an in-situ process makes the substitute hard; the mechanical strength in compression resembles that of cancellous bone, whereas the strength in bending and shear is lower. Over time, CP products undergo re-modelling through a cell-mediated process that seems to mimic the normal bone re-modelling, while CS products are dissolved through a faster process that is not cell-mediated. For CP, a number of clinical studies have shown that it can be useful for augmentation of metaphyseal fractures when a space is present. Randomized studies have verified that CP works especially well in tibial plateau fractures when compared with conventional bone grafting. So far the number of clinical studies on CS products is very low. Development at present seems to be heading towards premixed or directly mixed products as well as new compounds that contain fibers or other components to enhance bending and shear strength. Products that are based on combinations of CP and CS are also being developed to combine the fast-dissolving CS with the stronger and more slowly re-modelling CP. Injectable bone substitutes, and especially CS, have also been targeted as potentially good carriers for antibiotics and growth factors.

In summary, there is currently a lack of good quality RCTs on the use of ceramic-based products (e.g., b-TCP) bone void fillers.

Buchberg et al (2010) noted that treatment of complex anal fistulas presents an ongoing challenge to colorectal surgeons. The anal fistula plug is an attractive definitive option due to its minimal risk of incontinence, simple design, and easy application. These researchers compared the Cook Surgisis AFP plug and the newer Gore Bio-A plug in the management of complex anal fistulas. A retrospective chart review of patients treated with Cook and Gore fistula plugs between August 2007 and December 2009 was performed. Success was defined as closure of all external openings and absence of drainage and abscess formation. Twelve Cook patients underwent 16 plug insertions and 10 Gore patients underwent 11 plug insertions. The overall procedural success rate in the Gore group was 54.5 % (6 of 11) versus 12.5 % (2 of 16) in the Cook group. The reasons for failure were unknown in the majority of patients and plug dislodgement in 2 patients. These short-term results with the Gore fistula plug suggested a higher procedural success rate in comparison to the Cook plug. The authors concluded that patients should be cautioned regarding potentially high failure rates; moreover, they stated that longer follow-up and a larger patient population are needed to confirm significant differences in fistula plug efficacy.

O'Riordan et al (2012) summarized the anal fistula plug literature for Crohn's and non-Crohn's fistula-in-ano in a homogenous patient population. PubMed, MEDLINE, Embase, and Cochrane medical databases were searched from 1995 to 2011. Abstracts from the American Society of Colon and Rectal Surgeons, the Society for Surgery of the Alimentary Tract, the European Society of Coloproctology, and the Association of Coloproctology of Great Britain and Ireland meetings between 2007 and 2010 were also evaluated. Studies were included if results for patients with and without Crohn's disease could be differentiated. Patients with recto-vaginal, ano-vaginal, recto-urethral, or ileal-pouch vaginal fistulas were excluded as were studies where the mean or median follow-up was less than 3 months. Two researchers independently selected studies matching the inclusion criteria. The primary outcomes measured were the overall fistula closure rates and length of follow-up. A total of 76 articles or abstracts were identified from the title as being of relevance; 20 studies (2 abstracts, 18 articles) were

finally included. Study sample size ranged from 4 to 60 patients; 530 patients were included in all studies (488 non-Crohn's and 42 Crohn's patients). The plug extrusion rate was 8.7 % (46 patients). The proportion of patients achieving fistula closure varied widely between studies for non-Crohn's, ranging from 0.2 (95 % confidence interval [CI] 0.04 to 0.48) to 0.86 (95 % CI: 0.64 to 0.97). The pooled proportion of patients achieving fistula closure in patients with non-Crohn's fistula-in-ano was 0.54 (95 % CI: 0.50 to 0.59). The proportion achieving closure in patients with Crohn's disease was similar (0.55, 95 % CI: 0.39 to 0.70). The authors concluded that fistula closure is achieved by using the anal fistula plug in approximately 54 % of patients without Crohn's disease. The anal fistula plug has not been adequately evaluated in the Crohn's population.

Abrams et al (2013) stated that osteochondritis dissecans lesions occur frequently in children and adolescents. Treatment can be challenging and depends on the status of the articular cartilage and subchondral bone. Injection of calcium phosphate bone substitute into the area of subchondral bone edema (Subchondroplasty; Knee Creations, West Chester, PA) may be an option. These researchers presented a case of a lateral tibial plateau osteochondritis dissecans lesion treated with subchondral injection of nanocrystalline calcium phosphate. Pre-operative magnetic resonance imaging is used to determine the area of subchondral edema, and intra-operative fluoroscopy is used to localize this area with the injection cannula. Calcium phosphate is injected by use of a series of syringes until the appropriate fill is obtained. Treatment of concomitant cartilage defects may also be carried out at this time. The authors noted that potential challenges in using this technique are accurate localization of the lesion intra-operatively. Fluoroscopy is often not able to show the lesion at the time of the procedure. Because of this, these investigators regularly had the pre-operative MRI scan available in the operating room for reference to be able to properly match the location of the cannula with the area of maximal T2 signal intensity based on the MRI scan. In addition, when this technique is used in skeletally immature individuals, there is the potential for physeal injury because of the proximity of the calcium phosphate. The findings of this single case study need to be validated by well-designed studies.

Miscellaneous Interventions

AlloStem is partially demineralized allograft bone combined with adipose derived mesenchymal stem cells.

Shin et al (2014) noted that focal chondral lesions of the glenohumeral joint, though less common than chondral defects in the knee or ankle, can be a significant source of pain in an active population. For patients in whom non-surgical management fails, promising results have been reported after arthroscopic microfracture surgery to treat such lesions. However, microfracture leads to growth of fibrocartilage tissue and is biomechanically less durable than native hyaline cartilage. Recently, augmentation of the microfractured defect with micronized allogeneic cartilage and PRP has been described to restore hyaline-like cartilage and potentially protect the subchondral bone from post-surgical fracture biology within the base of the defect. In a single-case study, these investigators presented a simple arthroscopic technique of implanting dehydrated, micronized allogeneic cartilage scaffold to treat an isolated chondral lesion of the glenoid. The authors noted that “Data regarding outcomes of BioCartilage use in human subjects are limited to expert opinion, but controlled human trials examining outcome differences between standard microfracture and BioCartilage techniques are currently under way …. Ongoing clinical studies will determine the effectiveness of augmented microfracture compared with standard microfracture alone”.

In a case-report, Desai (2014) stated that although talar dome osteochondral lesions (OCLs) are common injuries, OCLs of the tibial plafond are relatively infrequent. These lesions have historically been managed in a similar manner to talar OCLs, with most treated with debridement and marrow stimulation. This treatment has had mixed results. The present case report described a patient who underwent an all-arthroscopic surgical technique consisting of debridement and marrow stimulation with application of micronized allograft cartilage matrix (BioCartilage™, Arthrex, Naples, FL). This was a single case study on the use of BioCartilage for a tibial osteochondral defect; not a glenoid defect.

Muller et al (2010) noted that grafts generated by cultivation of progenitor cells from the stromal vascular fraction of human adipose tissue have been proven to have osteogenic and vasculogenic properties in-vivo. However, in-vitro manufacture of such implants is challenged by complex, impractical and expensive processes, and requires implantation in a separate surgery. This study investigated the feasibility of an intra-operative approach to engineer cell-based bone grafts with tissue harvest, cell isolation, cell seeding onto a scaffold and subsequent implantation within a few hours. Freshly isolated adipose tissue cells from a total of 11 donors, containing variable fractions of mesenchymal and endothelial progenitors, were embedded at different densities in a fibrin hydrogel, which was wrapped around bone substitute materials based on beta-tricalcium phosphate (ChronOS), hydroxyapatite (Engipore), or acellular xenograft (Bio-Oss). The resulting constructs, generated within 3 hours from biopsy harvest, were immediately implanted ectopically in nude mice and analyzed after 8 weeks. All explants contained blood vessels formed by human endothelial cells, functionally connected to the recipient's vasculature. Human origin cells were also found within osteoid structures, positively immune-stained for bone sialoprotein and osteocalcin. However, even with the highest loaded cell densities, no frank bone tissue was detected, independently of the material used. These results provided a proof-of-principle that an intra-operative engineering of autologous cell-based vasculogenic bone substitutes is feasible, but highlighted that -- in the absence of in-vitro commitment -- additional cues (e.g., low dose of osteogenic factors or orthotopic environmental conditions) are likely needed to support complete osteoblastic cell differentiation and bone tissue generation.

Ondrus et al (2011) tested the hypothesis that the application of tricalcium phosphate (TCP) mixed with autologous bone marrow can achieve better and faster healing of benign bone lesions than the application of tricalcium phosphate granules alone. The prospective study included 2 groups, each consisting of 10 patients, treated for benign cystic bone lesions at the Department of Paediatric Surgery, Orthopaedics and Trauma Surgery from July 1, 2008 to June 30, 2010. The bone cysts involved non-ossifying fibroma, enchodroma, fibrous dysplasia, aneurysmal bone cyst and juvenile bone cyst. One group was treated using ChronOS(TM) Beta-Tricalcium Phosphate (Synthes GmbH, Switzerland) granules mixed with autologous bone marrow harvested

during surgery (BM group). The other (CH group) received treatment with ChronOS granules alone. Relevant clinical data were obtained from all 20 patients treated for one of the bone cyst forms mentioned above. The patients were followed up till the end of 2010. TCP application was a 1-step procedure in both groups. In the BM group, bone regeneration ad integrum (Neer 1) was achieved, with only an occasional very small residue of the cyst seen on radiographs (Neer 2). None of the patients reported any problems, not even at 6 months after surgery. In the CH group, 2 patients required further surgical treatment because of insufficient bone healing (Neer 3) and 2 other patients reported pain persisting at the site of the lesion at 6 months post-operatively. In these patients TCP was used to fill a defect after excochleation of an aneurysmal bone cyst or fibrous dysplasia. The rest of the patients showed satisfactory healing. The main objective of the use of synthetic biocompatible materials in surgical treatment of benign bone cysts requiring filling of the lesion is to reduce the post-operative stress of pediatric patients as much as possible. Although their first results were not statistically significant to give unambiguous support to the hypothesis that lesions would heal better with the use of synthetic tricalcium phosphate mixed with autologous bone marrow, there is plenty of evidence that further development of cell technologies will result in a more exact definition of bone substitute materials in both their components, i.e., well-defined cells and non-biological scaffolds close in structure to inorganic compounds of bone, i.e., biodegradable osteo-inductive materials. The authors concluded that patients with benign bone lesions treated by TCP mixed with autologous bone marrow showed neither recurrent disease nor complications. The group treated with TCP alone had recurrent lesions in 2 and persisting pain also in 2 patients. Other complications were not recorded.

Currently, there is insufficient evidence to support the use of ChronOS bone graft substitute.

Osteocel

Neman et al (2013) noted that arthrodesis is a critical component of spine surgery for both degenerative and oncologic pathologies, with durable clinical benefits requiring successful bony fusion. The gold standard for bone grafting remains the autograft, optimally from the iliac crest.

However, the effectiveness of an autograft varies due to the inconsistent quality of the bone procured as well as risks of donor site morbidity. Several technologies exist as alternatives to autograft, either as a graft extender or replacement. These include treatment with bone morphogenetic protein (BMP-2), use of synthetic ceramics, demineralized bone matrix (DBM), and allografts; all with varying strengths and weaknesses in terms safety and/or efficacy. Alternatively, stem cells have become increasingly popular as cell-based therapeutics for musculoskeletal applications. Mesenchymal stem cells (MSCs) have been obtained from adipose tissue, bone marrow, peripheral blood, and synovial fluid, then combined with various osteo-conductive scaffolds. The rationale for their use is to add an osteogenic component to enhance formation of new bone via differentiation into osteoblasts. However, despite the appeal of this approach, there is a paucity of data supporting the efficacy of using stem cells in a clinical setting for spinal surgery. Furthermore, the best method for incorporating this technology into spinal surgery has not yet been determined. One approach has been to process an allograft such that endogenous progenitor cells are retained during the processing of freshly procured cadaveric bone. This approach has the advantage that cells potentially benefit from micro-environmental cues derived from maintaining their attachment to the native cancellous bone scaffold. Indeed, signaling in terms of chemical and mechanical cues between the cell and its scaffold is critically important for new bone formation. While cellularized allografts are known to harbor endogenous cells, the identity of these cells remains obscure, largely due to the lack of bona fide markers for stem and progenitor cells. In this study, these investigators hypothesized that a cellular allograft bone matrix (Osteocel Plus) contains a population of mesenchymal stem and bone progenitor cells, the former capable of self-renewal and multi-lineage differentiation. Currently, no single cell marker can unequivocally distinguish stem cells from progenitor cells. The use of cell surface marker combinations allows for enrichment of the stem cell population but is inadequate for prospective isolation. A novel use of lineage mapping allowed identification of highly proliferative clones and permitted us to determine whether cells endogenous to a cellular allograft undergo extensive self-renewal-a functional hallmark of stem cells. Further, these researchers used genetic and proteomic profiling as well as functional assays to examine whether these cells in the Osteocel Plus allograft are capable of multi-potential differentiation (the second functional hallmark of stem

cells). They postulated that the use of these 2 functional hallmarks could enable us to establish corroborative evidence for the existence of a stem and progenitor cell population in cellular allografts. They also stated that "As of the date of this publication, there have been no well-controlled prospective clinical studies published on the effectiveness of Osteocel Plus …. Taken together, these data provide corroborative evidence that Osteocel Plus cellular allograft contains a heterogeneous cell population with some cells demonstrating extensive self-renewal and multipotential differentiation in vitro -- the hallmarks for progenitor/stem cell state. In-vivo investigation is constrained to the use of small immune- deficient animals, because cellular allografts have retained human cells that would be rejected in immune-competent models. Small animal models, such as rodents, have limited translation to human biology. Ultimately, determining whether allografts containing a viable population of stem cells function comparably to autograft will require further study".

In a prospective, multi-center, non-randomized, institutional review board-approved clinical and radiographic study, Eastlack et al (2014) evaluated and summarized the 2-year outcomes of patients treated with Osteocel Plus cellular allograft as part of an anterior cervical discectomy and fusion procedure. A total of 182 patients were treated with anterior cervical discectomy and fusion using Osteocel Plus in a PEEK (polyetheretherketone) cage and anterior plating at 1 or 2 consecutive levels. Clinical outcomes included visual analog scale (VAS) for neck and arm pain, neck disability index, and SF-12 physical and mental component scores. Computed tomography and plain film radiographic measures included assessment of bridging bone, disc height, disc angle, and segmental range of motion (ROM). A total of 249 levels were treated in 182 patients. Mean procedure time was 100 minutes, blood loss was less than 50 ml in 93 % of patients, and hospital stay was 1 day or less in 84 % of patients. Significant ($p < 0.05$) average improvements in clinical outcomes from pre-operatively to 24 months included the following: neck disability index: 21.5 %; VAS neck: 34 mm; VAS arm: 35 mm; SF-12 physical component score: 11.2; SF-12 mental component score: 6.8. At 24 months, 93 % of patients were satisfied with their outcome. In patients treated at a single level with a minimum of 24-month follow-up, 92 % (79/86) of levels achieved solid bridging and 95 % of levels demonstrated ROM of less than 3°. In combined single- and 2-level procedures, 87 % (157/180) of levels achieved solid bridging and 92 % (148/161) had ROM

of less than 3° at 24 months. No patient required revision for pseudarthrosis. The authors concluded that improvements in clinical results at 2 years, high patient satisfaction, and high radiographic and clinical fusion rates provided confidence in Osteocel Plus as an effective alternative to structural allograft or autograft in anterior cervical discectomy and fusion procedures. The level of evidence was “4”. These findings need to be confirmed in well-designed randomized controlled trials with longer follow-up periods.

McAnany and colleagues (2016) noted that live MSC allograft-containing allogeneic bone grafts have recently gained popularity and currently account for greater than 17 % of all bone grafts and bone graft substitutes used in spinal surgery. Although the claim of cellular bone matrices containing osteogenic cells with osteoinductive properties is attractive, little is known about their clinical success when used in anterior cervical discectomy and fusion (ACDF). In a retrospective review of prospectively matched cohort of patients with radiologic assessment of fusion as the primary end-point, these investigators reported on the radiographic fusion rates in 1- and 2-level instrumented ACDF using an MSC. Two matched cohorts of adult patients who underwent ACDF with MSC or standard allograft were included. The outcome measures included radiographic and clinical evidence of healing at 1 year. A consecutive series of 57 patients who underwent a 1- or 2-level instrumented ACDF procedure between 2010 and 2012 were retrospectively analyzed. All fusion constructs comprised an interbody allograft, an anterior plate, and Osteocel. These patients were matched to a control group of 57 patients. Of the 57 cases in both cohorts, 29 (50.9 %) were 1-level, and 28 (49.1 %) were 2-level instrumented ACDFs. There were no significant differences in patient age ($p = 0.71$), gender, co-morbidity burden (Charlson Comorbidity Index [CCI]: 1.95; 2.42, $p = 0.71$) or body mass index (BMI; $p = 0.79$). At the 1-year follow-up, 50 of 57 (87.7 %) patients in the Osteocel cohort demonstrated a solid fusion compared with 54 of 57 (94.7 %) in the control group ($p = 0.19$); 7 (12.3 %) patients in the Osteocel cohort were reported as having a failed fusion at 1 year. The authors concluded that this was the first non-industry sponsored study to analyze a matched cohort assessing the 1-year arthrodesis rates associated with a non-structural MSC allograft in 1- and 2-level ACDF procedures. They noted that although not statistically significant, patients treated with MSC allografts demonstrated lower fusion rates compared

with a matched non-MSC cohort. They stated that further long-term studies with larger patient samples are needed before incorporating MSC allografts into routine clinical practice.

In an assessment of bone graft substitutes for the International Society for the Advancement of Spinal Surgery, Abjornson, et al. (2018) noted that, as human cell or tissue products (HCT/P)-regulated products, clinical data are not required for cellular autografts (CBMs) prior to commercialization. The assessment cited the aforementioned study by McAnany et al. comparing Osteocel CBM to a retrospective paired cohort of allograft patients in an anterior cervical discectomy and fusion (ACDF) surgery, finding no statistically significant difference in fusion rates between Osteocel CPM and the allograft control arm. The assessment also discussed the aforementioned noncontrolled study of Osteocel in ACDF by Eastlack et al. The assessment found: "In conclusion, CBM represents a promising bone grafting technology, but the HCT/P regulatory classification allows for the introduction of products without FDA review of preclinical or human clinical data for safety and efficacy. In addition, the biological mechanism for new bone formation has not been well elucidated in the clinical setting. The postimplantation biology regarding cell viability, differentiation, immunogenicity, and the growth factor profile has not been established in rigorous clinical trials."

Bone Morphogenetic Proteins

Kim et al (2013) stated that no long-term studies beyond a 2-year follow-up have been performed comparing the use of bone morphogenetic protein (BMP) versus iliac crest bone graft (ICBG) for fusion rates in long fusions to the sacrum in adult spinal deformity (ASD). In a single-center study, these researchers compared the use of BMP or ICBG on the long-term outcomes in patients undergoing long fusions to the sacrum for ASD. A total of 63 consecutive patients, from 1997 to 2006, comprised of 31 patients in the BMP group and 32 patients in the ICBG group, operated on at a single institution with a minimum 4-year follow-up (4 to 14 years) were analyzed. Inclusion criteria were ambulators who were candidates for long fusions (thoracic as the upper level) to the sacrum. Exclusion criteria were revisions, neuromuscular scoliosis, ankylosing spondylitis, and patients who had both BMP and ICBG used for fusion. Oswestry Disability Index (ODI) and 3 domains of the Scoliosis Research

Society (SRS) score were used to examine outcomes. The 2 groups were similar with respect to age, sex, smoking history, co-morbidities, BMI, number of fusion levels and Cobb angles; 8 patients in the BMP group underwent a posterior only, whereas 23 underwent combined anterior and posterior (A/P) surgery. All 32 patients in the ICBG had A/P fusion. The average BMP level was 11.1 mg (3 to 36 mg). The rate pseudarthrosis was 6.4 % (2/31) in the BMP and 28.1 % (9/32) in the ICBG group (p = 0.04) using Fisher exact test and odds ratio (OR) = 5.67. The fusion rates for BMP group were 93.5 % and 71.9 % for the ICBG group; ODIs were similar between the 2 groups. However, the BMP group demonstrated superior sum composite SRS scores in pain, self-image and function domains (p = 0.02). The authors concluded that BMP was superior to ICBG in achieving fusion in long constructs in ASD surgery. The rate of pseudarthrosis was significantly higher in the ICBG group than BMP group. The concentration and dosage of rhBMP-2 used appeared to have an effect on the rate of fusion and pseudarthrosis rate because no patient receiving more than 5 mg per level had apparent or detected pseudarthroses (n = 20/20). Level of Evidence = III. Moreover, these researchers stated that further studies on its safety profile regarding tumorigenesis and dose-related effectiveness are needed to justify its use and to optimize the cost-effective utilization of BMP.

The authors stated that drawbacks to this study included the difference in time periods between the patients with ICBG (1997 to 2002) and the BMP group (2002 to 2006). This may lend the populations to selection bias. However, it was not until 2002 that the uniform use of BMP at the authors' institution was practiced and during the transition time-points, BMP was used in addition to ICBG. Because the use of ICBG alone was not practiced after 2002, these investigators had to use patients who underwent surgery before 2002 in order to be able to formulate a matched pairs comparison. The demographic analysis did not show any significant differences in their populations, suggesting that the 2 groups were well-matched despite the difference in time periods and potential identifiable bias was controlled for statistically. Another potential drawback of the study was that the 2 groups showed significant differences in implant density. The ICBG group had a lower implant density compared with the BMP group, which might portend itself to higher pseudarthrosis rates. However, the ICBG group also had a higher number of anterior fusion levels performed that should intuitively result in

lower pseudarthrosis rates. These researchers recognized this as a possible limitation in the matched cohort comparison, although the combination of these surgical details may level any potential biases. Furthermore, the ODI and SRS outcome scores at the final follow-up were after the time-point in which a revision surgery was carried out for the non-unions; thus, they did not accurately reflect the ODI as well as SRS scores the authors most likely would have observed when a pseudarthrosis was present. It could be argued that the ICBG group would have had worse ODI and SRS outcomes if those scores at the time of pseudarthrosis, before revision surgery, were used for comparison with the BMP groups.

Bess et al (2014) noted that off-label rhBMP-2 use is common; however, under-reporting of rhBMP-2 associated complications has been recently scrutinized. In a prospective, observational, multi-center study, these researchers examined complications associated with recombinant human bone morphogenetic protein-2 (rhBMP-2) use in ASD. Inclusion criteria were age of 18 years and older, ASD, spinal arthrodesis of more than 4 levels, and 3 or more months of follow-up. Patients were divided into those receiving rhBMP-2 (BMP) or no rhBMP-2 (NOBMP). BMP divided into location of use: posterior (PBMP), interbody (IBMP), and interbody + posterior spine (I + PBMP). Correlations between acute peri-operative complications and rhBMP-2 use including total dose, dose/level, and location of use were evaluated. A total of 279 patients (mean age of 57 years; mean spinal levels fused was 12.0; and mean follow-up of 28.8 months) met inclusion criteria. BMP (n = 172; average posterior dose = 2.5 mg/level, average interbody dose = 5 mg/level) had similar age, smoking history, previous spine surgery, total spinal levels fused, estimated blood loss (EBL), and duration of hospital stay as NOBMP (n = 107; $p > 0.05$). BMP had greater Charlson Co-morbidity Index (1.9 versus 1.2), greater scoliosis (43° versus 38°), longer operative time (488.2 versus 414.6 mins), more osteotomies per patient (4.0 versus 1.6), and greater percentage of antero-posterior fusion (APSF; 20.9 % versus 8.4%) than NOBMP, respectively ($p < 0.05$). BMP had more total complications per patient (1.4 versus 0.6) and more minor complications per patient (0.9 versus 0.2) than NOBMP, respectively ($p < 0.05$). NOBMP had more complications requiring surgery per patient than BMP (0.3 versus 0.2; $p < 0.05$). Major, neurological, wound, and infectious complications were similar for NOBMP, BMP, PBMP, IBMP, and I + PBMP

($p > 0.05$). Multi-variate analysis demonstrated small to non-existent correlations between rhBMP-2 use and complications. The authors concluded that RhBMP-2 use as well as location of rhBMP-2 use in ASD surgery, at reported doses, did not increase acute major, neurological, or wound complications. Moreover, these investigators stated that future research on rhBMP-2 use must focus on long-term outcomes, evaluate outcomes at higher doses of rhBMP-2, and should consistently employ advanced statistical techniques to quantify the strength of correlations between rhBMP-2 use and specific complications.

The authors stated that this study had several drawbacks. First, the lack of long-term follow-up. Second, no analysis of spinal fusion rates. Third, no reporting of patient health-related quality of life (HR-QOL) outcomes scores. The mean follow-up for all patients in this study was 28.8 months (range of 3.6 to 47.9 months). However, the objective of this trial was to address concerns surrounding previous reports that failed to accurately evaluate short-term complications associated with rhBMP-2 use; thus, the inclusion criteria for the analysis were patients with minimum 3 months of follow-up. These researchers recognized the importance of long-term complication analysis for rhBMP-2 and anticipated that as these data set matured, they would report on long-term complications. No attempt was made in this study to examine the effectiveness of rhBMP-2 in promoting spinal fusion or improving HR-QOL outcomes for patients with ASD. The safety, effectiveness, and cost-effectiveness of rhBMP-2 use in ASD must be demonstrated. As these data set matured, these investigators would examine fusion rates, patient outcomes, and cost-effectiveness of rhBMP-2 for ASD. Finally, the mean rhBMP-2 dose per level reported in this study (PSF dose/level = 2.5 mg, IBF dose/level = 5.0 mg) was below values previously reported for rhBMP-2 use in the spine. This study was a prospective, observational study; consequently, the dosages of rhBMP-2 reported in this study were chosen according to the discretion of the surgeon. The dosages of rhBMP-2 reported may account for these findings, and it was possible that if rhBMP-2 was used in higher doses, greater complication rates may arise.

The use of BMPs in the pediatric population is off-label. Allareddy et al (2015) estimated the prevalence of complications in children who had insertion of rhBMP at the time of spinal fusion procedures (SFP) and examined if the use of rhBMP is associated with an increased risk of

complications. Nationwide Inpatient Sample (NIS) (years 2004 to 2010) was used. All patients with age less than 18 years who had a SFP during hospitalization with or without insertion of rhBMP were selected. Complications were selected based on a literature review of studies examining outcomes of SFP. Association between insertion of rhBMP and occurrence of complications was examined by multi-variable logistic regression models. Of the 72,898 children who underwent SFP, 7.1 % children had insertion of rhBMP. Overall complication rate was 14.34 % (15.2 % in rhBMP group and 14.3 % in no-rhBMP group). There was no statistically significant difference in the overall complication rate [odds ratio (OR)=1.08, 95 % CI: 0.89 to 1.30] or among 14 different complications between rhBMP and no-rhBMP groups. Children who had rhBMP were associated with higher odds for "other infections" (OR=2.09, 95 % CI: 1.26 to 3.48, p=0.004) when compared with their counterparts. The authors concluded that despite the lack of FDA approval, rhBMP was not infrequently used in pediatric SFP. In this large retrospective study using administrative data, the use of rhBMP in children during SFP was not associated with higher risks for majority of assessed complications with the exception of "other infections". The main drawbacks of this study were: (i) it did not include delayed complications necessitating emergency department visitation or out-patient setting, which might have under-estimated the present findings, and (ii) the effect of rhBMP on skeletal growth in children or the increased risk of cancer was not addressed. The authors stated that future studies must examine the long-term impact of use of rhBMP in skeletally immature children with SFP.

Paul et al (2016) stated that randomized controlled trials (RCTs) have suggested that BMP may increase the likelihood of solid arthrodesis in spinal surgery. This would imply fewer re-operations for pseudarthrosis; however, small cohort sizes were inadequate to monitor these events. In a retrospective study, followed the inpatient-stay administrative data that were collected for a cohort of thousands of patients who had spine fusion surgery in the state of New York (NY). These researchers examined ASD for re-operation events with and without the use of BMP. The 2008 to 2011 NY State Inpatient Database was queried using International Classification of Diseases, 9th Revision, Clinical Modification codes. Patients aged 21 years and older with a diagnosis of scoliosis and an

index fusion of greater than 2 spinal motion segments were included. Patient identifiers and linkage variables were used to identify revisits. The relative risk (RR) of re-operation was calculated. The use of BMP at the initial inpatient stay was used to define the 2 cohorts for RR assessment. A total of 3,751 patients of ASD were identified in 2008. The use of BMP at the initial visit was performed at a rate of 37.6 % for ASD. For posterior fusion cases longer than 8 levels, the rate of re-operation for a pseudarthrosis was 23.4 %. For ASD fusions greater than 8 levels, the rate of re-operation for pseudarthrosis after using BMP at the index surgery was 5 % and 33.9 % when BMP was not used, a RR of 7.5 ($p < 0.001$). The authors concluded that this study showed re-operation rates were decreased over time when BMP was used. These researchers theorized that the decreased re-operation rates were caused by the improved long fusion mass needed for successful treatment in ASD. Moreover, the authors cautioned use of BMP in off-label usages, in particular in women of child-bearing age and in children as its long-term impact on the fetus because of circulating antibodies and its carcinogenic potential in the skeletally immature individual is not known. Furthermore, these investigators stated that if subsequent unnecessary hospitals stay could be avoided, the economics of BMP use should be re-examined.

The authors stated that this study had several drawbacks. This was a limited time frame to follow a patient for pseudarthrosis, because this could be reported many years after surgery. The scope of this analysis concerned relatively early pseudarthrosis. In this analysis, there was no way to determine if the re-operation involved the same spinal levels previously operated. Moreover, precise pseudarthrosis rates were not determinable by their assessment, only re-operations, which likely under-estimated asymptomatic pseudarthrosis. Ultimately, patient-based outcomes were most important and HR-QOL assessments were not available in the database employed in this study. There were no data on pain, function, or symptomatic outcomes. These data did not allow direct measure of fusion rates, only of re-operation for pseudarthrosis. These researchers could not elucidate the reasons for revision, describe patient-reported outcomes, or determine if there were subtle differences between the population receiving BMP and the population who did not. The pseudarthrosis that appeared for re-operation were captured in the data; however, the actual fusion rate was not available and patients who did not re-appear for pseudarthrosis should not be assumed to have solid fusion.

The diagnostic coding scheme did not indicate transition syndrome (adjacent level disease). Therefore, these investigators could not examine the contribution of this phenomenon to the re-operation rate. The accuracy-related limitations of ICD-9-based coding of peri-operative adverse events (AEs) have been evaluated previously. This study showed that retrospective review may under-estimate the complication incidence. Under-representing pseudarthrosis was indeed possible in this study; however, it was unlikely that this under-representation would favor 1 cohort of patients over the other. Furthermore, this study showed an increased rate of re-operations compared to previous studies.

Poorman et al (2017) stated that since its introduction BMP has been used in populations with higher rates of malunion, such as ASD patients. Contradictory conclusions exist in spinal literature regarding the safety and effectiveness of the use of BMP in this setting. Previous studies, however, did not distinguish deformity cases from spondylolisthesis or stenosis. In a meta-analysis, these investigators examined the safety and effectiveness of BMP use in spinal fusion surgery for ASD. A total of 166 papers were screened after database search; and 40 full texts were assessed for eligibility. A total of 5 studies were included for meta-analysis; 3 were comparative studies between a BMP and non-BMP group, and the other was used to supplement dose-effect analysis. The current meta-analysis found increased odds of developing radiculitis or neurological complications (OR = 2.18, 95 % CI, p = 0.02, i2 = 0); however, no other significant relationship between complications commonly attributed to BMP use (tumorigenesis, infections, seroma formation, or osteolysis) and BMP use. BMP patients had decreased rates of pseudarthrosis (OR = 0.23, 95 % CI, p = 0.002, i2 = 0). There was an average dose of 8.75 mg/level in the 417 patients studied, lower than the advised dosage of 12 mg/level. The authors concluded that the current literature showed BMP to be a safe and effective grafting technique in the treatment of ASD. Spine surgeons may currently be using sub-optimal doses of BMP. The benefit of increasing the rate of fusion must be weighed against the increased risk of radiculitis and neurologic complications in this patient population.

The authors stated that this meta-analysis was limited by the lack of randomized clinical control trials. Without randomization it was very difficult to control for confounding variable, as there may have been

patient variables that led spine surgeons to choose to use rhBMP-2. One such variable was that patients receiving rhBMP-2 were usually older and have increased co-morbidities, due to the lack of autologous graft availability. Furthermore, the surgeons performing the operation in the included studies were spine surgeons operating in large volume hospitals with familiarity of the technical aspects of rhBMP-2. The reported results may be different when performed by surgeons with less familiarity with rhBMP-2.

Wetzell et al (2021) built upon previously reported 12-month findings by retrospectively comparing 24-month follow-up hospitalization charges and potentially-relevant re-admissions in U.S. lumbar fusion surgeries that employed either rhBMP-2 or a cellular bone allograft comprised of viable lineage-committed bone cells (V-CBA) via a nationwide healthcare system database. A total of 16,172 patients underwent lumbar fusion surgery using V-CBA or rhBMP-2 in the original study, of whom 3,792 patients (23.4 %) were identified in the current study with all-cause re-admissions during the 24-month follow-up period. Confounding baseline patient, procedure, and hospital characteristics found in the original study were used to adjust multi-variate regression models comparing differences in 24-month follow-up hospitalization charges (in 2020 US$) and lengths of stay (LOS; in days) between the groups. Differences in potentially relevant follow-up re-admissions were also compared, and all analyses were repeated in the subset of patients who only received treatment at a single level of the spine. The adjusted cumulative mean 24-month follow-up hospitalization charges in the full cohort were significantly lower in the V-CBA group ($99,087) versus the rhBMP-2 group ($124,389; $p < 0.0001$), and this pattern remained in the single-level cohort (V-CBA = $104,906 versus rhBMP-2 = $125,311; $p = 0.0006$). There were no differences between groups in adjusted cumulative mean LOS in either cohort. Differences in the rates of follow-up re-admissions aligned with baseline co-morbidities originally reported for the initial procedure. Subsequent lumbar fusion rates were significantly lower for V-CBA patients in the full cohort (10.12 % versus 12.00 %; $p = 0.0002$) and similar between groups in the single-level cohort, in spite of V-CBA patients having significantly higher rates of baseline co-morbidities that could negatively impact clinical outcomes, including bony fusion. The authors concluded that the findings of this study suggested that use of V-CBA for lumbar fusion surgeries performed

in the U.S. was associated with substantially lower 24-month follow-up hospitalization charges versus rhBMP-2, with both exhibiting similar rates of subsequent lumbar fusion procedures and potentially relevant re-admissions.

Chang et al (2022) examined the current evidence on graft materials used in fusion procedures for spinal deformity corrections. These investigators searched PubMed, Embase, and Cochrane Library for relevant published observational studies and clinical trials using osteo-biologics and biomaterials in spinal deformity surgery. The use of autograft in deformity correction surgeries has been reported in a limited number of studies, with the harvest sites including iliac crest, ribs, and local bone. Various allografts and biologics have been used in the treatment of spinal deformities including idiopathic and degenerative scoliosis, either as stand alone or in combination with autograft. Limited number of studies reported no differences in fusion rates or outcomes. Use of rh-BMP2 in anterior, posterior or front/back approaches showed higher fusion rates than other graft materials in patients with spinal deformities. Due to the limited number of quality studies included in the review, as well as alternative factors, such as costs, availability, and surgeon expertise/preference, no definitive conclusion or recommendations can be made as to the ideal graft choice in spinal deformity surgery. The authors concluded that most commonly used grafts included autograft, allograft and rh-BMP2, with new biologics and biomaterials constantly emerging in the market. Limited number of high-quality comparative studies and heterogeneity in study design prevented direct comparisons that can lead to meaningful recommendations. These researchers stated that further studies are needed to prove superiority of any single graft material and/or biologic that is also cost-effective and safe.

In an investigator-initiated, prospective RCT, Choi et al (2022) examined the safety and effectiveness of the combined use of rhBMP-2 and a hydroxyapatite (HA) carrier in multi-level fusion in patients with ASD. A total of 30 patients underwent posterolateral fusion for lumbar spinal deformities at 3 to 5 segments between L1 and S1. Subjects received rhBMP-2+HA or HA on the left or right side of the transverse processes. They were followed-up regularly at 1, 3, 6, and 12 months post-operatively. Fusion was defined according to the bone bridging on

computed tomography (CT) scans. The fusion rate per segment was sub-analyzed. Function and QOL as well as pain in the lower back and lower extremities were evaluated. The union rate for the rhBMP-2+HA group was 100 % at 6 and 12 months. The union rate for the HA group was 77.8 % (21 of 27) at 6 months and 88.0 % (22 of 25) at 12 months ($p = 0.014$ at 6 months; not significant at 12 months). All segments were fused at 6 and 12 months in the rhBMP-2+HA group ($p < 0.001$). In the HA group, 108 of 115 segments (93.5 %) were fused at 6 months and 105 of 109 segments (96.3 %) at 12 months. Other clinical parameters (VAS, 36-item Short Form Health Survey, and Scoliosis Research Society-22 scores) improved compared to baseline. The authors concluded that combining rhBMP-2 and an HA carrier was a safe and effective method to achieve multi-level fusion in patients with ASD.

The author stated that this study had several drawbacks. First, the number of patients was too small (n = 30) to sufficiently detect differences in effects between groups. Second, the follow-up period was short (12 months); thus, it is necessary to verify long-term clinical results with a larger cohort in the future. Third, these researchers employed 2 interventions simultaneously in each subject that could have mutually influenced the outcomes of each treatment group. The split-body trial had an advantage that the confounding factors between groups could be removed because the subjects of the test group were the same. However, since half of the human body is not completely independent, the difference in effects between groups may be under-estimated, and systemic side effects may be masked. Thus, this trial was able to directly compare the difference in effects according to BMP-2 without confounding factors between individuals; however, the degree of difference may have been under-estimated. Furthermore, safety could not be evaluated separately between groups.

OsteoAMP

Roh et al (2013) noted that since the introduction of rhBMP-2 (Infuse) in 2002, surgeons have had an alternative substitute to autograft and its related donor site morbidity. Recently, the prevalence of reported adverse events (AEs) and complications related to the use of rhBMP-2 has raised many ethical and legal concerns for surgeons. Additionally, the cost and decreasing reimbursement landscape of rhBMP-2 use have

required identification of a viable alternative. Osteo allogeneic morphogenetic protein (OsteoAMP) is a commercially available allograft-derived growth factor rich in osteoinductive, angiogenic, and mitogenic proteins. These researchers compared the radiographic fusion outcomes between rhBMP-2 and OsteoAMP allogeneic morphogenetic protein in lumbar interbody fusion spine procedures. A total of 321 patients from 3 centers underwent a transforaminal lumbar interbody fusion (TLIF) or lateral lumbar interbody fusion (LLIF) procedure and were assessed by an independent radiologist for fusion and radiographically evident complications. The independent radiologist was blinded to the intervention, product, and surgeon information; 226 patients received OsteoAMP with autologous local bone, while 95 patients received Infuse with autologous local bone. Patients underwent radiographs (x-ray and/or CT) at standard post-operative follow-up intervals of approximately 1, 3, 6, 12, and 18 months. Fusion was defined as radiographic evidence of bridging across endplates, or bridging from endplates to interspace disc plugs. Osteobiologic surgical supply costs were also analyzed to ascertain cost differences between OsteoAMP and rhBMP-2. OsteoAMP produced higher rates of fusion at 6, 12, and 18 months ($p \leq 0.01$). The time required for OsteoAMP to achieve fusion was approximately 40 % less than rhBMP-2 with approximately 70 % fewer complications. Osteobiologic supply costs were 80.5 % lower for OsteoAMP patients (73.7 % lower per level) than for rhBMP-2. The authors concluded that the findings of this study indicated that OsteoAMP is a viable alternative to rhBMP-2 both clinically and economically when used in TLIF and LLIF spine procedures. The drawbacks of this study were: (i) each center used different instrumentation and fixation devices, which may influence some of the results, (ii) the study did not evaluate the clinical outcomes, and (iii) the 3 surgeons have unique, surgical techniques that may have contributed to some variability within the results. The authors stated that multi-center RCTs are needed to confirm the effectiveness and cost-effectiveness of osteoAMP.

Field et al (2014) stated that the initial success of rhBMPs in lumbar spine surgery led to its use outside the initial indication. As complications from the use of rhBMP-2 in cervical spine surgery continued to rise, the need for a safer alternative was evident. The discovery of a new allogeneic tissue processing technique has provided a way to access growth factors

naturally found within bone marrow cells. These investigators evaluated the clinical outcomes associated with the use of allogeneic morphogenetic protein in cervical spine fusion. They performed a retrospective analysis of 140 consecutive patients (228 levels) who underwent cervical spine fusions between C3 and T3. Patients received radiographs (x-ray and/or CT) at standard post-operative follow-up time-points, which were generally at 3, 6, 12 and 18 months post-surgical intervention. Fusion was defined as any radiographic evidence of bridging across endplates, or bridging from endplates to interspace disc plugs; 80 % of patients had evidence of fusions at 6 months, 98 % of patients had evidence of fusions at 12 months, and 100 % of patients had evidence of fusions at 18 months. The authors concluded that high fusion rate resulted in this report demonstrated the benefits of using an array of growth factors in cervical spine surgery and supported allogeneic morphogenetic protein as a possible alternative option to rhBMP-2. The main drawbacks of this study were its retrospective design, lack of a control group, and clinical outcomes were not assessed. The authors stated that multi-center RCTs are needed to confirm the clinical effectiveness and results of this analysis.

Yeung and colleagues (2014) stated that donor-to-donor variation has long been a concern of the allograft industry. Demineralized bone matrix and stem cell products have been particularly susceptible to inter-variability between donors, regardless of the process used to manufacture these products. Manufacturers of allograft based products have often utilized in-vitro or small animal models to help predict reliability, yet little data are available correlating pre-clinical outcomes with clinical efficacy. OsteoAMP is a commercially available allograft-derived growth factor rich in osteoinductive, angiogenic, and mitogenic proteins. These researchers carried out an analysis of radiographic results comparing fusion outcomes for 285 consecutive cervical and lumbar spinal fusion patients utilizing OsteoAMP bone grafts from 114 donors of varying ages. A blinded radiological fusion assessment, performed by an independent radiologist, showed all patients, except 1, fused within 18 months (average time to fusion was 189.9 days). The authors concluded that this evidentiary analysis showed that OsteoAMP fusion success did not show donor inter-variability and that fusion rate/time is not dependent on donor age. In addition, the implant retained bioactivity over time and terminal sterilization via low-dose gamma irradiation did not impair the

bioactivity of the grafts. The main drawbacks of this study were its retrospective design, lack of a control group, and clinical outcomes (ODI, VAS, etc.) were not assessed. The authors stated that future study is needed to further track any donor-dependent, gender-based differences in the time to fusion.

An et al (2014) noted that postero-lateral lumbar fusions have been successfully used to surgically treat mechanical back pain, low-grade spondylolisthesis and other degenerative spinal conditions. The addition of biological grafts to augment available autologous bone has further improved fusion rates, yet, some of these biologics have been found to cause deleterious post-operative clinical situations and sometimes are used in an off-label manner. A biological alternative that provides equivalent fusion rates with a similar, or lower, risk profile is desirable. These investigators reported on fusion rates associated with the use of OsteoAMP to assist with lumbar spinal arthrodesis with and without augmentation with bone marrow aspirate as compared to rhBMP-2 used with and without the allogeneic growth factor. Patients having postero-lateral lumbar fusion were evaluated for fusions at clinically relevant time-points. A total of 302 patients (146 growth factor with BMA, 81 growth factor without BMA, 50 rhBMP-2 alone, 25 rhBMP-2 with allogeneic growth factor) were retrospectively reviewed. The growth factor with BMA group had approximately an 88 % fusion rate by 12 months and 99 % by 24 months. The growth factor non-BMA group had a fusion rate of 35 % by 12 months and exceeding 98 % at the 2 year follow-up. The OsteoAMP augmented rhBMP-2 group had fusion rates of 33 % at 12 months and 100 % at 24 months, while the rhBMP-2 alone group only attained a 14 % fusion rate at 12 months and a 32 % fusion rate at 24 months. The authors concluded that the allogeneic growth factor appeared to provide a viable option to assist with the development of postero-lateral spinal arthrodesis. Moreover, they stated that longer follow-up and increased patient sample size are needed to confirm these initial findings. The main drawbacks of this study were its retrospective design, the number of patients in each group vary in sample size, and the reporting clinical outcomes (relief of pain via VAS, restoration of function via ODI, patient satisfaction, etc.) was not available in all patients at all time-points.

Fallucco and Carstens (2009) stated that a novel method for primary alveolar cleft bony reconstruction avoids donor site morbidity and does not require close approximation of alveolar segments as necessitated by gingivo-periosteoplasty. These researchers provided radiographic evidence of de-novo synthesis of bone using rhBMP-2. This institutional review board-approved class IV study retrospectively evaluated primary alveolar cleft patients from 2004 to 2006. Subjects chose an off-label application of rhBMP-2 impregnated on an absorbable collagen sponge carrier to reconstruct alveolar clefts, all greater than 3 mm in width. The surgical technique used for soft tissue closure, developmental field re-assignment, is not a gingivo-periosteoplasty and is applicable to alveolar clefts of virtually any size. Developmental field re-assignment produced a periosteal "pocket" containing mesenchymal stem cells sensitive to cytokines, such as BMP-2. Inductive conversion of mesenchymal stem cells to osteoblasts by BMP-2 is known as in-situ osteogenesis (ISO). A total of 17 cleft sites treated with ISO were evaluated at 6 months post-operative using low-dose spiral computed tomography with 1-mm cuts limited to the maxilla. Alveolar bone density (Hounsfield units) was assessed by 3 radiologists; trabecular bone was defined as Hounsfield units of more than 226. In 16 of 17 cleft sites, ISO produced trabecular bone that filled the implantation site both transversely and vertically. The authors concluded that stem cell stimulation using rhBMP-2/absorbable collagen sponge offered highly effective radiographic and clinical unification of the dental arch without donor site morbidity. Moreover, they stated that long-term follow-up studies for this initial cohort are under way to examine information on orthodontic relationships and cephalometrics using cone-beam computed tomography technology. This was a small (n = 17) short-term (6 months follow-up) that examined the use of rhBMP-2 for alveolar cleft bony reconstruction; not for ankle fusion.

Fourman and colleagues (2014) noted that although its FDA-approved applications are limited, the pro-osteogenic benefits of recombinant human BMP-2 (rhBMP-2) administration have been shown in off-label surgical applications. However, the effects of rhBMP-2 on ankle fusions are insufficiently addressed in the literature, which fails to include a case-control study of adequate sample size to evaluate the efficacy of rhBMP-2 treatment. In this study these researchers examined if rhBMP-2

treatment (i) would increase the rate of successful ankle fusion in complex patients (patients with co-morbidities associated with poor surgical healing) compared with a control group of patients undergoing ankle fusion who did not receive rhBMP-2; (ii) would reduce total time wearing a frame when compared with the control group; (iii) would result in a difference in the percentage of bone bridging between the group treated with rhBMP-2 and the control group, as determined by CT scans 3 months after surgery; and (iv) would encounter an equal rate of complications different from untreated patients. A retrospective chart study was performed on 82 patients who, because of a host of co-morbidities associated with poor healing, required a complex ankle arthrodesis with the Ilizarov technique. The first 40 patients did not receive rhBMP-2, whereas the subsequent 42 patients received intraoperative rhBMP-2. Time wearing the frame was determined by chart review; decision to remove the frame was made by the surgeon based on quantitative bone bridging measured using a CT scan taken 3 months after fusion. Patients treated with rhBMP-2 were more likely to obtain fusion after the initial surgery (93 % versus 53 %, $p < 0.001$; OR, 11.76; 95 % CI: 3.12 to 44.41), spent less total time wearing the frame (124 versus 161 days, $p < 0.01$), and showed more bone bridging on CT scans (48 % versus 32 %, $p < 0.05$). All patients with greater than 30 % bone bridging observed on CT scans 3 months post-operatively achieved successful union without further intervention. The authors concluded that the findings of this study suggested that rhBMP-2 is a beneficial adjunct for selected groups of patients undergoing complex ankle arthrodesis; and CT is a promising modality in the assessment of bone healing in ankle fusion. Moreover, they stated that a proper randomized controlled trial is needed to ascertain the effectiveness of rhBMP-2 in accelerating bone healing. (This study provided Level III evidence).

Anti-Microbial Bone Graft Substitute for the Treatment of Osteomyelitis

van Vugt and colleagues (2016) noted that osteomyelitis is a common occurrence in orthopedic surgery, which is caused by different bacteria. Treatment of osteomyelitis patients aims to eradicate infection by debridement surgery and local and systemic antibiotic therapy. Local treatment increases success rates and can be performed with different

anti-microbial bone graft substitutes. These investigators evaluated the level of evidence of synthetic bone graft substitutes in osteomyelitis treatment. According to the PRISMA statement for reporting systematic reviews, different types of clinical studies concerning treatment of osteomyelitis with bone graft substitutes were included. These studies were evaluated on their methodological quality as level of evidence and bias and their clinical outcomes as eradication of infection. In the 15 included studies, the levels of evidence were weak and in 10 out of the 15 studies there was a moderate-to-high risk of bias. However, first results of the eradication of infection in these studies showed promising results with their relatively high success rates and low complication rates. The authors concluded that as a consequence of the low levels of evidence and high risks of bias of the included studies, these results were inconclusive and no conclusions regarding the performed clinical studies of osteomyelitis treatment with anti-microbial bone graft substitutes could be drawn.

Progenitor Cells

McLain et al (2005) stated that connective tissue progenitor cells aspirated from the iliac crest and concentrated with allograft matrix and demineralized bone matrix provide a promising alternative to traditional autograft harvest. The vertebral body, an even larger reservoir of myeloproliferative cells, should provide progenitor cell concentrations similar to those of the iliac crest. The authors conducted a study in twenty-one adults (eleven men and ten women with a mean age of 59 +/- 14 years) who were undergoing posterior lumbar arthrodesis and pedicle screw instrumentation, and who underwent transpedicular aspiration of connective tissue progenitor cells. The methodology included collection of the cell count, progenitor cell concentration (cells/cc marrow), and progenitor cell prevalence (cells/million cells) were calculated and aspirates of vertebral marrow demonstrated comparable or greater concentrations of progenitor cells compared with matched controls from the iliac crest. Progenitor cell concentrations were consistently higher than matched controls from the iliac crest ($p = 0.05$). The concentration of osteogenic progenitor cells was, on the average, 71% higher in the vertebral aspirates than in the paired iliac crest samples ($p = 0.05$). With the numbers available, there were no significant differences relative to vertebral body level, the side aspirated, the depth of aspiration, or gender.

An age-related decline in cellularity was suggested for the iliac crest aspirates. The authors concluded that the vertebral body is a suitable site for aspiration of bone marrow for graft augmentation during spinal arthrodesis.

Kartogenin-Treated Autologous Tendon Graft

Huang and colleagues (2017) stated that the meniscus is one of the most commonly injured parts of the body, and meniscal healing is difficult. Kartogenin (KGN) induces tendon stem cells (TSCs) to differentiate into cartilage cells in-vitro and form meniscus-like tissue in-vivo. A damaged meniscus can be replaced with a KGN-treated autologous tendon graft. In a controlled laboratory study, TSCs were isolated from rabbit patellar tendons and cultured with various concentrations of KGN, from 0 to 1,000 μM . The effect of KGN on the chondrogenesis of TSCs in-vitro was investigated by histochemical staining and quantitative real-time reverse transcription polymerase chain reaction (qRT-PCR). The in-vivo experiments were carried out on 6 New Zealand White rabbits by removing a meniscus from the rabbit knee and implanting an autologous tendon graft treated with KGN or saline. The meniscus formation i- vivo was examined by histological analysis and immune staining. The proliferation of TSCs was promoted by KGN in a concentration-dependent manner. Both histochemical staining and qRT-PCR showed that the chondrogenic differentiation of TSCs was increased with KGN concentration. After 3 months of implantation, the tendon graft treated with KGN formed a meniscus-like tissue with a white and glistening appearance, while the saline-treated tendon graft retained tendon-like tissue and appeared yellowish and unhealthy. Histochemical staining showed that after 3 months of implantation, the KGN-treated tendon graft had a structure similar to that of normal meniscus. Many cartilage-like cells and fibrocartilage-like tissues were found in the KGN-treated tendon graft. However, no cartilage-like cells were found in the saline-treated tendon graft after 3 months of implantation. Furthermore, the KGN-treated tendon graft was positively stained by both anti-collagen type I and type II antibodies, but the saline-treated tendon graft was not stained by collagen type II. The authors concluded that the findings of this study indicated that KGN can induce the differentiation of TSCs into cartilage-

like cells in-vitro and in-vivo; the results suggested that KGN-treated tendon graft may be a good substitute for meniscal repair and regeneration.

Nacre (Mother-of-Pearl)

Zhang and colleagues (2017) stated that during the past 20 years, with a huge and rapidly increasing clinical need for bone regeneration and repair, bone substitutes are more and more seen as a potential solution. Major innovation efforts are being made to develop such substitutes, some having advanced even to clinical practice. It is now time to turn to natural biomaterials. Nacre, or mother-of-pearl, is an organic matrix-calcium carbonate coupled shell structure produced by mollusks. In-vivo and in-vitro studies had revealed that nacre is osteoinductive, osteoconductive, biocompatible, and biodegradable. With many other outstanding qualities, nacre represents a natural and multi-use biomaterial as a bone graft substitute. This review aimed at summarizing the current needs in orthopedic clinics and the challenges for the development of bone substitutes; most of all, the authors systematically reviewed the physiological characteristics and biological evidence of nacre's effects centered on osteogenesis, and finally these researchers put forward the potential use of nacre as a bone graft substitute.

Gerhard and associates (2017) noted that the field of tissue engineering and regenerative medicine relies heavily on materials capable of implantation without significant foreign body reactions and with the ability to promote tissue differentiation and regeneration. The field of bone tissue engineering in particular requires materials capable of providing enhanced mechanical properties and promoting osteogenic cell lineage commitment. While bone repair has long relied almost exclusively on inorganic, calcium phosphate ceramics such as hydroxyapatite and their composites or on non-degradable metals, the organically derived shell and pearl nacre generated by mollusks has emerged as a promising alternative. Nacre is a naturally occurring composite material composed of inorganic, calcium carbonate plates connected by a framework of organic molecules. Similar to mammalian bone, the highly organized microstructure of nacre endows the composite with superior mechanical properties while the organic phase contributes to significant bioactivity. Studies, both in-vitro and in-vivo, have demonstrated nacre's

biocompatibility, biodegradability, and osteogenic potential, which are superior to pure inorganic minerals such as hydroxyapatite or non-degradable metals. Nacre can be used directly as a bulk implant or as part of a composite material when combined with polymers or other ceramics. While nacre has demonstrated its effectiveness in multiple cell culture and animal models, it remains a relatively under-explored biomaterial. This review introduced the formation, structure, and characteristics of nacre, and discussed the present and future uses of this biologically-derived material as a novel biomaterial for orthopedic and other tissue engineering applications. The authors concluded that nacre is a highly promising yet overlooked biomaterial for orthopedic tissue engineering with great potential in a wide variety of material systems. It is the authors' hope that publication of this article will lead to increased community awareness of the potential of nacre as a versatile, bioactive ceramic capable of improving bone tissue regeneration and will elicit increased research effort and innovation utilizing nacre.

Stem Cell Therapy

Miguita and colleagues (2017) noted that several non-biological materials are currently being used to increase the alveolar bone volume to support dental implants. Recently, stem cell therapy (SCT) has emerged as a promising biological substitute or adjuvant to enhance bone healing. In order to determine if SCT has enough clinical evidence to bone ridge augmentation in humans, a systematic review and meta-analysis were carried out. Two independent investigators searched the Entrez PubMed, SCOPUS and Web of Science databases for eligible randomized clinical trials that describe SCT for alveolar bone formation. The included studies were evaluated for risk of bias. A random-effects meta-analysis model was used to evaluate the percentage of bone formation in the selected studies. Heterogeneity was evaluated using the Cochrane Chi 2 and I 2; a total of 9 eligible trials were included. These studies presented an overall unclear risk of bias. A comparison between the lower heterogeneity studies and the long-term observational outcomes showed a slight tendency to enhance bone formation. High heterogeneity between the included studies was observed. The lack of outcome standardization made a wide-ranging comparison difficult. The authors concluded that application of stem cells in oral surgery and implantology

appeared to be promising although more standardized study designs, increased samples and long-term observations are needed to strength the clinical evidence that SCT is effective for alveolar bone formation.

Tooth-Bone Graft

In a systematic review, Gharpure and Bhatavadekar (2018) evaluated the clinical efficacy of the tooth-bone graft as a bone substitute in the oral and maxillofacial region in humans as compared to un-grafted sites and other bone substitutes. Databases were electronically and manually searched up to January 2017 to identify animal and human studies and a risk of bias analysis and descriptive statistics was performed. A total of 18 animal controlled trials (401 animals), 4 human RCTs, 1 cohort study, and 3 controlled trials (184 patients) were included. Graft processing was highly heterogeneous; 71.42 % clinical and 55.56 % animal studies reported no significant difference between tooth-bone graft and controls. Histologically, a dentin-bone complex was reported. A low risk of bias was noted in only 50 % of the RCTs and 63.33 % animal study entries. An independent analysis of 6 high-quality case reports (350 patients) revealed complications in 18.86 % cases. The authors concluded that tooth-bone graft demonstrated no added benefits over conventional graft materials. They stated that in the absence of standardized processing and heterogeneous study results limited its use in clinical practice; until long-term studies determine its success, clinicians are recommended to use it with caution because of high variability in resorption time (2 to 24 weeks) and a risk of graft dehiscence (12.96 % to 34.38 %).

I-Factor Bone Graft

Technology: P-15 bone putty (i-Factor) is a synthetic osteoconductive bone substitute that is approved for use in cervical fusion procedures. The first step in the bone formation process is cell attachment. Osteogenic precursor cells bind to P-15, then a natural signaling cascade occurs that leads to new bone formation (CMS, 2016).

i-Factor is labeled for use in skeletally mature patients for reconstruction of a degenerated cervical disc at one level from C3-C4 to C6-C7 following single-level discectomy for intractable radiculopathy (arm pain and/or neurological deficit), with or without neck pain, or myelopathy due to a

single-level abnormality localized to the disc space, and corresponding to at least one of the following conditions confirmed by radiographic imaging (CT, MRI, X-rays): herniated nucleus pulposus, spondylolysis (defined by the presence of osteophytes), and/or visible loss of disc height as compared to adjacent levels,after failure of at least 6 weeks of conservative treatment. i-Factor Peptide Enhanced Bone Graft must be used inside an allograft bone ring and with supplemental anterior plate fixation.

i-Factor peptide enhanced bone graft should not be used in situations where there is:

- Acute or chronic infections, systemic or at the operative site
- An absence of load bearing structural support at the graft site
- Compromised renal or hepatic function
- Metabolic or systemic disorders that affect bone or wound healing
- Sensitivity to any components of i-Factor peptide enhanced bone graft.

Yang et al (2004) noted that type I collagen provides a structural framework for connective tissues and plays a central role in the temporal cascade of events leading to the formation of new bone from progenitors. These researchers examined the ability of the cell-binding domain of type I collagen (P-15 peptide [i-Factor]) to promote human bone marrow stromal cell adhesion, proliferation, and differentiation on 3D scaffolds. Human bone marrow stromal cells were selected, expanded, and cultured on particulate microporous ABM ("pure" hydroxyapatite) phase adsorbed with or without P-15 under basal or osteogenic conditions. Immobilized P-15 increased alkaline phosphatase activity and bone morphogenetic protein 2 (BMP-2) gene expression after 1 and 5 days as determined by real-time PCR. P-15 promoted human bone marrow stromal cell attachment, spreading, and alignment on ABM as well as alkaline phosphatase-specific activity in basal and osteogenic cultures. The presence of mineralized bone matrix, extensive cell ingrowth, and cellular bridging between 3D matrices adsorbed with P-15 was confirmed by confocal microscopy, scanning electron microscopy, and alizarin red staining. Negligible cell growth was observed on ABM alone. In-vivo diffusion chamber studies using MF1-nu/nu mice showed bone matrix formation and organized collagen formation after 6 weeks. The authors

concluded that these studies indicated the potential of P-15 to generate appropriate biomimetic microenvironments for osteoblasts and demonstrated the potential for the exploitation of extracellular matrix cues for osteogenesis and, ultimately, bone regeneration.

Mobbs et al (2014) determined the safety and efficacy of i-FACTOR bone graft composite used in patients who underwent anterior lumbar interbody fusion (ALIF) by evaluating fusion rates and clinical outcomes. A non-blinded cohort of patients who were all referred to a single surgeon's practice was prospectively studied. A total of 110 patients with degenerative spinal disease underwent single or multi-level ALIF using the ABM/P-15 bone graft composite with a mean of 24 months (minimum 15 months) of follow-up were enrolled in the study. Patient's clinical outcomes were assessed using the ODI for low-back pain, the 12-Item Short Form Health Survey, Odom's criteria, and a VAS for pain. Fine-cut CT scans were used to evaluate the progression to fusion. All patients who received i-FACTOR demonstrated radiographic evidence of bony induction and early incorporation of bone graft. At a mean of 24 months of follow-up (range of 15 to 43 months), 97.5 %, 81 %, and 100 % of patients, respectively, who had undergone single-, double-, and triple-level surgery exhibited fusion at all treated levels. The clinical outcomes demonstrated a statistically significant ($p < 0.05$) difference between pre-operative and post-operative ODI, 12-Item Short Form Health Survey, and VAS. The authors concluded that the use of i-FACTOR bone graft substitute demonstrated promising results for facilitating successful fusion and improving clinical outcomes in patients who undergo ALIF surgery for degenerative spinal pathologies. Moreover, they stated that further studies comparing rate of arthrodesis, clinical outcome, and cost between ABM/P-15 and other graft alternatives are needed.

The authors stated that limitations of this study were the lack of direct control and the potential for respondent bias. Patients who participated in research were more likely to comply with post-operative medications and physiotherapy, thus resulting in better health outcomes. For instance, a few patients declined completing the post-operative outcome data because they were discontent with their outcome. These patients were subsequently excluded from statistical analysis, which may have positively skewed results.

Lauweryns and Raskin (2015) examined the safety and efficacy of bone graft material ABM/P-15 (iFACTOR) for use in posterior lumbar interbody fusion (PLIF). A total of 40 patients underwent PLIF surgery, with each patient as control. Assessments up to 24 months included radiographs, CT scan, VAS, and ODI. Primary success criteria were fusion and safety. Intra-cage bridging bone occurred earlier with ABM/P-15 than autograft (97.73 % versus 59.09 % at 6 months). On average pain decreased 29 points and function improved 43 points. Radio dense material outside the disk space occurred more frequently with ABM/P-15 than autograft, without clinical consequence. The authors concluded that the findings of this study suggested that ABM/P-15 had equal or greater efficacy at 6 and 12 months. Pain improvements exceeded success criteria at all time-points. Functional improvement exceeded success criteria at all time-points. Moreover, they stated that additional studies, with separate interventional and control groups and larger sample sizes, are needed to further investigate and clearly delineate differences between the safety and efficacy of ABM/P-15 and autograft.

The authors stated that study limitations included the small sample size (n = 40) and lack of a separate control group, which did not offer as “clean” a comparison as 2 separate patient groups. These researchers could not rule out the potential for “biological crosstalk,” with either autograft or ABM/P-15 migrating and benefitting the other. To reduce the risk of any occurrence influencing interpretation of outcomes, these investigators had an independent radiologist reported fusion only inside each of the cages.

In a prospective, randomized, controlled, parallel, single-blinded, non-inferiority multi-center pivotal FDA investigational device exemption (IDE) trial, Arnold et al (2016) examined the safety and efficacy of i-Factor Bone Graft (i-Factor) compared with local autograft in single-level anterior cervical discectomy and fusion (ACDF) for cervical radiculopathy. Patients randomly received either autograft (n=154) or i-Factor (n=165) in a cortical ring allograft. Study success was defined as non-inferiority in fusion, neck disability index (NDI), and neurological success end-points, and similar AEs profile at 12 months. At 12 months (follow-up rate 87 %), both i-Factor and autograft subjects demonstrated a high fusion rate (88.97 % and 85.82 %, respectively, non-inferiority p=0.0004), significant improvements in NDI (28.75 and 27.40, respectively, non-inferiority p<

0.0001), and high neurological success rate (93.71 % and 93.01 %, respectively, non-inferiority p<0.0001). There was no difference in the rate of AEs (83.64 % and 82.47 % in the i-Factor and autograft groups, respectively, p=0.8814). Overall success rate consisting of fusion, NDI, neurological success and safety success was higher in i-Factor subjects than in autograft subjects (68.75 % and 56.94 %, respectively, p=0.0382). Improvements in VAS pain and SF-36v2 scores were clinically relevant and similar between the groups. A high proportion of patients reported good or excellent Odom outcomes (81.4 % in both groups). The authors concluded that i-Factor met all 4 FDA mandated non-inferiority success criteria and demonstrated safety and efficacy in single-level ACDF for cervical radiculopathy; i-Factor and autograft groups demonstrated significant post-surgical improvement and high fusion rates.

The authors stated that there were limitations of their study. This study included patients who met detailed inclusion and exclusion criteria and were willing to participate in RCT. Patients in clinical practice may differ from the patients enrolled in this study. Next, some patients were not unavailable at 12 months follow-up. The authors accounted for such data by using pre-specified imputation approaches. They also performed multiple statistical sensitivity analyses to address this issue and had arrived to same conclusions suggesting that missing follow-ups did not bias the results. For practical reasons, surgeons were not blinded to the treatment assignment. However, assessment of fusion and neurological success was performed by independent blinded adjudicators and assessment of functional and quality of life outcomes was self-reported by blinded subjects.

Arnold et al (2018) reported 2-year follow-up of their afore-mentioned 2016 study. Subjects randomly received either autograft (n = 154) or i-Factor (n = 165) in a cortical ring allograft and followed using radiological, clinical, and patient-reported outcomes. At 2 years, the fusion rate was 97.30 % and 94.44 % in i-Factor and autograft subjects, respectively (p = 0.2513), and neurological success rate was 94.87 % (i-Factor) and 93.79 % (autograft; p = 0.7869); NDI improved 28.30 (i-Factor) and 26.95 (autograft; p = 0.1448); VAS arm pain improved 5.43 (i-Factor) and 4.97 (autograft) (p = 0.2763); VAS neck pain improved 4.78 (i-Factor) and 4.41 (autograft; p = 0.1652), Short Form-36 (SF-36v2) Physical Component Score improved 10.23 (i-Factor) and 10.18 (autograft; p = 0.4507), and

SF36v2 Mental Component Score improved 7.88 (i-Factor) and 7.53 (autograft; p = 0.9872). The composite end-point of overall success (fusion, NDI improvement greater than 15, neurological success, and absence of re-operations) was greater in i-Factor subjects compared to autograft subjects (69.83 % and 56.35 %, respectively, p = 0.0302); 12 (7.45 %) i-Factor subjects and 16 (10.53 %) autograft subjects underwent re-operation (p = 0.3411). There were no allergic reactions associated with i-Factor. The authors concluded that use of i-Factor in ACDF was safe and effective, and resulted in similar outcomes compared to local autograft bone at 2 years following surgery.

The authors noted that this study had several drawbacks. First, this trial involved subjects who met detailed inclusion and exclusion criteria and were willing to participate in a RCT; thus, subjects in clinical practice may differ from the subjects enrolled in this study. Second, some subjects were not available at 2-year follow-up. These researchers accounted for such data by using pre-specified imputation approaches. They also performed multiple statistical sensitivity analyses to address this issue and arrived at the conclusion that missing follow-ups did not bias their results. Third, for practical reasons, surgeons were not blinded to the treatment assignment. However, assessment of fusion and neurological success was performed by independent blinded adjudicators, and assessment of functional and quality of life outcomes was self-reported by blinded subjects. Finally, criteria for fusion, determined by the FDA, were less stringent than in some other studies. Nevertheless, the presence of continuous bridging bone was an absolute requirement determining whether fusion had occurred. Fusion assessment was blinded, and the criteria were the same in both arms of the study.

Ortega et al (2020) provided information on characteristics and use of various ceramics in spine fusion and future directions. These researchers stated that in most recent years, focus has been shifted to the use of ceramics in minimally invasive surgeries or implementation of nano-structured surface modification features to promote osteo-inductive properties. Furthermore, effort has been placed on the development of bioactive synthetics. Core characteristic of bioactive synthetics is that they undergo change to simulate a beneficial response within the bone. This change is based on chemical reaction and various chemical elements present in the bioactive ceramics. Recently, a synthetic 15-

amino acid polypeptide bound to an anorganic bone material (ABM) that mimics the cell-binding domain of type-I collagen opened a possibility for osteogenic and osteo-inductive roles of this hybrid graft material. Ceramics have been present in the spine fusion arena for several decades; however, their use has been limited. The major obstacle in published literature is small sample size resulting in low evidence and a potential for bias. In addition, different physical and chemical properties of various ceramics further contribute to the limited evidence. The authors concluded that although ceramics have several disadvantages, they still hold a great promise as a value-based graft material with being easily available, relatively inexpensive, and non-immunogenic. These researchers stated that well-designed prospective studies on new ceramics and hybrids are needed to determine their clinical impact and utility in various spine fusion procedures.

These researchers noted that recently, P-15, a synthetic 15-amino acid polypeptide that mimics the cell-binding domain of type-I collagen, has emerged as a bone graft alternative in the setting of spinal fusion. Serving as an attachment site for osteogenic cells to type I collagen, P-15 also stimulates osteoblastogenesis by promoting the differentiation of bone marrow stromal cells to osteoblasts. Furthermore, it enhances the cellular production and subsequent secretion of various growth factors, cytokines, and bone morphogenetic proteins that signal for further osteoblastogenesis. i-Factor Bone Graft is a bone graft substitute that consists of P-15 bound to an ABM; thus, it contains both osteo-conductive and osteo-inductive properties. To-date, favorable fusion rates with use of P-15 have been demonstrated in a limited number of clinical studies in the setting of cervical fusion (the 2018 study by Arnold et al was discussed). In the setting of lumbar fusion surgery, preliminary animal studies have additionally shown P-15 to be a useful fusion adjunct. While high-quality clinical data is limited, in their series of 110 patients who underwent either single or multi-level ALIF utilizing ABM/P-15, Mobb et al (2014) demonstrated high fusion rates (97.5 %, 81 %, and 100 % in 1-level, 2-level, 3-level fusions, respectively) and favorable patient-reported outcome measures at 2-year follow-up. (The study by Mobb et al 2014 is discussed in CPB 0411 -- The authors stated that limitations of this study were the lack of direct control and the potential for respondent bias. Patients who participated in research were more likely to comply with post-operative medications and physiotherapy, thus resulting in better

health outcomes. For instance, a few patients declined completing the post-operative outcome data because they were discontent with their outcome. These patients were subsequently excluded from statistical analysis, which may have positively skewed results). Ortega et al (2020) concluded that while P-15 may prove to be a useful adjunct in the setting of spinal fusions, additional high-quality and independently conducted clinical studies are needed to further delineate its clinical utility.

Arnold et al (2021) carried out secondary analysis of data from the multi-center, randomized, parallel-controlled FDA IDE study. These investigators stated that studies on outcomes following ACDF in patients with diabetes are scarce. They compared 24-month radiological and clinical outcomes in individuals with and without diabetes undergoing single-level ACDF with either i-FACTOR or local autologous bone. Between 2006 and 2013, a total of 319 individuals with single-level DDD and no previous fusion at the index level underwent ACDF. The presence of diabetes determined the 2 cohorts. Data collected included radiological fusion evaluation, neurological outcomes, NDI, VAS scores, and the SF-36v2 Physical and Mental component summary scores. There were 35 individuals with diabetes (11.1 %; average BMI = 32.99 kg/m2; SD = 5.72) and 284 without (average BMI = 28.32 kg/m2; SD = 5.67). The number of non-diabetic smokers was significantly higher than diabetic smokers: 73 (25.70 %) and 3 (8.57 %), respectively. Pre-operative scores of NDI, VAS arm pain, and SF-36v2 were similar between the diabetic and non-diabetic participants at baseline; however, VAS neck pain differed significantly between the cohorts at baseline ($p = 0.0089$). Maximum improvement for NDI, VAS neck and arm pain, and SF-36v2 PCS and MCS scores was observed at 6 months in both cohorts and remained stable until 24 months. The authors concluded that ACDF was effective for cervical radiculopathy in patients with diabetes; these investigators stated that diabetes is not a contraindication for patients requiring single-level surgery for cervical DDD.

Thaci et al (2021) noted that the safety and effectiveness of traditional ACDF surgery has improved with the introduction of new implants and compounds. Cost-effectiveness of these innovations remains an often-overlooked aspect of this effort. These investigators examined the cost-effectiveness of i-FACTOR compared to autograft for patients undergoing ACDF surgery. They carried out decision analytical modeling using a

Markov model to determine the incremental cost-effectiveness ratio (ICER) of i-FACTOR compared to autograft in ACDF surgery. The patient cohort was extracted from a prospective, multi-center RCT from 22 North American medical centers. Patients randomly received either autograft (n = 154) or i-FACTOR (n = 165). These researchers analyzed various real-world scenarios, including inpatient and outpatient surgical settings as well as private versus public insurances. Two primary outcome measures were assessed: cost and utility. In the base-case analysis, both health and societal system costs were evaluated. Health-related utility outcome was expressed in quality-adjusted life years (QALYs). Cost-effectiveness was expressed as an ICER. In all scenarios, i-FACTOR reduced costs within the 1st year by 1.4 % to 2.1 %. The savings proved to be incremental over time, increasing to 3.7 % over an extrapolated 10 years. The ICER at 90 days was $13,333 per QALY and became negative ("dominated") relative to the control group within 1 year and onwards. In a threshold sensitivity analysis, the cost of i-FACTOR could theoretically be increased 70-fold and still remain cost-effective. The authors concluded that the novel i-FACTOR was not only cost-effective compared to autograft in ACDF surgery, but was the dominant economic strategy.

The authors stated that this analysis had several drawbacks. First, as with other complex statistical approaches, the Markov model is conditional on the present state alone; future and past events are independent. With disease processes, it is rarely plausible to assume that a patient's transition to another health state was entirely independent of a previous health state. The model also assumed that surgical cohorts began in similar health states, which is likely acceptable because of the trial randomization and non-significant differences in baseline characteristics. Second, these investigators recognized that some cost data were not ascertainable. As it is problematic to use hospital charge data to perform a cost-effective analysis, these researchers used Medicare and Humana DRG/APC rates as representations of public and private payers. As a consequence, differences in parameters (such as operating room time and length of stay [LOS]) were not captured. Third, medication-related costs were estimated from the average wholesale price (0.85), as updated Medicare average sales prices are not publicly available. Although this estimate was considered appropriate, it was impossible to determine if it over-estimated or under-estimated costs for

both groups. Fourth, productivity loss was also a significant contributor to cost; however, this analysis was unable to include factors such as transportation costs, caregiver time/responsibilities, as well as educational days missed. Fifth, the authors have disclosed a conflict of interest since the analysis of the IDE RCT data that contributed to this study was funded by industry. The authors felt confident, however, that their analysis, interpretation of the data, and creation of this manuscript was independent of industry, and that this ultimately mitigated bias and provided appropriate equipoise.

Sathe et al (2022) noted that ABM/P-15 is a commercially available synthetically manufactured P-15 collagen peptide fragment, that is adsorbed on anorganic bone matrix. In a retrospective, observational, single-center study, these investigators examined the effectiveness of ABM/P-15 in achieving fusion in the lumbar spine and compared it with that of bone morphogenic protein (rhBMP-2) and demineralized bone matrix (DBM). They collected data of 140 patients who underwent lumbar spinal fusion surgeries in a single specialty spine hospital between 2016 and 2020, with a minimum 6-months follow-up was conducted. Based on the material used for the augmentation of the bone graft at the fusion site, the patients were divided into 3 categories namely ABM/P-15, rhBMP-2, and DBM group. ABM/P-15, rhBMP-2, and DBM were used in 46, 44, and 50 patients, respectively. Patient characteristics like age, gender, bone mineral density (BMD), smoking history, and presence of diabetes mellitus were comparable among the 3 groups. Average follow-up was 11.65 ± 3.57, 18.34 ± 9.87, and 26.2 ± 14.9 months, respectively in ABM/P-15, rhBMP-2, and DBM groups. The fusion was achieved in 97.9 %, 93.2 %, and 98 % patients while the average time-to-union was 4.05 ± 2.01, 10 ± 4.28, and 9.44 ± 3.49 months ($p < 0.001$) for ABM/P-15, rhBMP-2, and DBM groups, respectively. The average pre-operative VAS-score was 6.93 ± 2.42, 7.14 ± 1.97, 7.01 ± 2.14 ($p = 0.900$) for ABM/P-15, rhBMP-2 and DBM groups, respectively, which reduced to 1.02 ± 0.80, 1.21 ± 0.96, 0.54 ± 0.70 ($p = 0.112$), respectively at the last follow up. Pre-operative ODI scores were 52.7 ± 18.02, 55.4 ± 16.8, 53.56 ± 19.6 ($p = 0.751$) in ABM/P-15, rhBMP-2 and DBM groups, which post-operatively reduced to 33.77 ± 15.52, 39.42 ± 16.47, 38.3 ± 15.89 ($p < 0.001$) and further to 15.74 ± 8.3, 17.41 ± 10.45, 16.76 ± 9.81 ($p= 0.002$), respectively at the last follow-up. The authors concluded that ABM/P-15 appeared to achieve union significantly earlier than rhBMP-2

and DBM in lumbar spinal fusion cases while maintaining a comparable clinical and complication profile. This was a retrospective, observational, single-center study, with a relatively small sample size (n = 46 for the ABM/P-15 group); and the follow-up duration for the ABM/P-15 group was only about 12 months. These findings need to be validated by well-designed studies with larger sample size and longer follow-up.

Jacobsen et al (2020) noted that as a consequence of poor bone stock in the elderly, a non-instrumented fusion is often performed in Scandinavia when instability is present. Allograft bone is commonly employed as graft extender with consequent low fusion rates. The use of 15 amino acid residue (ABM/P-15) has shown superior fusion rates in dental and cervical spinal surgery; however, no clinical studies have been carried out in non-instrumented lumbar fusion surgery. In a randomized, double-blind study, these researchers examined PROs and the inter-transverse fusion rate in non-instrumented posterolateral fusion with either ABM/P-15 or allograft. Participants were patients 60 years of age or older with degenerative spondylolisthesis undergoing decompression and non-instrumented posterolateral fusion. Outcome measures included VAS for back and leg pain, ODI, and EuroQoL-5D. A total of 101 patients were enrolled in the study. Subjects were randomized 1:1 to either ABM/P-15 (mixed 50/50, 5cc/level) or allograft bone (30 g/level), both mixed with local bone graft. PROs were collected at baseline and at 12 and 24 months after surgery. Subjects underwent 1-year post-operative fine cut CT-scans (0.9 mm) with reconstructions, independently evaluated by 3 reviewers. Fusion status was concluded by consensus of 2 of the 3 as "fusion" or "no fusion". There were 49 patients available for analysis in both cohorts. The 2 groups were similar in terms of sex distribution, age, and number of levels fused. The fusion rate was significantly higher in the ABM/P-15 group with 50 % fused compared with 20 % in the allograft group. PROs at baseline and at all follow-up time-points were similar between the 2 groups. The authors concluded that patients undergoing non-instrumented posterolateral fusion augmented with ABM/P-15 had a statistically significantly higher fusion rate compared with allograft when evaluated with post-operative fine cut CT-scans (0.9 mm) with reconstructions; however, this did not translate to better clinical outcomes in this 2-year follow-up study. These researchers stated that longer follow-up is needed to examine the impact of ABM/P-15 compared with allograft on patient reported outcomes.

The authors stated that this study had several drawbacks. First, although non-instrumented fusion is standard of care (SOC) in Scandinavia, fusion with instrumentation is more widely used in other countries such as the U.S. and may limit the generalizability of this study. Second, using a non-instrumented fusion created a more challenging environment for fusion compared with instrumented fusion. This may account for the lower rates of fusion compared with previously published reports. Third, other criteria such as demineralized bone or autograft were not included in the study design. The status of the fusion was determined using fine-cut CT scans, which still carry inaccuracies compared with the gold-standard of surgical exploration.

In a commentary on the afore-mentioned study by Jacobsen et al (2020), Demetriades and Shekhar (2020) stated that the reported degree of fusion achieved was significantly lower than previous reports of non-instrumented posterolateral lumbar fusions, which merited further discussion. Moreover, Demetriades and Shekhar (2020) stated that accepting that elderly patients with grade I spondylolisthesis needing a decompression for stenosis randomized control trial data showed that instrumentation might not be needed. Furthermore, the findings from the study by Jacobsen et la (2020) showed that a 50 % fusion (with ABM/P-15 + local bone graft) did not result in better PROMs than a 20 % fusion (with cadaveric allograft + local bone graft), which raised some additional questions:

- Is achieving 50 % fusion good enough?
- What level of minimum fusion is clinically relevant (versus failed fusion)?
- Is this a cost-effective fusion?

Demetriades and Shekhar (2020) stated that they hoped that commercial marketing does not promote one graft over the other in the absence of clinical benefit.

Arnold et al (2023) noted that previous analyses of the FDA IDE trial showed the superiority of i-FACTOR compared with local autograft bone in single-level ACDF surgery at 12 and 24 months post-operatively in a composite endpoint of overall success. These investigators reported the final, 6-year clinical and radiological outcomes of the FDA post-approval

study. Of the original 319 subjects enrolled in the IDE study, 220 participated in the post-approval study (n = 106 for i-FACTOR, and n = 114 for control). The study met statistical non-inferiority success for all 4 coprimary endpoints. Radiographic fusion was achieved in 99 % (103/104) and 98.2 % (109/111) in i-FACTOR and local autograft subjects, mean NDI improvement from baseline was 28.6 (24.8 to 32.3) in the i-FACTOR and 29.2 (25.6 to 32.9) in the control group, respectively (non-inferiority $p < 0.0001$). The neurological success rate at 6 years was 95.9 % (70/73) in i-FACTOR subjects and 93.7 % (70/75) in local autograft subjects (non-inferiority $p < 0.0001$). Safety outcomes were similar between the 2 groups. Secondary surgery on the same or different cervical levels occurred in 20/106 (18.9 %) i-FACTOR subjects and 23/114 (20.2 %) local autograft subjects ($p = 0.866$). Secondary outcomes (pain, SF-36 physical component score and mental component score) in i-FACTOR subjects were similar to those in local autograft subjects. The authors concluded that i-FACTOR met all 4 FDA-mandated non-inferiority success criteria and showed safety and effectiveness in single-level ACDF for cervical radiculopathy through 6 years post-operatively. Safety outcomes were acceptable, and the clinical and functional outcomes observed at 12 and 24 months remained at 72 months.

The authors stated that this study had several drawbacks. First, this trial involved subjects who met the detailed inclusion and exclusion criteria and were willing to participate in a RCT. Patients in clinical practice may differ from the patients enrolled in this study. This is a common drawback of randomized evaluations. Second, some sites did not participate in the PAS. Randomization was stratified by site, and the randomized nature of the study was therefore maintained. Third, some participants were not available at 6-year follow-up. The 6-year follow-up rate approached 80 % in both groups, and missing data were accounted for using pre-specified imputation approaches. Fourth, surgeons were unable to be blinded to the assigned treatment. Nevertheless, assessment of fusion and neurological success was carried out by independent blinded adjudicators, and the assessment of pain, functional, and QOL outcomes was self-reported by blinded subjects. Fifth, criteria for fusion, determined by the FDA, were less stringent than in some other studies. Criteria for fusion used in this study were common at the time the study was initiated but differ from current criteria enacted by the Cervical Spine

Research Society. Nevertheless, the presence of continuous bridging bone was an absolute requirement determining whether fusion had occurred. Sixth, it was possible these researchers under-estimated fusion rates in both groups because many patients with pseudoarthrosis are asymptomatic.

Khan and Shahzad (2023) stated that autologous bone grafts, sourced from the iliac crest, are the gold standard for bone substitution in spine surgery; however, harvesting autografts increases the risk of post-operative complications. Bone allografts are another popular source of graft material; however, their use is rapidly surpassing their availability. There has been considerable interest in manufactured bone graft substitutes, commonly referred to as osteo-biologics that mimic the properties of autologous bone, and may be osteo-conductive, osteo-inductive, osteogenic, or a combination. Osteo-biologics have been developed to mimic the properties of autologous bone; however, their high cost and variable effectiveness raise questions regarding their value. These researchers examined the challenges and opportunities associated with the use of osteo-biologics used to aid in bone healing in spinal fusion surgery within a value-based care framework. Spinal fusion treatments such as MPs, PRP, autologous conditioned serum, DBM, biomaterial scaffolds, stem cells, and cellular bone matrices were compared.

The authors noted that BPMs are highly effective but often associated with serious risks; PRP shows promising results but lacks standardization in research protocols. Autologous conditioned serum is inconclusive and cost-effective, while DBM has variable effectiveness and limited data to use in anterior spinal fusions. Biomaterial scaffolds have limited application in the anterior spine but show high effectiveness when it comes to spinal fusion. Stem cells show improved post-surgical outcomes but have low yield from bone marrow, and potential risks associated with genetic engineering and cell therapy. Cellular bone matrices show promising results and have high fusion rates, yet there is currently no FDA requirement for pre-clinical or clinical data before commercial usage. These researchers stated that although osteo-biologics have considerable potential, their high price and uncertain efficiency raise questions regarding their usefulness in spinal fusion

surgery. These investigators stated that to ensure better patient outcomes, extensive research is needed to examine their use within a value-based care framework.

In a systematic review, Hasan et al (2024) examined the effectiveness and overall outcomes of i-Factor/ABM/P-15 following lumbar spine surgery. These investigators carried out a search of the Cochrane Library, Medline Ovid, PubMed, and SCOPUS databases from inception until August 2023. Eligible studies included outcomes of patients receiving I Factor following lumbar spine surgery. The primary outcomes of interest were fusion rates, and i-Factor effectiveness following lumbar surgery in patients who received i-Factor. Secondary outcomes included patient-reported outcomes and complication rates. A total of 766 titles were initially screened. After inclusion criteria were applied, 5 studies (388 patients) were included, which measured overall outcomes of i-Factor/ABM/P-15 following lumbar spine surgery. These studies showed acceptable reliability for inclusion based on the Methodical Index for Non-Randomized studies and Critical Appraisal Skills Program assessment tools. I-Factor/ABM/P-15 facilitated significantly faster bone development in various procedures while maintaining favorable clinical outcomes compared to traditional grafts. The authors concluded that this systematic review found that i-Factor/ABM/P-15 use for lumbar spine surgery maintained similar managing patient-reported outcomes relative to other grafting methods. In regard to rates of fusion, i-Factor/ABM/P-15 showed a significantly faster rate of fusion when compared to traditional grafts including allograft, autograft, DBM, and recombinant human BMP-2 (rhBMP-2). Moreover, these researchers stated that future financially-independent, multi-center RCTs with larger sample sizes are recommended to further examine i-Factor/ABM/P-15 effectiveness in lumbar spine surgery.

The authors noted that a significant drawback of this systematic review was the restriction to English-only publications, which potentially omitted valuable research carried out in other languages and regions. The heterogeneity observed across the included studies further complicated the analysis. This heterogeneity spanned different surgical procedures and varied outcome measurements, making direct comparisons and meta-analyses challenging. Financial disclosures presented another layer of complexity. Notably, 4 out of the 5 studies included in this review

were sponsored, either directly or indirectly, by Cerapedics, the manufacturers of i-Factor. Such financial ties raise concerns regarding potential biases in the study outcomes and interpretations. Specifically, the study by Mobbs et al (2014) scored low on the MINORS bias criteria, indicating potential methodological shortcomings that could influence the study's conclusions. In addition, the low sample sizes observed in some of the studies limited the power of the findings. Such small cohorts could result in over-estimations or under-estimations of the true effect sizes. The predominance of single-center studies in this review further narrowed the generalizability of the results, as findings from a single center might not be representative of broader clinical practices. Furthermore, i-Factor grafting methods were also variable among studies, posing another drawback. Some studies used i-Factor independently while others combined it with either allograft or autograft. It was also worth noting that studies failed to comment on whether the volume of the local allograft used was sufficient in both single-level and multi-level cases.

Gordon et al (2024) stated that despite successful fusion rates with ICBG, donor-site morbidity and increased operating time remain a considerable limitation and drive the search for alternatives. In a systematic review, these investigators compared grafts with additional cellular supplementation with ICBG for spinal arthrodesis. They examined the safety, effectiveness, and long-term outcomes; thereby, providing the current and relevant evidence for orthopedic surgeons to make informed choices regarding this rapidly developing field. Two independent reviewers carried out an electronic literature search according to the PRISMA guidelines looking for articles published up to March 1, 2023 using PubMed, Embase, and the Cochrane Central Register of Controlled Trial. Cellular allografts were not included. The following data were extracted: Number of patients, type of graft, fusion assessment method, follow-up duration, fusion rates, clinical outcomes, and complications. The methodological quality of evidence (MQOE) was assessed using the Risk of Bias 2 (RoB-2) tool and Risk of Bias In Non-Randomised Studies (ROBINS) tool developed by Cochrane for evaluating bias in randomised and non-randomised studies. A total of 10 studies met the inclusion criteria, including 465 patients. The mean number of patients per study was 43.8 (S.D. 28.81, range of 12 to 100); 2 studies reported cell-based therapy to be significantly more successful in terms of fusion rates compared to ICBG. However, the remaining 8 showed equivocal results.

No study found that cell-based therapy was inferior. No difference was observed between the 2 groups in 3 studies that focused on degenerative cohorts. No difference in functional outcome scores was observed between the groups. A number of different preparation techniques for cell-based grafts were used throughout the studies. The authors concluded that cell-based therapy offers a promising alternative to ICBG in spinal fusion surgery, which could help reduce the associated morbidity to patients. This review found that cell-based therapy is non-inferior to ICBG and may offer patients an alternative therapeutic option with fewer complications and reduced post-operative pain. However, the literature to-date is limited by heterogeneity of the cell preparation and grafting process. These researchers stated that further investigations with a unified approach to the cell preparation process are needed to fully delineate the potential advantages of this technology.

Actifuse

In a prospective, phase-IV, multi-center, randomized study, Coughlan and colleagues (2018) compared vertebral fusion success rates following postero-lateral fusion [(PLF)/posterolateral inter-transverse fusion (PITF)] surgery. The surgical procedure combined posterior lumbar interbody fusion (PLIF) and PLF with internal fixation over 1 or 2 levels using silicated calcium phosphate (SiCaP; n = 51) or BMP-2 (n = 52) as graft material in patients with a degenerative disorder of the lumbar spine. Following randomization, the surgical procedure consisting of PLIF and PLF with internal fixation over 1 or 2 levels was performed using SiCaP or BMP-2. No other osteoconductive/osteoinductive graft materials were permitted. Spinal fusion was assessed radiographically at less than or equal to 24 months. Clinical outcomes (pain on VAS, ODI, SF-36) and AEs were monitored. A total of 103 patients were enrolled. At 12 months, fusion was achieved in 25 of 35 (71.4 %) of the SiCaP and 20 of 27 (74.1 %) of the BMP-2 group, respectively (p = 1.000). At 24 months, the fusion rate was 78.6 % and 84.8 % for SiCaP and BMP-2, respectively (p = 0.5613). Clinical outcomes improved similarly in both groups over time; AEs were consistent with this surgical population. The authors concluded that on the basis of historical control data, SiCaP may be as useful as iliac crest bone autograft in the context of spine fusion surgery, with less

risk of unwanted donor site morbidity associated with that commonly performed procedure. This study provided only short-term follow-up data (24 months).

This was a 2-arm, pilot, proof-of-concept study aimed at obtaining estimates of successful fusion from both SiCaP and BMP-2. The authors stated that this study had several drawbacks. It was under-powered due to the number of patients who could not be included in the per protocol population. This was in part due to the numbers of protocol deviations, mostly arising from the omission of bulking agent to the BMP-2 material. The success of lumbar fusion and magnitude of clinical improvement was also dependent on patient baseline characteristics. A recent study compared the clinical outcomes following interbody fusion surgery and demonstrated that by stratifying patients into severity of initial diagnosis, the results could be correlated with success of clinical outcomes. This should be considered when drawing conclusions between treatment groups, and furthermore, the results in this study should be treated with caution when making comparisons with other fusion techniques.

Osteoset Resorbable Mini-Bead (Calcium Sulfate Antibiotic Beads)

Humm and colleagues (2014) noted that surgical debridement and prolonged systemic antibiotic therapy are an established management strategy for infection after tibial fractures. Local antibiotic delivery via cement beads has shown improved outcome but requires further surgery for extraction of beads. Osteoset-T is a resorbable bone void filler composed of calcium sulphate and 4 % tobramycin that is packed easily into bone defects. This was a review of the outcomes of 21 patients treated with Osteoset-T for osteomyelitis of the tibia. This was a retrospective case note and clinical review. In all cases, the strategy was debridement, with removal of any implants, with excision back to bleeding bone. Osteoset-T pellets were packed into any contained defects or the intra-medullary canal with further bony stabilization (n = 9) and soft tissue reconstruction (n = 7) undertaken as needed. Intravenous vancomycin and meropenem were administered after sampling with substitution to targeted antibiotic therapy for between 6 weeks and 6 months; average follow-up was 15 months. Union rate after tibial reconstruction was 100 %. Wound complications were encountered in 52 %: a wound discharge in the early post-operative period was noted in 7 patients (33 %)

independent of site of pellet placement. In the 14 cases without a wound leak, 5 developed wound complications (p = 0.06, Fisher's exact test) either from delayed wound-healing or pin-site infections; 1 patient developed a transient acute kidney injury (AKI) and 1 refractory osteomyelitis. The authors concluded that Osteoset-T was an effective adjunct in the treatment of chronic tibial osteomyelitis following trauma based on the low incidence of relapse of infection within the period of follow-up in this study, but significant wound complications and 1 transient nephrotoxic event were also recorded. These researchers stated that further prospective and controlled studies are needed to evaluate the role of local antibiotic delivery systems in the treatment of chronic osteomyelitis.

The authors stated that this case series was limited by the following: the lack of a control group, small numbers (n = 21) and the minimum follow-up period of 6 months. These investigators also noted that a retrospective case-control study, which contained a mixed group of long-bone infections, found favor in the use of Osteoset-T; however, prospective data on Osteoset-T were limited, with published work concentrating on post-traumatic osteomyelitis in long bones but without comparison with controls.

Trujillo and associates (2017) noted that surgical site infections (SSIs) account for about 17 % of all nosocomial infections, second only to urinary tract infections (UTIs). Antibiotic beads deliver high local antibiotic concentrations and maintain low systemic levels. These researchers evaluated the efficacy of calcium sulfate absorbable antibiotic beads (CSAAB) in the prevention of SSIs for complex wound closures. Patient records from the University of New Mexico Hospital (UNMH; Albuquerque, NM) and Dartmouth-Hitchcock Medical Center (DHMC; Lebanon, NH) were retrospectively analyzed from 2004 to 2015. Each patient received CSAAB prophylaxis during operations performed by the principle investigator. Charts were grouped by wound location and category. Outcomes were defined solely by re-admission within 30 days for repeat intervention; 0 of the 38 UNMH; and 15 of the 104 DHMC patients were re-admitted. Data reached statistical significance based on 95 % CIs using the binomial distribution. The authors concluded that the findings of this brief retrospective chart review showed promising use for CSAAB in the prevention of soft tissue SSIs. These investigators stated

that these data may be useful for promoting further research and establishing protocols for their utilization in the management of colonized complex wounds in reconstructive surgery.

Furthermore, the authors stated that currently, antibiotic beads are most commonly prescribed for orthopedic applications, such as in the treatment or prevention of osteomyelitis. Also, PMMA beads have been shown to provide adequate wound sterilization and graft preservation in the management of extra-cavitary prosthetic vascular graft infections. Moreover, infected breast implants as well as left ventricular assist devices have been salvaged with antibiotic-impregnated PMMA. These studies, in addition to the present results, high-lighted the importance of conducting a controlled prospective study to further assess their novel use in decreasing infection rates and bioburden of complex surgical wounds.

Menon and co-workers (2018) stated that the use of antibiotic impregnated biodegradable synthetic high purity calcium sulfate (SHPCS) beads is frequently reported as they offer increased concentration of antibiotics locally, without need for removal. However some wound discharge following their use has been noted. In a retrospective study, these researchers determined any correlation between wound discharge and infection remission. This trial entailed 39 cases of osteo-articular infections from April 2013 to November 2016 in whom SHPCS beads were used. All patients underwent the standard staged protocol of aggressive debridement, deep tissue biopsy, implant removal where indicated and early soft tissue cover; SHPCS beads were used locally in the 2nd stage combined with appropriate antibiotics based on tissue culture. All patients received systemic antibiotics for a period of 6 weeks and followed-up for a minimum period of 6 months. The study analyzed the patient demographics, etiology, surgical procedures, culture patterns, local antibiotics used, radiological status of beads, incidence and characteristics of wound discharge and outcome. There were 25 cases of chronic osteomyelitis, 8 infected non unions, 3 peri-prosthetic joint infections, 2 soft tissue infections, and 1 case of acute osteomyelitis; 17 of these infections were following osteo-synthesis. The cultures were negative on 8 occasions in 7 patients. A total of 40 organisms were isolated in the other patients; commonest being Staphylococcus aureus (n = 16) and E coli (n = 7); SHPCS beads were mixed with vancomycin in

17 cases, colistin in 11, vancomycin with colistin in 8 and vancomycin with gentamicin in 4. Voriconazole was used in 1 case with fungal infection; 8 cases (20.51 %) developed discharge from the wound at an average of 6 days after inserting the beads. The discharge was serous with no foul smell in 6 and purulent in 2 inflamed wounds; 4 cases underwent re-debridement; 2 cases with purulent discharge and subsequent positive cultures; 2 with serous discharge early in the series and no evidence of infection on re-exploration with negative cultures. The remaining 4 patients with serous wound discharge were observed without any further surgical intervention, with the discharge stopping spontaneously between 15 to 36 days post-operatively. There was no correlation between antibiotic used and wound discharge. Radiographic analysis showed dissolution of all the beads at an average of 36 days in the 39 cases. Heterotrophic ossification was not observed. Clinical and radiological remission of infection was observed in 37 cases (94.9 %); 2 patients died during the course of hospitalization, secondary to septicemia and multi organ failure; 3 patients had an infection recurrence within 6 months, managed successfully by re-debridement and appropriate antibiotics. Radiological union was achieved in 7 of the 8 infected non unions. The authors concluded that with the encouraging rates of infection remission observed, they continued to use antibiotic loaded SHPCS as an alternative for local antibiotic delivery in the treatment of osteo-articular infections. However, wound discharge is a known potential observation following implantation of calcium sulfate beads, subsiding typically within 4 to 6 weeks. The appearance of wound discharge could vary, ranging from purulent discharges to non-purulent, serous/sero-sanguineous fluid wound discharges. The presence of a wound discharge alone did not necessarily imply a failure to treat the infection. It was important to be aware of this side effect and guard against unnecessary re-operations, by careful consideration and monitoring all of the available clinical signs of infection, in addition to blood test results and radiographic evidence. These researchers stated that further research is needed to determine the relationship between the implantation of antibiotic loaded calcium sulfates and the incidence and duration of drainage.

An UpToDate review on “Overview of control measures for prevention of surgical site infection in adults” (Anderson and Sexton, 2019) does not mention calcium sulfate antibiotic beads as a management tool.

On the other hand, an UpToDate review on “Osteomyelitis in adults: Treatment” (Osmon and Tande, 2019) states that “Surgical debridement also allows placement of local antimicrobials, either applied directly to the site of infection or mixed with absorbable (e.g., calcium sulfate) or non-absorbable (e.g., polymethyl methacrylate cement) carriers … Local antimicrobials mixed with absorbable carriers (e.g., calcium sulfate beads) or non-absorbable carriers (e.g., polymethyl methacrylate beads or cement) placed in and around the fracture site may be useful as an adjunct to systemic antibiotics for prevention or treatment of infection. Data are most robust for polymethylmethacrylate beads; these are typically removed within 2 to 4 weeks. Antibiotic-impregnated calcium sulfate beads are absorbable; they have been used as a bone substitute and delivery system for antibiotics repairing bony defects in open fractures caused by combat-related injuries”.

Genetically Corrected Autologous Epidermal Grafts in Patients With Recessive Dystrophic Epidermolysis Bullosa

Siprashvili and colleagues (2016) stated that recessive dystrophic epidermolysis bullosa (RDEB) is a devastating, often fatal, inherited blistering disorder caused by mutations in the COL7A1 gene encoding type VII collagen. Support and palliation are the only current therapies. In a single-center, phase I clinical trial, these researchers evaluated the safety and wound outcomes following genetically corrected autologous epidermal grafts in patients with RDEB. This study was conducted in 4 patients with severe RDEB with a measured area of wounds suitable for grafting of at least 100 cm2. Patients with undetectable type VII collagen keratinocyte expression were excluded. Autologous keratinocytes isolated from biopsy samples collected from 4 patients with RDEB were transduced with good manufacturing practice-grade retrovirus carrying full-length human COL7A1 and assembled into epidermal sheet grafts. Type VII collagen gene-corrected grafts (approximately 35 cm2) were transplanted onto 6 wounds in each of the patients (n = 24 grafts). The primary safety outcomes were recombination competent retrovirus, cancer, and autoimmune reaction. Molecular correction was assessed as type VII collagen expression measured by immunofluorescence and immuno-electron microscopy. Wound healing was assessed using serial photographs taken at 3, 6, and 12 months after grafting. The 4 patients (mean age of 23 years [range of 18 to 32]) were all male with an

estimated body surface area affected with RDEB of 4 % to 30 %. All 24 grafts were well-tolerated without serious AEs. Type VII collagen expression at the dermal-epidermal junction was demonstrated on the graft sites by immunofluorescence microscopy in 9 of 10 biopsy samples (90 %) at 3 months, in 8 of 12 samples (66 %) at 6 months, and in 5 of 12 samples (42 %) at 12 months, including correct type VII collagen localization to anchoring fibrils. Wounds with recombinant type VII collagen graft sites displayed 75 % or greater healing at 3 months (21 intact graft sites of 24 wound sites; 87 %), 6 months (16/24; 67 %), and 12 months (12/24; 50 %) compared with baseline wound sites. The authors concluded that in this preliminary study of 4 patients with RDEB, there was wound healing in some type VII collagen gene-corrected grafts, but the response was variable among patients and among grafted sites and generally declined over 1 year. They stated that long-term follow-up is needed for these patients, and controlled trials are needed with a broader range of patients to better understand the potential long-term effectiveness of genetically corrected autologous epidermal grafts.

Bioglass (45S5)

Li and colleagues (2017) stated that 45S5 Bioglass (45S5) is one of the most widely used biomaterials in ceramic-based bone graft substitutes by virtue of its excellent biocompatibility and bioactivity. However, the fracture toughness and wear resistance of 45S5 have to be improved to extend its applications in load bearing orthopedic implants. These investigators reported the first use of graphene nano-platelet (GNP) to enhance the fracture toughness and wear resistance of 45S5. Composite powders with 4 different loadings of graphene oxide (GO), i.e., 0, 0.1, 0.5 and 1 wt %, were sintered by spark plasma sintering (SPS) at a relatively low temperature of 550° C, during which in-situ thermal reduction of GO took place. It was found that by adding 0.5 wt % GO to the 45S5 powder, the fracture toughness of the sintered pellets was increased by 130.2 % while friction coefficient and specific wear rate were decreased by 21.3 % and 62.0 %, respectively. Furthermore, the viability of MG63 cells grown on the GNP-incorporated pellets was comparably high to that of the cells grown on the pure 45S5 pellets. As compared with the pure 45S5 leachates, the media conditioned by the GNP/45S5 pellets fabricated from the composite powder with 1 wt % GO could enhance both the proliferation and viability of MG63 cells. The authors concluded that it is

thus envisioned that the GNP-reinforced 45S5 is a highly promising material for fabricating mechanically strong and biocompatible load-bearing bone implants.

OsteoVive

OsteoVive is a next generation viable cell allograft combining osteogenic cells with an osteoconductive, osteoinductive scaffold. OsteoVive processing preserves a cell population that includes marrow-isolated adult multilineage-inducible (MIAMI) cells.

Synthetic Bone Graft for Spinal Fusion

Buser and colleagues (2016) compared the safety and effectiveness of synthetic bone graft substitutes versus autograft or allograft for the treatment of lumbar and cervical spinal degenerative diseases. Multiple major medical reference databases were searched for studies that evaluated spinal fusion using synthetic bone graft substitutes (either alone or with an autograft or allograft) compared with autograft and allograft; RCTs and cohort studies with more than 10 patients were included. Radiographic fusion, patient-reported outcomes, and functional outcomes were the primary outcomes of interest. The search yielded 214 citations with 27 studies that met the inclusion criteria. For the patients with lumbar spinal degenerative disease, data from 19 comparative studies were included: 3 RCTs, 12 prospective, and 4 retrospective studies. Hydroxyapatite (HA), HA+collagen, β-tricalcium phosphate (β-TCP), calcium sulfate, or polymethylmethacrylate (PMMA) were used. Overall, there were no differences between the treatment groups in terms of fusion, functional outcomes, or complications, except in 1 study that found higher rates of HA graft absorption. For the patients with cervical degenerative conditions, data from 8 comparative studies were included: 4 RCTs and 4 cohort studies (1 prospective and 3 retrospective studies). Synthetic grafts included HA, β-TCP/HA, PMMA, and biocompatible osteoconductive polymer (BOP). The PMMA and BOP grafts led to lower fusion rates, and PMMA, HA, and BOP had greater risks of graft fragmentation, settling, and instrumentation problems compared with iliac crest bone graft. The authors concluded that the overall quality of evidence evaluating the potential use and superiority of the synthetic biological materials for lumbar and cervical fusion in this systematic

review was low or insufficient, largely due to the high potential for bias and small sample sizes. Thus, definitive conclusions or recommendations regarding the use of these synthetic materials should be made cautiously and within the context of the limitations of the evidence.

Kadam and associates (2016) reviewed clinically relevant bone substitutes and bone expanders for spinal surgery in terms of effectiveness and associated clinical outcomes, as reported in contemporary spine literature. A Medline search of English language literature published through March 2016 discussing bone graft substitutes and fusion extenders was performed. All clinical studies reporting radiological and/or patient outcomes following the use of bone substitutes were reviewed under the broad categories of allografts, DBM, ceramics, BMPs, autologous growth factors (AGFs), stem cell products and synthetic peptides. These were further grouped depending on their application in lumbar and cervical spine surgeries, deformity correction or other miscellaneous procedures, namely, trauma, infection or tumors; wherever data were forthcoming. Studies in animal populations and experimental in-vitro studies were excluded. Primary end-points were radiological fusion rates and successful clinical outcomes. A total of 181 clinical studies were found suitable to be included in the review. More than 1/3 of the published articles (62 studies, 34.25 %) focused on BMP. Ceramics (40 studies) and allografts (39 studies) were the other 2 highly published groups of bone substitutes. Highest radiographic fusion rates were observed with BMPs, followed by allograft and DBM. There were no significant differences in the reported clinical outcomes across all classes of bone substitutes. The authors concluded that there is a clear publication bias in the literature, mostly favoring BMP. Based on the available data, BMP is however associated with the highest radiographic fusion rate. Allograft is also very well corroborated in the literature. The use of DBM as a bone expander to augment autograft is supported, especially in the lumbar spine. Ceramics are also utilized as bone graft extenders and results are generally supportive, although limited. The use of autologous growth factors is not substantiated at this time. Cell matrix or stem cell-based products and the synthetic peptides have inadequate data. They stated that more comparative studies are needed to evaluate the effectiveness of bone graft substitutes overall.

Magnus and colleagues (2018) noted that hind-foot arthrodesis is a frequently performed procedure by foot and ankle surgeons. The relatively high nonunion rate associated with these procedures has led surgeons to use adjunctive bone graft to help augment osseous union. Cellular bone allografts (CBAs) are a specific type of graft that incorporates osteo-conductive, osteo-inductive, and osteogenic properties while also eliminating the common disadvantages of autografts and traditional allografts. These investigators discussed the role of CBAs in hind-foot arthrodesis procedures, a review of current literature, and a comparison of available products. ViviGen (DePuy Synthes) was developed with only osteogenic cells (osteoblasts, osteocytes, bone lining cells). In contrast with undifferentiated MSCs, bone cells are fully differentiated and are committed to the formation of osseous tissue. Thus, ViviGen avoids the erroneous production of cartilage and muscle by providing lineage committed bone cells. These researchers stated that perhaps the most critical concern when deciding between products is the number of active cells available prior to and following the preparation process. Although several companies emphasize vast quantities of MSCs and other osteogenic cells, only a select few products actually disclose the number of surviving cells after the thawing and preparation processes. Unfortunately, the current literature lacks an accepted standard number of MSCs needed to promote bony healing in the setting of arthrodesis, adding another layer of complexity to the issue. These investigators stated that future studies are needed to provide CBAs guidelines regarding product contents, which will allow transparency between the manufactures and surgeons. The authors concluded that although this is a fast developing field, there is a need for further research regarding the use and efficacy of these products; prospective RCTs are needed, as well as studies that compare successful fusion rates between CBAs and traditional grafting techniques. However, the available literature showed promise with the use of CBAs to enhance bony healing. For the time being, current studies demonstrated that CBAs appeared to be safe and effective in hind-foot and ankle arthrodesis procedures.

Hall et al (2019) noted that low back pain (LBP) is the leading cause of absence from work, disability, and impaired quality of life (QOL). Fusion surgery may be indicated when non-operative treatments have failed to

provide relief. Surgery may include the use of fusion-enhancing implants, such as cellular bone allografts (CBAs). In a retrospective study, these researchers examined the safety and efficacy of one CBA (ViviGen cellular bone allograft [V-CBA]) in patients who underwent instrumented posterolateral fusion (IPLF). Retrospective data were collected from 150 consecutive patients who had undergone IPLF surgery between January 1, 2015, and March 31, 2018, in which V-CBA was used. All surgeries were performed by 1 surgeon. V-CBA was mixed with local autograft bone. Patient diagnoses included degenerative disc disease (DDD), spondylosis, spondylolisthesis, or spondylolysis with or without stenosis. Standing antero-posterior (AP) and lateral images were collected prior to surgery and again at the terminal visit, which took place between 6 and 33 months post-operatively. De-identified images were assessed radiologically. Adverse events (AEs) were documented. The primary composite end-point of fusion status was dependent upon 2 main criteria: bridging bone per the Lenke scale (classified as "A" definitely solid or "B" possibly solid) and posterior hardware status (intact). Lenke scale C or D were categorized as pseudarthrosis. A total of 87 male and 63 female patients (613 levels total) underwent IPLF in which V-CBA was implanted. An average of 4.1 levels was treated, with 59.3 % of patients having undergone treatment for more than 3 levels; 29 % of patients had diabetes; 52 % of patients had previously used nicotine products, and 12 % were current smokers. A total of 16 serious AEs were recorded and included lumbar seroma, cerebrospinal fluid (CSF) leak, wound dehiscence, pneumonia, urinary tract infection, and myocardial infarction. Successful fusion (Lenke scale "A" or "B") was recorded in 148 out of 150 patients (98.7 %), or 608 out of 613 levels. The total pseudarthrosis rate was 0.8 %. The authors concluded that the use of V-CBA combined with local autograft in multi-level IPLF resulted in successful fusions in 98.7 % of patients. The researchers stated that these findings were particularly robust given the complex nature of many of these cases: 89 patients had 4 or more surgical levels, and many patients had multiple co-morbidities. Moreover, these investigators stated that future studies should examine the relationship between fusion status and the ratio of V-CBA to the local bone. Level of Evidence = IV.

The authors stated that this study had several drawbacks. The data were collected from 1 hospital and represent the experience of 1 surgeon. The results may not be applicable to other study centers. The data were

collected retrospectively, without a comparative control or competitive arm. The retrospective nature of the study did allow for calculation of fusion rates across multi-level procedures from radiographs using a standardized classification, listing of complications, and basic patient demographics with a medical history. Additionally, VAS pain scores were not charted prior to surgery and fusion was calculated at the last visit with the surgeon instead of at the industry standard of 12 or 24 months. Another limitation was the inability to differentiate the relative contributing factors of V-CBA and local bone since a combination was used in these surgeries. An average of 8.7 cc of V-CBA was used with 32.4 g of local bone across all subject, with these values varying according to the number of surgical levels, quantity and quality of local bone, and patient co-morbidities, especially diabetes and tobacco use. However, insight can still be gained comparing the results presented here to the studies mentioned above, which suggested a much lower rate of fusion when only the local bone was used.

Shahrdar and colleagues (2020) stated that although use of cellular bone allografts (CBA) in orthopedic surgery has become increasingly common, little information is available regarding their short-term clinical performance. In these 2 case reports of 2-stage hip arthroplasties, ViviGen Formable CBA (V-CBA) was used in stage-I to fill voids left by previous metal implants. The 2 patients had distinctly different health profiles, but each of them had previous metal implants due to a hip fracture. In the otherwise healthy 49-year old man, total hip arthroplasty (THA) was performed 7 weeks after nail removal and V-CBA backfill. In the 64-year old woman with type 1 diabetes mellitus and severe osteoporosis, stage-II was performed after 12 weeks. At the time of THA for each patient, bone containing some V-CBA was removed to accommodate the hip implant. The explants were histologically analyzed for bone matrix, mineralization, and neo-vascularization. Histological staining showed substantial new bone formation and neo-vascularization in both explants albeit at different levels of maturity. The authors concluded that although limited, these results suggested that V-CBA may facilitate new bone formation in healthy as well as in metabolically challenged patients. This was the first publication showing the efficacy of V-CBA cellular bone allograft via histological results in human subjects; these preliminary findings need to be validated by well-designed studies. Level of Evidence = V.

Bone Micro-Indentation Testing

Microindentation has the potential to measure the stiffness of an individual patient's bone (Arnold, et al., 2017). Bone stiffness plays a crucial role in the press-fit stability of orthopaedic implants. Arming surgeons with accurate bone stiffness information may reduce surgical complications including periprosthetic fractures.

Coutts and associates (2015) stated that reference point indentation (RPI) has been proposed as a new clinical tool to aid the diagnosis of osteoporosis. These researchers examined the performance of the tool within entire femurs to improve the understanding of the mechanical properties of bone and to guide future RPI testing to optimize repeatability of results obtained using the technique. Human, bovine, porcine and rat femurs were indented along 3 longitudinal axes: anterior and posterior: medial and lateral, as well as around the circumference of the femoral head and neck. Cortical and subchondral bone thickness was measured using CT and radiography. The study showed that in some samples, bone was too thin to support the high loads applied with the technique and in these cases, RPI values were highly influenced by thickness. The technique would be useful in the mid-shaft region where cortical thickness was greatest, providing previously established guidelines were followed to optimize measurement repeatability, including performing multiple measurements per sample and investigating multiple samples. The study has also provided evidence that RPI values varied significantly with test site, hence mechanical properties should not be inferred from RPI findings alone away from the test site, even within the same bone. The authors concluded that RPI appeared to be a useful tool for scientific investigation; however further work is needed to examine the feasibility of using RPI for assessing differences between healthy and diseased bone in a clinical setting.

Hunt and Donnelly (2016) In a stated that in a micro-indentation test, the sample is subjected to a static load, and the resulting geometry of the indentation impression after load removal is used to calculate hardness. Hardness is defined as the force applied divided by the area of the residual indentation, which characterizes the material's resistance to plastic deformation. Micro-indentation allows for hardness testing at the length scales of individual trabeculae and osteons of bone. The sampling

volume includes micro-porosity associated with lacunae. Within the past 10 years, a new type of indentation testing known as RPI was developed; it measures bone material properties on a scale similar to that of micro-indentation ex-vivo. There are 2 main RPI devices, the OsteoProbe (Active Life Scientific, Santa Barbara) and the BioDent (Active Life Scientific, Santa Barbara), which operate under different mechanical loading conditions resulting in dissimilar outcome measurements. Empirical evidence for the clinical utility of these techniques is bolstered by several studies using both the OsteoProbe and the BioDent. Studies using the OsteoProbe showed differences in bone material strength index (BMSi) between patient populations, including a decrease in BMSi in type 2 diabetic patients compared to non-diabetic patients, as well as an increase in BMSi of glucocorticoid-treated patients compared to non-treated patients. Using the BioDent, the maximum indentation distance and indentation distance increase between the 1st and last cycle were higher in femoral fracture patients than in non-fracture controls, suggesting the BioDent can distinguish a clinical fracture population. Key advantages of these techniques include the ability, for the first time, to directly measure mechanical properties of bone tissue in-vivo. However, the outcome variables await rigorous validation, and standardization of RPI techniques between studies is lacking, complicating comparisons of RPI data across studies.

Herrera and Diez-Perez (2017) noted that densitometry and imaging techniques are currently used in clinical settings to measure bone quantity and spatial structure. Recently, RPI has opened the possibility of directly assessing the mechanical characteristics of cortical bone in living individuals, adding a new dimension to the assessment of bone strength. Impact micro-indentation was specifically developed for clinical studies and has been tested in several populations where there were discrepancies between bone density and fracture propensity, such as type 2 diabetes, atypical femoral fracture, stress fractures, glucocorticoid treatment, patients with osteopenia and fragility fractures, and individuals infected with HIV, among others. Micro-indentation will complement, not replace, existing bone analysis methods, especially where bone mineral density (BMD) does not fully explain fracture propensity. The authors concluded that available evidence provided solid proof-of-concept; future studies will fully define the role of micro-indentation for the assessment of bone health both in clinics and in research.

Arnold and colleagues (2017) stated that micro-indentation, which has 2 main categories: depth-sensing micro-indentation; and RPI; it has the potential to measure the stiffness of an individual patient's bone. Bone stiffness plays a crucial role in the press-fit stability of orthopedic implants. Arming surgeons with accurate bone stiffness information may reduce surgical complications including peri-prosthetic fractures. In a systematic review, these researchers examined if micro-indentation could accurately measure cortical bone stiffness. They carried out a systematic review of all English language articles using Medline, Embase, PubMed, Scopus and Cochrane databases. Studies that only used nano-indentation, cancellous bone or animal tissue were excluded. A total of 1,094 abstracts were retrieved and 32 papers were included in the analysis, 20 of which used reference point indentation, and 12 of which used traditional depth-sensing indentation. There were several factors that must be considered when using micro-indentation, such as tip size, depth and method of analysis. Only 2 studies validated micro-indentation against traditional mechanical testing techniques. Both studies used reference point indentation (RPI), with 1 showing that RPI parameters correlated well with mechanical testing, but the other suggested that they did not. The authors concluded that micro-indentation has been used in various studies to evaluate bone stiffness, but only 2 studies with conflicting results compared micro-indentation with traditional mechanical testing techniques. Thus, it is currently unclear whether micro-indentation can accurately measure cortical bone stiffness. These investigators stated that further research, including more studies comparing micro-indentation with other mechanical testing methods, is needed before micro-indentation can be used reliably to calculate cortical bone stiffness.

Rufus-Membere and co-workers (2018) noted that the OsteoProbe measures BMSi of cortical bone in living humans using impact micro-indentation (IMI). Research using this minimally invasive technique is expanding yet, to-date, there have been no reports regarding its feasibility in the research setting. In a cross-sectional analysis of data collected in a population-based study, these researchers examined the feasibility and tolerability of using the OsteoProbe in men enrolled in the Geelong Osteoporosis Study. For 252 of 345 consecutive participants (aged 33 to 96 years), BMSi was measured using the OsteoProbe at the mid-tibia. Immediately following measurement, each subject used a visual analog

scale (0 to 10) to rate the level of discomfort that was anticipated and experienced, their initial reluctance towards the measurement and their willingness to repeat measurement. Reasons for non-measurement in 92 men were needle phobia (n = 8), discomfort after 1st indentation (n = 5), skin infections (n = 21), excessive soft tissues around the mid-tibia region (n = 56), inability to provide informed consent (n = 2). Among 252 men who had IMI measures, the expectation for pain during measurement was low (1.54 ± 1.56), as was actual pain experienced (0.38 ± 0.71). Reluctance to undergo measurement was low (0.34 ± 0.93). All subjects indicated a willingness to have the measurement performed again. Mean (± SD) BMSi was 83.0 ± 6.4 (range of 62.3 to 93.0). The authors concluded that the procedure was well-accepted by subjects suggesting that IMI testing with the OsteoProbe was feasible in a research setting. These investigators stated that further assessment of the clinical utility of this technology for evaluating fracture risk is needed and is currently in progress.

The authors stated that this study had several drawbacks. First, IMI could not be performed in individuals with substantial amounts of soft tissue around the mid-tibia region, nor in individuals with skin disorders or infections in at the site of measurement and this may have biased these findings. Second, there were also drawbacks with other technologies. For example, a weight limitation (typically 120 kg or 300 lb) and narrow bed width (approximately 60 cm) necessitated exclusion of large individuals from evaluation with DXA. Third, individuals with spinal abnormalities and those affected by devices such as plates, screws, silicone implants and prostheses could compromise the interpretation of DXA scans. Fourth, in this study, these investigators examined men only; thus these observations may not be generalized to women or other populations.

Adipose-Derived Stromal Vascular Fraction (SVF) Cells

There is insufficient evidence to support the use of adipose-derived stromal vascular fraction (SVF) cells for any indication.

van Dongen et al (2018) noted that intraoperative application of the stromal vascular fraction (SVF) of adipose tissue requires a fast and efficient isolation procedure of adipose tissue. These investigators

evaluated and compared procedures currently used for the intraoperative isolation of cellular SVF (cSVF) and tissue SVF (tSVF) that still contain the extra-cellular matrix. PubMed, Embase and the Cochrane central register of controlled trials databases were searched for studies that compare procedures for intraoperative isolation of SVF (searched September 28, 2016). Outcomes of interest were cell yield, viability of cells, composition of SVF, duration, cost and procedure characteristics. Procedures were sub-divided into procedures resulting in a cSVF or tSVF; 13 out of 3,038 studies, evaluating 18 intraoperative isolation procedures, were considered eligible. In general, cSVF and tSVF intraoperative isolation procedures had similar cell yield, cell viability and SVF composition compared to a non-intraoperative (i.e., culture laboratory-based collagenase protocol) control group within the same studies. The majority of intraoperative isolation procedures were less time-consuming than non-intraoperative control groups, however. The authors concluded that intraoperative isolation procedures were less time-consuming than non-intraoperative control groups with similar cell yield, viability of cells and composition of SVF, and therefore more suitable for use in the clinic. Nevertheless, none of the intraoperative isolation procedures could be designated as the preferred procedure to isolate SVF.

Bateman et al (2018) described the safety and efficacy of adipose stem/stromal cells (ASC) and stromal vascular fraction (SVF) in treating common diseases and the next steps in research that must occur prior to clinical use. PubMed, Ovid Medline, Embase, Web of Science, and the Cochrane Library were searched for articles about use of SVF or ASC for disease therapy published between 2012 and 2017; 1 meta-analysis, 2 randomized controlled trials (RCTs), and 16 case series were included, representing 844 human patients; 69 studies were performed in pre-clinical models of disease. ASCs improved symptoms, fistula healing, remission, and recurrence rates in severe cases of inflammatory bowel disease (IBD). In osteoarthritis (OA), ASC and SVF improved symptom-related, functional, radiographic, and histological scores. ASC and SVF were also shown to improve clinical outcomes in ischemic stroke, multiple sclerosis (MS), myocardial ischemia, chronic obstructive pulmonary disease (COPD), idiopathic pulmonary fibrosis (IPF), chronic liver failure, glioblastoma, acute kidney injury (AKI), and chronic skin wounds. These effects were primarily paracrine in nature and mediated through reduction

of inflammation and promotion of tissue repair. In the majority of human studies, autologous ASC and SVF from liposuction procedures were used, minimizing the risk to recipients. Very few serious, treatment-related adverse events (AEs) were reported. The main AE was post-procedural pain. The authors concluded that SVF and ASC are promising therapies for a variety of human diseases, especially for patients with severe cases refractory to current medical treatments. These researchers stated that further RCTs must be performed to elaborate potential safety and efficacy prior to clinical use.

Pak et al (2018) stated that ASCs in the form of SVF and cultured expansion have been applied in clinical settings in some countries to treat OA of knees. Since the 1st report of successful cartilage-like tissue regeneration with autologous adipose SVF containing ASCs, there has been a gradual increase in the number of publications confirming such results. Thus far, most of the reports have been limited to treatments of OA of knees. Recently, successful applications of adipose SVF in treating OA of ankles and hips have been reported. In addition, several groups have reported modified methods of applying adipose SVF, such as combining bone marrow stimulation with adipose SVF or adding additional extra-cellular matrix (ECM) in treating OA. These researchers presented an updated, systematic review on the safety and effectiveness in treating OA of knees, ankles, and 1 hip since 2016 using ASCs in the form of adipose SVF or in cultured expansion, along with a description and suggestion of potential biological mechanisms of cartilage regeneration. The authors concluded that better-designed studies are needed to elucidate the true mechanism of action of the therapy and for the potential general application of these stem cells to treat OA/degenerative joint disease by cartilage regeneration.

Moon et al (2019) stated that beneficial effects of human adipose-derived SVF cell injection on microcirculation have been recently reported in in-vitro and in-vivo studies. However, no clinical studies have reported its effect in diabetic patients who commonly experience compromised tissue perfusion, regardless of the status of intravascular blood flow. This pilot study was designed to clinically examine the possibility of SVF cell injection to accelerate microcirculation, particularly in ischemic diabetic feet. A total of 10 diabetic feet were included to receive subcutaneous injection of SVF cells around wounds. Transcutaneous partial oxygen

pressure (TcPO2) and cutaneous microvascular blood flow were measured before and every 4 weeks after cell injection until the 12th week visit. TcPO2 values increased from 31.3 ± 7.4 before injection to 46.4 ± 8.2 mmHg at 12 weeks after SVF injection (1.5-fold, $p < 0.05$). Cutaneous microvascular blood flow levels increased from 34.0 ± 21.1 before injection to 76.1 ± 32.5 perfusion unit at 12 weeks after SVF injection (2.2-fold, $p < 0.05$). There were no AEs related to SVF cell injection. The authors concluded that findings this pilot study showed that adipose-derived SVF cell injection had the possibility to provide beneficial effects on microcirculation in ischemic diabetic feet.

Mehranfar et al (2019) stated that OA is a major cause of disability across the world. Current accepted therapies such as exercise, anti-inflammatory drugs and intra-articular inoculation of corticosteroids are aimed at controlling symptoms in the affected patients. Surgical options including arthroplasty, osteotomy and joint replacement are other choices of treatment, which are invasive and can be applied in case of failure of conventional therapies. In the last few decades, efforts to treat musculo-skeletal diseases are being increasingly focused on regenerative cellular therapies; SVF, obtained from adipose tissue, contains a variety of cells include mesenchymal stem cells (MSCs) and has shown to be effective in cartilage repair. Autologous blood products such as platelet-rich plasma (PRP) act as an adjuvant of surgical treatment and its intra-articular delivery has shown beneficial effects for OA treatment. The authors discussed both pre-clinical and clinical evidence with major focus on clinical trials. These researchers stated that the efficacy of such treatments, as well as new stem cell-based approaches in OA, need to be confirmed by further controlled long-termed studies.

Polly et al (2019) noted that adipose-derived SVF is a heterogeneous population of cells that yields a homogeneous population of plastic-adherent adipose-derived multipotent stromal cells (ASC) when culture-expanded. SVF and ASC have been used clinically to improve tendon healing, yet their mechanism of action is not fully elucidated. These investigators examined the potential for ASC to act as trophic mediators for tendon healing. Flexor digitorum superficialis tendons and adipose tissue were harvested from adult horses to obtain SVF, ASC and tenocytes. Growth factor gene expression was quantified in SVF and ASC in serial passages and growth factors were quantified in ASC-

conditioned medium (CM). Micro-chemotaxis assays were performed using ASC-CM. Tenocytes were grown in co-culture with autologous ASC or allogeneic SVF. Gene expression for insulin-like growth factor-1 (IGF-1), stromal cell-derived factor 1α (SDF-1α), transforming growth factor (TGF)-β1 and -β3 was significantly higher in SVF compared to ASC. Concentrations were significantly increased in ASC-CM compared to controls for IGF-1 (4-fold) and SDF-1α (6-fold). Medium conditioned by ASC induced significant cell migration in a dose-dependent manner. Gene expression for collagen types I and III, decorin, and cartilage oligomeric matrix protein was modestly, but significantly increased following co-culture of tenocytes with autologous ASC. The authors concluded that these findings supported the ability of SVF and ASC to act as trophic mediators in tendon healing, particularly through chemotaxis, which stands to critically impact the intrinsic healing response. These researchers stated that in-vivo studies to further delineate the potential for SVF and/or ASC to improve tendon healing are needed.

Arthrex Quickset (Calcium Phosphate Cement)

Quickset (calcium phosphate cement) is a macro-porous, injectable, hardening and resorbable bone filler provided in an easy-to-use, closed mixing system. There is insufficient evidence to support the clinical value of Quickset (calcium phosphate cement).

Scordino et al (2017) noted that there has been a resurgence in the use of opening wedge high tibial osteotomy (owHTO). Calcium phosphate cement has been shown to improve strength in compression for augmentation of tibial plateau and owHTO fixation. However, knee kinematics includes a torsional load during ambulation, which is as yet unstudied in this model. These investigators examined the effect of injectable calcium phosphate cement on the biomechanical stability of standard high tibial osteotomy defect with applied torsional load and ultimate stiffness of the supporting construct. Testing was performed on 22 bone mineral density-matched and age-matched cadaver specimens. Intact specimens were treated with 10° opening wedge osteotomies, identical surgical techniques as clinically used and fixation provided by iBalance PEEK implant (Arthrex, Naples FL); 9 specimens were augmented with calcium phosphate injectable cement, Quickset (Arthrex Inc., Naples Fl). Constructs were for construct stiffness, torsional loads to

failure, and mechanisms of failure. As a gold-standard comparison group, 4 samples were tested with a titanium, fixed angle device alone: Contourlock plate (Arthrex Inc., Naples Fl). Peak torque to failure was significantly greater in samples augmented with calcium phosphate bone cement (23.0 ± 9.6 Nm) compared with specimens fixed with PEEK implant alone (18.1 ± 7.3). Construct stiffness in torsion was also significantly improved with bone cement application (349.0 ± 126.8 Nm/°) compared with PEEK implant alone (202.2 ± 153.4 Nm/°) and fixed angle implant system (142.9 ± 74.7 Nm/°). The authors concluded that injectable calcium phosphate cement improved the initial maximal torsional strength and stiffness of high tibial osteotomy construct.

Ollivier et al (2018) stated that injectable cements have been developed to improve fixation's stability and thus obtain early return to adequate joint function. These researchers compared post-operative radiographic and clinical outcomes of patients suffering from a complex tibial plateau fracture (TPF) fixed with calcium-phosphate bone substitutes (CPBS) augmentation to a matched group of patients with identical fracture pattern, treated with the same fixation's type, but augmented with bone grafting. After local ethic committee approval, these investigators retrospectively identified in a prospectively collected database, patients with complex comminuted metaphyseal and epiphysial bicondylar TPF (Schatzker type VI) admitted in the authors' emergency department between January 2011 and December 2013. From those, 23 patients (14 males, 9 females) were treated with CPBS (Quickset-CP, Graftys, Aix-en-Provence, France) fixation augmentation. Patients' mean age were 44.4 years. These investigators then created a control group using a 1:1 matching process on gender, age, fracture pattern, and method of fixation. Patients were evaluated prospectively at 3, 6, and then every 6 months using radiographic (AP/ML views) and clinical criteria (knee osteoarthritis outcomes score (KOOS) and EuroQOL-5D). Articular step-off and variation of articular step-off were significantly lower in the CPBS groups (mean step-off 1.4 ± 1.9 (0.5-6.5 mm) and mean step-off Δ = 0.3 ± 0.4 (0.5-2.2 mm)) than in the control group (mean step-off 3.6 ± 2.1 (1-7.5 mm) and mean step-off Δ = 2.2 ± 2 (0.5-7 mm) $p < 0.01$). At last follow-up, patients of the control group presented a higher rate of step-off greater than 2 mm and step-off Δ greater than 2 mm (respectively, 56 and 35 %) than patients of the CPBS group (26 and 9 %). Odd ratio of, respectively, 3.6 (95 % confidence interval [CI]: 1.08 to 12.7) and $p = 0.03$) and 5.6 (95

% CI: 1.04 to 30.1) and p = 0.03). At mean follow-up of 29 months, KOOS pain subscore was significantly better in patients of the CPBS group (85.3 ± 12.1) than in control patients (74.2 ± 10.4 and p = 0.03). The authors concluded that the findings of the present study demonstrated that calcium-phosphate bone substitute used as synthesis augmentation improved mid-term radiological outcomes of patients suffering from complex tibial plateau fracture. These researchers stated that series reporting outcomes from a larger number of patients and longer follow-up must confirm safety and benefits of this method.

Brueckner et al (2019) noted that bone substitutes are essential in orthopedic surgery to fill up large bone defects. These researchers compared diverse bone fillers biomechanically to each other in a clinical-relevant test set-up and to detect differences in stability and handling for clinical use. This study combined compressive strength tests and screw pullout-tests with dynamic tests of bone substitutes in a clinical-relevant biomechanical fracture model. Beyond well-established bone fillers (ChronOSTM Inject and Graftys Quickset [non-drillable]), 2 newly designed bone substitutes, a magnesium phosphate cement (MPC) and a drillable hydrogel reinforced calcium phosphate cement (CPC), were investigated. The drillable CPC revealed a comparable displacement of the fracture and maximum load to its commercial counterpart (Graftys Quickset) in the clinically relevant biomechanical model, even though compressive strength and screw pullout force were higher using Graftys. The authors concluded that the in-house-prepared cement allowed unproblematic drilling after replenishment without a negative influence on the stability. A new, promising bone substitute is the MPC, which showed the best overall results of all 4 cement types in the pure material tests (highest compressive strength and screw pullout force) as well as in the clinically relevant fracture model (lowest displacement and highest maximum load). The low viscosity enabled a very effective interdigitation to the spongiosa and a complete filling up of the defect, resulting in this demonstrated high stability. The authors concluded that the 2 in-house-developed bone fillers revealed overall good results and are budding new developments for clinical use.

DeMill and colleagues (2014) stated that cryo-preserved amnion tissues derived from amniotic membrane/umbilical cord (AM/UC) have been widely used in ophthalmology for minimizing post-operative inflammation, pain, and adhesion formation following various surgical procedures. There is limited data in the current literature regarding the use of amnion tissue products in foot and ankle surgery. In a retrospective study, these researchers examined the short-term safety profile after in-vivo application of cryo-preserved AM/UC tissue use in foot and ankle surgery. They evaluated cases where amnion tissue was used with a minimum follow-up of 120 days between 2011 and 2012. The clinical outcomes of interest were post-operative infections, delayed or non-healing wounds, adverse surgical site reactions, and repeat surgery for formal irrigation and debridement. A total of 124 patients qualified for inclusion. Cryo-preserved AM/UC tissue was introduced into the surgical wound and placed adjacent to the compromised and repaired tendons, most frequently in peroneal and Achilles tendon repairs. In this level IV consecutive case-series cohort, there was an overall wound complication rate of 5.64 %, with a re-operation rate of 1.6 % (2/124). In each of these cases, patients had an irrigation and debridement with ultimate successful wound healing. The authors concluded that the findings of this study showed that the use of amnion tissue in the foot and ankle setting was safe with a decreasing trend in overall complication rates compared with historically published norms.

Esquivel and associates (2020) stated that extensor tendon repairs, although common, can be difficult injuries to treat. Their treatment is tailored to the zone of the hand that is affected since varying biomechanical forces are applied to the tendon at each zone. Prompt treatment is needed to prevent potential complications associated with these injuries. This is particularly true of Zone V extensor tendon injuries, as their mechanism is commonly a highly infectious human bite. These investigators presented a case of a human fight bite resulting in a Zone V extensor tendon injury. The delayed presentation of this case resulted in an untreated infection that caused an abscess with associated extensor tendon necrosis and rupture. Given the large gap length between the ends of the tendons, tendon repair was performed using a palmaris longus autograft. Even when these were performed in a controlled

setting, adhesions were common. The compromised wound bed caused irritation, erosion, and subsequent rupture of the extensor tendon of the hand. In an effort to avoid common complications such as adhesion, the repair was then wrapped with human umbilical membrane (AVIVE Soft Tissue Membrane, AxoGen Inc., Alachua, FL) to separate adjacent tissue and reduce inflammation. The authors concluded that even without access to formal physical therapy, [their] patient had excellent functional outcomes at his final follow-up visit. The patient was able to make a loose composite fist, had no extensor lag at the MCP joints, and had extensor lag of 15 degrees at the PIP joints of digits 4 to 5. Moreover, thee researchers stated that further studies with a larger cohort are needed to examine the outcomes for Zone V extensor tendon repairs using this approach.

CartiMax Allograft

Albrecht et al (1983) noted that in 75 knee joints of 46 adult rabbits osteochondral defects of 4 mm diameter were placed by a drill reaching the cancellous bone; 23 defects were left untreated, or closed by collagen foam or fibrin adhesive, or a combination of both; 52 defects were closed with very small autologous cartilage fragments and a special fibrin adhesive. The adhesive differed from commercially supplied types by digested alpha chain of fibrinogen for increase in concentration, and by the addition of alpha-2-macroglobulin as protease-inhibitor. In most cases, small pieces of collagen foam were added for hemostasis. In the 1st group of 23 joints observed over up to 40 weeks, no hyaline cartilage was found histologically in any of the defects. In the 2nd group a rapid proliferation of chondrocytes appeared with development of hyaline cartilage with alcianblue-positive matrix. It resembled juvenile cartilage in its histologic appearance and with regard to the induction of ossification. The phenomenon was interpreted as a "second adolescence" of the adult cartilage induced by the rich nutritional and oxygen supply from the cancellous vessels, which resembled the environmental conditions before the forming of subchondral cortical bone at the end of the growth period. This method enabled these researchers to achieve a complete closure of defects by hyaline cartilage on the very level of the surrounding articular surface.

Lu et al (2006) stated that traumatic articular cartilage injuries heal poorly and may predispose patients to the early onset of osteoarthritis (OA). One current treatment relies on surgical delivery of autologous chondrocytes that are prepared, prior to implantation, through ex-vivo cell expansion of cartilage biopsy cells. The requirement for cell expansion, however, is both complex and expensive and has proven to be a major hurdle in achieving a widespread adoption of the treatment. These investigators presented evidence that autologous chondrocyte implantation (ACI) can be delivered without requiring ex-vivo cell expansion. The proposed improvement relies on mechanical fragmentation of cartilage tissue that is sufficient to mobilize embedded chondrocytes via increased tissue surface area. The authors' outgrowth study, which was used to demonstrate chondrocyte migration and growth, indicated that fragmented cartilage tissue is a rich source for chondrocyte re-distribution. The chondrocytes outgrown into three-dimensional (3D) scaffolds also formed cartilage-like tissue when implanted in severe combined immunodeficiency (SCID) mice. Direct treatment of full-thickness chondral defects in goats using cartilage fragments on a resorbable scaffold produced hyaline-like repair tissue at 6 months; therefore, delivery of chondrocytes in the form of cartilage tissue fragments in conjunction with appropriate polymeric scaffolds provided a novel intra-operative approach for cell-based cartilage repair.

Frisbie et al (2009) stated that current ACI techniques require 2 surgical procedures: 1 for cell harvest and 1 for re-implantation of cultured cells. A 1-step procedure is more desirable. In a controlled laboratory study, a 1-step surgical procedure using autologous cartilage fragments on a polydioxanone scaffold, or CAIS (cartilage autograft implantation system), in a clinically relevant defect (15-mm diameter) within equine femoral trochlea was compared with a 2-step ACI technique as well as with empty defects and defects with polydioxanone foam scaffolds alone. A total of 10 skeletally mature horses were used. Articular cartilage from the lateral trochlea of the femur was harvested arthroscopically (n = 5), and chondrocytes were cultured on small intestinal submucosa to produce ACI constructs. The CAIS procedure had cartilage harvested during defect creation to prepare minced cartilage on polydioxanone-reinforced foam. The ACI and CAIS constructs were placed in defects using polydioxanone/polyglycolic acid staples. Defects were examined arthroscopically at 4, 8, and 12 months and with gross, histological, and

immunohistochemical examination at 12 months. Arthroscopic, histologic, and immune-histochemistry results showed superiority of both implantation techniques (ACI and CAIS) compared with empty defects and defects with polydioxanone foam alone, with CAIS having the highest score. The authors concluded that this was the 1st demonstration of long-term healing with strenuous exercise using ACI and CAIS in a critically sized defect. Moreover, these researchers stated that given these results with the CAIS procedure, testing in human patients is the next logical step; and a phase-I human clinical trial has proceeded from this work.

Cerament Bone Void Filler

Karr et al (2011) noted that several absorbable and non-absorbable antibiotic carrier systems are available in the adjunctive surgical management of osteomyelitis of the foot, ankle, and lower leg. These carrier systems have significant limitations regarding which antibiotics can be successfully incorporated into the carrier vehicle. The calcium sulfate and hydroxyapatite Cerament Bone Void Filler is a biocompatible, absorbable ceramic bone void filler that can successfully deliver multiple heat-stable and heat-unstable antibiotics that have not been generally used before with antibiotic beads in treating musculoskeletal infections. In this study, Cerament Bone Void Filler discs with the antibiotics rifampin, vancomycin, tobramycin, cefazolin, cefepime hydrochloride, vancomycin-tobramycin, piperacillin-tazobactam, ceftazidime, and ticarcillin-clavulanate were tested in-vitro against methicillin-resistant Staphylococcus aureus. The zones of inhibition for the Cerament Bone Void Filler antibiotic discs plated against Staphylococcus aureus obtained were 33 % to 222 % greater than the minimum zones of inhibition breakpoints for bacteria susceptibility as defined by the standard set by the Clinical and Laboratory Standards Institute (CLSI). Cerament Bone Void Filler discs with the antibiotics plated against Pseudomonas aeruginosa produced zones of inhibition of 93 % to 200 % greater than the minimum zones of inhibition breakpoints for bacteria susceptibility as defined by the standard set by the CLSI. The authors concluded that the calcium sulfate and hydroxyapatite Cerament Bone Void Filler was an excellent carrier vehicle for multiple antibiotics creating in-vitro significant

zones of inhibition, thus demonstrating susceptibility against Staphylococcus aureus and Pseudomonas aeruginosa, which holds tremendous promise in treating osteomyelitis.

Karr (2011) noted that several non-biodegradable and biodegradable antibiotic cement delivery systems are available for the delivery of antibiotics for adjunctive therapy in the management of osteomyelitis. A major non-biodegradable delivery system is polymethylmethacrylate beads. Antibiotics that can be incorporated into this delivery system are limited to the heat-stable antibiotics vancomycin and aminoglycosides, tobramycin being the most popular. Calcium sulfate and hydroxyapatite (Cerament Bone Void Filler) is a unique biocompatible and biodegradable ceramic bone void filler that can successfully deliver heat-stable and heat-unstable antibiotics in musculoskeletal infections. The use of Cerament as antibiotic beads has not been previously reported. The author presented an off-label case of diabetic foot osteomyelitis successfully managed with surgical bone resection and vancomycin Cerament antibiotic beads. Subsequent surgery for the bone infection and staged removal of the antibiotic beads was not necessary.

Nusselt et al (2014) stated that bone graft substitutes are widely used for reconstruction of post-traumatic bone defects. However, their clinical significance in comparison to autologous bone grafting, the gold-standard in reconstruction of larger bone defects, still remains under debate. This prospective, randomized, controlled clinical study examines the differences in pain, quality of life (QOL), and cost of care in the treatment of tibia plateau fractures-associated bone defects using either autologous bone grafting or bioresorbable hydroxyapatite/calcium sulphate cement (Cerament bone void filler [CBVF]). CERTiFy (CERament Treatment of Fracture defects) is a prospective, multi-center, controlled, randomized trial. These researchers planned to enroll 136 patients with fresh traumatic depression fractures of the proximal tibia (types AO 41-B2 and AO 41-B3) in 13 participating centers in Germany. Patients will be randomized to receive either autologous iliac crest bone graft or CBVF after reduction and osteosynthesis of the fracture to reconstruct the subchondral bone defect and prevent the subsidence of the articular surface. The primary outcome is the SF-12 Physical Component Summary at week 26. The co-primary endpoint is the pain level 26 weeks after surgery measured by a visual analog scale (VAS). The SF-

12 Mental Component Summary after 26 weeks and costs of care will serve as key secondary endpoints. The study is designed to show non-inferiority of the CBVF treatment to the autologous iliac crest bone graft with respect to the physical component of QOL. The pain level at 26 weeks after surgery is expected to be lower in the CBVF treatment group. The authors concluded that CERTiFy is the 1st randomized, multi-center clinical trial designed to compare QOL, pain, and cost of care in the use of the CBVF and the autologous iliac crest bone graft in the treatment of tibia plateau fractures. The results are expected to influence future treatment recommendations.

Iundusi et al (2015) stated that reduction of tibial plateau fractures and maintain a level of well aligned congruent joint is key to a satisfactory clinical outcome and is important for the return to pre-trauma level of activity. Stable internal fixation support early mobility and weight-bearing. The augmentation with bone graft substitute is often needed to support the fixation to maintain reduction. For these reasons there has been development of novel bone graft substitutes for trauma applications and in particular synthetic materials based on calcium phosphates and/or apatite combined with calcium sulfates. Injectable bone substitutes can optimize the filling of irregular bone defects. These investigators examined the potential of a novel injectable bone substitute CBVF in supporting the initial reduction and preserving alignment of the joint surface until fracture healing. From June 2010 through May 2011 adult patients presenting with acute, closed and unstable tibial plateau fractures that needed both grafting and internal fixation, were included in a prospective study with percutaneous or open reduction and internal fixation (ORIF) augmented with an injectable ceramic biphasic bone substitute CBVF to fill residual voids. Follow-up was performed at 1, 3, 9 and 12 months and any subsequent year; including radiographic analysis and Rasmussen system for knee functional grading. A total of 24 patients, balanced male-to-female, with a mean age of 47 years, were included and followed with an average of 44 months (range of 41 to 52 months). Both Schatzker and Müller classifications were used and was type II or 41-B3 in 7 patients, type III or 41-B2 in 12 patients, type IV or 41-C1 in 2 patients and type VI or 41-C3 in 3 patients, respectively. The joint alignment was satisfactory and maintained within a range of 2 mm, with an average of 1.18 mm. The mean Rasmussen knee function score was 26.5, with 14 patients having an excellent result and the remaining

10 with a good result. The authors concluded that radiological and clinical outcome was satisfactory and obtained in all cases without complications. This injectable novel bi-phasic hydroxyapatite and calcium sulfate ceramic material is a valuable armamentarium in the treatment of trauma where bone graft is required.

The authors stated that this study had several drawbacks. The study population was relatively small (n = 24), and did not effectively represent all types of tibial plateau fractures; and 3 out of the 5 authors performed the surgical procedures. Also, a control group using other bone substitutes or grafting was not enrolled. Lastly, patients were followed for a mid-period (average of 44 months), thus, further studies will be needed to verify the long-term clinical and radiological outcomes.

Kotrych et al (2018) noted that CBVF is an injectable and moldable ceramic bone substitute material intended for bone voids. The material consists of hydroxyapatite and calcium sulfate hemihydrate. These researchers presented the first long-term results following open curettage of benign bone tumors and tumor-like lesions and void filling with this novel injectable and synthetic bone graft. A total of 33 patients were enrolled into the study between June 2013 and October 2014. These investigators treated 24 women and 9 men with a median age of 47 years (range of 22 to 74). All patients suffered from primary musculoskeletal system disorders (enchondroma 63.6 %, giant cell tumor 18 %, aneurysmal bone cyst 9 %, fibrous dysplasia 9 %, Gaucher disease 3 %). they performed curettage of pathological lesions, then the bone substitute was administered by means of needle to the void. The average follow-up was 13 months (range of 2 to 13 months, median of 10 months). No metastasis or recurrence had been detected. They received significant clinical improvement relating to VAS, MSTS, and oncological results. The authors preliminary concluded that CBVF can be successfully used as a bone substitute in patients with various bone diseases, as well as benign bone tumors; CBVF can provide an effective and long-term solution for reconstructive procedures following curettage of bone tumors and tumor like lesions. The main drawbacks of this study were its relatively small sample size (n = 33) and short-term follow-up (13 months). These preliminary findings need to be validated by well-designed studies.

Niazi et al (2019) stated that diabetic foot ulcers are associated with a high morbidity and are common cause of non-traumatic lower limb amputations. In a retrospective, case-series study, these researchers examined the effect of debridement and the use of an adjuvant local antibiotic carrier in the treatment of diabetic foot ulcers with osteomyelitis. Patients with diabetic foot ulceration and osteomyelitis treated by debridement with adjuvant local antibiotic were included in this trial -- 70 patients with Texas Grade 3B & 3D lesions were included, with a mean age of 68 years. Cerament G, an antibiotic-loaded absorbable calcium sulphate/hydroxyapatite bio-composite was used along with intra-operative multiple bone sampling and culture-specific systemic antibiotics. Patients were followed up until infection eradication or ulcer healing. Mean follow up was 10 months (4 to 28 months); 9 patients had Charcot foot deformity, 14 had peripheral vascular disease. 62 % of patients had fore-foot, 5 % mid-foot and 33 % hind-foot involvement; 53 patients (87 %) had polymicrobial infection. Staphylococcus aureus was the most common microorganism isolated. Infection was eradicated in 63 patients (90 %) with mean time to ulcer healing of 12 weeks; 7 patients were not cured and needed further treatment; 5 patients had below knee amputation. The authors concluded that adjuvant, local antibiotic therapy with an absorbable bio-composite could help achieve up to 90 % cure rates in diabetic foot ulceration with osteomyelitis. Cerament G can act as effective void filler allowing dead space management after excision and preventing reinfection and the need for multiple surgical procedures. Level of Evidence = IV.

Collagen-Based Bio-Inductive Implants (e.g., Regeneten) for Repair of Rotator Cuff Tears

Thon and co-workers (2019) stated that failure of repair of large and massive rotator cuff tears is a challenging problem within orthopedics. Poor tendon tissue and vascularity are known causes for failure of rotator cuff repairs. In a prospective, single-arm, proof-of-principle study, these researchers examined the safety, outcomes, and healing rates when large and massive rotator cuff repairs are augmented with a bio-inductive collagen scaffold patch. A total of 23 patients undergoing repair of full-thickness large (2-tendon) or massive (3-tendon) rotator cuff tears augmented with a bio-inductive collagen patch were enrolled in this trial. No partial repairs were performed, and a complete rotator cuff repair was

successfully achieved in each case; 16 patients underwent revision rotator cuff repairs versus 7 primary repairs. Safety was determined by any implant-related AE. A single magnetic resonance imaging (MRI) scan was used to confirm tendon healing and thickness at a minimum of 6 months post-operatively. Post-operative ultrasound (US) was used in office by the treating surgeon to examine tendon thickness at 3-, 6-, 12-, and 24-month intervals. American Shoulder and Elbow Surgeons (ASES) scores were collected at final follow-up. Overall, a 96 % (22 of 23) healing rate was confirmed on MRI and US; however, incidence of treatment clinical failure was 9 % (2 of 23), as 1 patient's tendon healed but eventually underwent additional surgery. There were no AEs attributed to the implant reported. Final US rotator cuff thickness was 7.28 ± 0.85 mm (mean ± SD), and final MRI rotator cuff thickness was 5.13 ± 1.06 mm. The mean ASES score at final follow-up was 82.87 ± 16.68 (range of 53.33 to 100). The authors concluded that no complications attributed to the implant were reported, and new tendon formation was apparent on MRI and US, with relatively high healing rates at 2-year follow-up. The authors concluded that arthroscopic application of this bio-inductive collagen scaffold when combined with rotator cuff repair was a safe and effective treatment for healing of large and massive rotator cuff repairs. Level of Evidence = IV.

McIntyre et al (2019) collected outcomes data on patients treated with a bio-inductive collagen implant designed to induce rotator cuff healing in partial- and full-thickness cuff tears and examined the safety and effectiveness of the device. A total of 15 surgeons in 15 centers in the U.S. enrolled patients between April 2016 and August 2017 and collected standardized outcomes data. Patients aged 21 years and older, able to read and speak English, and with partial- or full-thickness tears of the rotator cuff documented by (MRI were included in the study. Patients were assessed pre-operatively with (VAS, single-assessment numeric evaluation (SANE), Veterans RAND 12-Item (VR-12), ASES, and Western Ontario Rotator Cuff (WORC) outcomes measures. Post-operative assessment was made at 2, 6, and 12 weeks, 6 months, and 1 year. Patients underwent a standardized operative procedure with the implant. Patient demographics, co-morbidities, tear types, and concomitant operative procedures were recorded. Patients in both groups experienced statistically significant improvement in VAS, SANE, VR-12 PCS, ASES, and WORC scores (mean values of 1.1, $p < 0.001$;

86.0, $p < 0.001$; 49.7, $p < 0.001$; 85.6, $p < 0.001$; and 84.4, $p < 0.001$ for partial tears and 1.2, $p < 0.001$; 80.7, $p < 0.001$; 45.7, $p < 0.001$; 83.8, $p < 0.0001$; and 80.1, $p < 0.001$ for full-thickness tears, respectively). For the partial tear group, average times for return to driving, work, and non-overhead athletic activity were 14.6, 37.3, and 65.6 days, and for the full-thickness group, 24.5, 50.7, and 119.2 days, respectively. In the partial-thickness group, 84 % and 83 % of patients reported improvement in their VAS pain and ASES scores, respectively, that met or exceeded each measure's minimal clinically important difference. In the full-thickness group, 72 % and 77 % of the patients met or exceeded the minimal clinically important differences for VAS pain and ASES, respectively. The authors concluded that outcomes after repair of partial- and full-thickness rotator cuff tears using a bio-inductive implant showed safety and effectiveness at 1-year follow-up. Moreover, these researchers stated that the value in full-thickness tears, especially in settings where healing is problematic, is promising and worthy of further study. Level of Evidence = IV.

The authors stated that drawbacks of this study included the selection bias associated with a level IV study design and non-consecutive patient inclusion. As such, the added value of the implant will require future higher-level comparative analysis. Lack of a control group made analysis of results comparing the implant to more traditional methods of repair without augmentation difficult. Furthermore, there was considerable heterogeneity in both the partial- and full-thickness tear groups involving extent of cuff disease, history of trauma, duration of symptoms, and concomitant pathology. Rotator cuff repair echniques were at the discretion of the treating surgeon and were not standardized, introducing additional bias. Drawing specific conclusions concerning application of the implant in any specific patient can be based only on the overall improvement of the groups. In addition, these researchers did not separate the partial-thickness group according to tear location, which limited delineation of results between partial tear locations. There was evidence in the literature that partial-thickness bursal-sided tears do not respond as well to surgical treatment as articular sided tears. There was a considerable body of literature documenting the results of in-situ repair, take-down and repair, and comparison of the two, but there were only 2 studies that showed increase in tendon thickness in partial-thickness tears treated surgically without instrumentation of the partially torn cuff.

Both studies employed the bio-inductive implant used in this report. These investigators noted that younger, active patients with partial cuff tears might be better managed with early surgical intervention with biologics to disrupt the natural history of their disease progression.

In an editorial that accompanied the afore-mentioned study, Yoo (2019) stated that there is still plenty of room for improvement in surgical rotator cuff repair healing as well as in non-surgical cases. Adding an implant or patch to improve results has been a subject of much discussion for a couple of decades. The main problem has been that the surgical procedures are so difficult, with added time and its effect, and therefore are controversial. Many surgeons are reluctant to go the surgical route. As technology has advanced, this field is at a turning point, with newer implants that can be easily augmented. Therefore, recently many surgeons have been using implants, and preliminary reports appeared promising. New implants must withstand vigorous challenges, and future extensive and long-term studies can enable improved implants using human tissue.

Dai and colleagues (2020) noted that partial thickness rotator cuff tears (PTRCT) have low healing potential and tend to progress over time if not addressed surgically. There is a relative paucity of literature discussing optimal treatment for symptomatic PTRCT as compared to full thickness rotator cuff tears. The available data supports a treatment course of debridement with or without acromioplasty for symptomatic patients with tears less than 50 % in thickness combined with the use of repair (conversion to full thickness or transtendinous) for symptomatic patients with tears greater than 50 % in thickness. These researchers examined functional and radiographic outcomes following surgical implantation of a collagen-based bio-inductive implant for the treatment of PTRCT. Patients with PTRCT who underwent implantation of a collagen-based bio-inductive implant over the bursal surface of the rotator cuff were identified. Patients who had an implant placed to augment a standard full thickness repair were excluded from analysis. These investigators administered questionnaires to patients with a minimum of 6 months since surgery and collected patient data including demographics, pre-operative and post-operative ASES scores, pre-operative and post-operative VAS pain scores, complications, and satisfaction level; MRI was obtained as needed, at 6 months, or at 12 months post-operatively, and

tendon thickness was measured on coronal views. A total of 30 patients were identified who met the inclusion and exclusion criteria; 24 patients (80 % response rate) were available for follow-up; 19 men and 5 women. Mean age and BMI were 54.5 ± 11.6 years and 28.6 ± 5.9 kg/m2, respectively. Mean tear thickness as measured intra-operatively was 56.6 %. There were 16 (66.7 %) articular-sided, 5 (20.8 %) bursal-sided, and 3 (12.5 %) intra-substance tears. Mean survey follow-up time was 19.1 months. Mean ASES scores increased significantly from 45.6 pre-operatively to 68.1 post-operatively ($p = 0.001$). Mean VAS pain scores decreased significantly from 8.3 pre-operatively to 3.8 post-operatively ($p < 0.001$). Mean patient satisfaction level was 7.5; 10 patients had both a pre-operative and post-operative MRI available for comparison. Tendon thickness at the tear site increased significantly from 5.7 mm pre-operatively to 6.5 mm at mean 9.9 months follow-up ($p = 0.007$). There were no implant-related complications; 1 patient suffered a traumatic re-tear 4 months post-operatively. The authors concluded that highly porous collagen-based bio-inductive implants were safe and effective for reducing pain and improving shoulder function in patients with PTRCT of approximately 50 %, with radiographic evidence of new tissue formation. Moreover, these researchers stated that RCTs are needed to evaluate efficacy relative to debridement and standard repair techniques.

Schlegel and associates (2021) stated that current surgical options for partial-thickness tears (e.g., takedown and repair, in-situ repair) are limited by the degenerative nature of the underlying tendon and may require extensive intervention that can alter the anatomic footprint. The complexity of available techniques to address these issues led to the development of a resorbable collagen implant, which can be used to create a bio-inductive repair of partial-thickness tears. In a prospective, multi-center study, these researchers prospectively enrolled 33 patients with chronic, degenerative, intermediate-grade (n = 12) or high-grade (n = 21) partial-thickness tears (11 articular, 10 bursal, 4 intra-substance, and 8 hybrid) of the supraspinatus tendon. Following arthroscopic subacromial decompression without a traditional rotator cuff repair, a bio-inductive implant was secured over the bursal surface of the tendon. Clinical outcomes were evaluated using ASES and Constant-Murley scores (CMS) pre-operatively and at 3 months, 1 year, and 2 years post-operatively; MRI was performed to examine post-operative tendon healing and thickness at the original tear site. At 2-year follow-up, mean

ASES and CMS scores improved both clinically and statistically at 1 and 2 years, compared with baseline, for intermediate- and high-grade tears. There was MRI evidence of new tissue fill-in within the original baseline tear in 100 % of the intermediate-grade tears and 95 % of the high-grade tears. In 90.9 % of intermediate-grade tears and 84.2 % of high-grade tears, this new tissue fill-in represented at least an additional 50 % of the volume of the initial lesion. From baseline to 2-year follow-up, the mean tendon thickness increased by 1.2 mm (SD, 1.3; p = 0.012) and 1.8 mm (SD, 2.2; p = 0.003) in the intermediate- and high-grade tears, respectively. Analysis of tear grade and location revealed no statistically significant difference in the change in mean tendon thickness at any time-point; 1 patient with a high-grade articular lesion showed progression to a full-thickness tear; however, subject was non-compliant and the injury occurred while shoveling snow 1 month after surgery. Neither tear location nor treatment of bicep pathology affected the ASES or CMS scores at any follow-up point. No serious AEs related to the implant were reported. The authors concluded that final results from this 2-year prospective study indicated that the use of this resorbable bovine collagen-based implant for isolated bio-inductive repair of intermediate- and high-grade PTRCTs of the supraspinatus was safe and effective, regardless of tear grade and location. Level of Evidence = IV.

McIntyre et al (2021) prospectively collected safety and effectiveness data in a large group of patients undergoing arthroscopic repair of full-thickness rotator cuff tears augmented with a resorbable bio-inductive bovine collagen implant designed to promote healing. A total of 17 centers across the U.S. enrolled patients in an institutional review board (IRB)-approved registry to collect outcomes data on the implant. Patients undergoing surgical management of full-thickness rotator cuff tears augmented with the implant were enrolled. Inclusion criteria were age of greater than or equal to 21 years, willingness to participate and the ability to read and speak English. Exclusion criteria included hypersensitivity to bovine-derived products. Patients were assessed before and after surgery at up to 1 year with outcomes including the SANE, VR-12 PCS, ASES, and WORC outcome measures. Ad-hoc analyses were carried out to compare these outcomes at all time-points depending on tear size (small/medium versus large/massive). Serious complications were collected. Of 210 patients enrolled, 192 had 1-year follow-up data available. The patients experienced statistically significant improvement

between baseline and 1 year for mean SANE, VR-12 PCS, ASES, and WORC scores (40.0 to 82.0, 33.5 to 47.3, 46.2 to 87.8, and 36.2 to 81.0, respectively; $p < 0.001$ for all results). Ad-hoc analysis demonstrated that similar results were obtained at 1 year regardless of tear size. A total of 20 patients (10.4 %) experienced serious complications (10.4 %), including revision surgery (n = 18), proximal humerus fracture/partial subscapularis tear resulting from multiple falls (n = 1), and adhesive capsulitis (n = 1). The authors concluded that the safety and effectiveness of a bio-inductive implant in the surgical management of full-thickness rotator cuff tears at 1 year was shown in this study. Implant effectiveness appeared to be comparable regardless of the underlying tear size. Level of Evidence = IV.

The authors stated that drawbacks of the study included a level IV design with inherent selection bias and lack of a control population. The use of the implant was at the discretion of the practicing surgeon without evidence-based indications for use and no comparison group available for analysis. As such, no conclusions regarding the use of the implant compared to current techniques with or without augmentation can be drawn. They stated that future high-level comparison studies are needed to outline indications and clinical situations where the implant will add value to healing rates and patient outcomes. The registry design of this study was intended as a real-world evidence capture activity, reflecting the typical clinical practices of surgeons treating patients with rotator cuff tears. As such, patients were followed out to 1 year, and additional outcomes of interest (e.g., ROM, strength, radiographic data) were not formally collected, all of which can be viewed as limitations. These findings also reflected different surgeon preferences regarding rehabilitation protocols and the optimal time to return to certain activities, potentially adding unforeseen bias. Finally, the ad-hoc analysis comparing PROs based on tear size at baseline was not sufficiently powered; thus, should only be considered hypothesis-generating.

Bushnell et al (2021) noted that surgical treatment of partial-thickness rotator cuff tears remains challenging and controversial, with several traditional options including debridement with acromioplasty, trans-tendon or in-situ repair, and take-down and repair. A resorbable bio-inductive bovine collagen implant has shown promise as an alternative treatment option for partial-thickness tears. In a case-series study, these

researchers analyzed data from a registry to further establish that the implant contributes to improved patient-reported outcome (PRO) scores across a large number of patients treated for partial-thickness rotator cuff tears. A total of 19 centers in the U.S. enrolled patients 21 years of age or older with partial-thickness tears of the rotator cuff in a comprehensive prospective multicenter registry. PRO scores were recorded pre-operatively and post-operatively at 2 and 6 weeks, 3 and 6 months, and 1 year: ASES, SANE, VR-12 Health Survey (physical and mental component scores), and WORC scores. Revisions were reported throughout the study. The registry included 272 patients with partial-thickness tears (49 grade 1 tears, 101 grade 2 tears, and 122 grade 3 tears), 241 who underwent isolated bio-inductive repair (IBR; collagen implant placed after bursectomy without a traditional rotator cuff repair), and 31 who had takedown and repair with bio-inductive augmentation. Patients experienced statistically significant and sustained improvement from baseline for all PRO scores beginning at 3 months. Among patients with grade greater than or equal to 2 tears, those with take-down and repair had significantly inferior scores at 2 and 6 weeks for most PRO scores as compared with those who underwent IBR, but the difference was no longer significant at 1 year for all but the physical component score of the VR-12 Health Survey. There were 11 revisions, which occurred at a mean ± SD of 188.7 ± 88.0 days after the index surgery; and there were no infections. The authors concluded that this registry analysis further established across a large data set that this resorbable bio-inductive bovine collagen implant improved PROs in all grades of partial-thickness tears, whether used as IBR or in conjunction with take-down and repair. These investigators stated that IBR may offer improved early clinical outcomes (less than or equal to 6 weeks) and comparable outcomes at 1 year when compared with a more invasive "take-down and repair" approach. Level of Evidence = IV.

The authors stated that this study had several drawbacks. There was no control group to compare outcomes with existing therapeutic interventions for this tear type. Although the lack of a well-defined standard of care for partial-thickness tears would make choosing the optimal control group difficult, it would nonetheless be valuable for isolating the effects of the implant. For example, it would be of interest to compare outcomes for take-down and repair with versus without the implant or to compare the isolated bio-inductive repair (IBR) technique versus subacromial

decompression alone. There were also limitations from the registry design, which was intended as a real-world evidence capture activity. It employed minimal inclusion and exclusion criteria. This was performed to reflect the patients encountered in real-world clinical practices but could play an unforeseen role in biasing the results. Another registry limitation was the lack of full-score data capture at every time-point, attributed to incomplete form fulfillment by various patients. Furthermore, this study did not call for the recording of tear location, tear type, or the number of tears, which may influence outcomes in partial-thickness tears, and MRI assessments were not performed to correlate tendon healing/thickness with PROs. In additional, the study was not powered for assessing specific outcomes, such as a single score or a single risk factor. Instead, data were extracted and analyzed in an ad-hoc manner. Given the relatively large size of this patient population, it was thought that this did not diminish the ability of the analysis to gauge the implant's ability to produce clinically meaningful improvements in PROs over time. Findings from subgroup analyses, however, likely require a structured and controlled design to reduce the potential for bias and cofounders. Also, there was a chance of bias introduced by the surgeons having the ability to choose whether or not to take down the cuff and do a repair versus doing an IBR with the implant alone. These researchers did not test for immunologic responses in those cases of inflammation or stiffness after surgery, wherein a potential negative "reaction" or "rejection" might have been involved. Prior histologic and clinical studies, however, have not shown any sort of immunologic rejection, so analysis of it was not part of the study protocol. Finally, this trial provided only 1-year follow-up data, which was reflective of the typical follow-up period in everyday clinical practice but may lack the scope of a 2-year window to evaluate outcomes.

In a triple-blinded, multi-center, randomized controlled trial (n = 57), Ruiz Iban et al (2022) reported interim (12 months) results indicating that repair of full-thickness rotator cuff tears augmented with the REGENETEN Implant led to significantly lower re-tear rates compared to standard repair alone. These are interim results, the final results are planned to be presented at the International Society of Arthroscopy, Knee Surgery and Orthopedic Sports Medicine (ISAKOS) in 2023.

In a poster presentation on “Early economic analysis of the REGENETEN Bioinductive Implant in the treatment of full-thickness rotator cuff tears”, Nherera (2022) concluded that in the augmentation of rotator cuff FT tear repair, the REGENETEN Implant was cost-effective from a payer’s perspective and cost-saving from a societal perspective in patients with a mean age of 58 years. Patients at higher risk of re-tear or with larger tear sizes had the greatest economic benefit. Moreover, these researchers stated that direct comparative evidence is needed to validate this early economic evaluation, which used non-comparative data. The mean age was chosen to mirror the age of most patients included in the studies upon which the clinical data in this study was based.

In a prospective study, Camacho-Chacon et al (2022) examined the clinical outcomes, MRI imaging and histological characteristics of biopsy samples of the tendon from patients in whom rotator cuff repair was previously performed with a bioinductive type I bovine collagen implants. This trial included 30 patients with partial or complete rotator cuff tears who underwent arthroscopic repair and augmentation with a resorbable type I bovine collagen implant. Pre-operatively and at 6 and 12 months after surgery, the visual analog scale (VAS), American Shoulder and Elbow Surgeons Standardized Shoulder (ASES) and Constant-Murley scores (CMS) were examined and an MRI study was performed. At 6 months, biopsies of the resulting tissue were obtained and examined histologically. Patients experienced statistically significant and sustained improvement from baseline for all scores and the mean tendon thickness increased by 1.84 mm. Magnetic resonance imaging (MRI) evidence of complete healing was found in 27 patients and a considerable reduction in defect size, greater than 50 %, was shown in 3. In all samples obtained, the new tissue generated had the histological appearance of a tendon; and was indistinguishable from the native tendon. There was no evidence of any remaining collagen implant. The authors concluded that biopsies of tissue formed from bioinductive type I bovine collagen implants showed, 6 months after surgery, the generation of a neotendon indistinguishable from the native one. Both histology and MRI imaging revealed complete integration of the implant and absence of inflammatory or foreign body reactions. The clinical parameters, thickness and MRI signal of the tendon improved significantly at 6 months, regardless of the type and size of the tear, and remained unchanged until 12 months.
Level of Evidence = IV.

The authors stated that this study had several drawbacks, especially a short follow-up (6 months), a small sample size (n = 30) and the lack of a control group. The patients were not randomized, nor was their treatment blind. In addition, a possible drawback, the location of the biopsy areas, was avoided by using ultrasound (US) to determine the donor site. The US visibility of the PEEK staples used for implant fixation offered these researchers great security for the selection of the extraction area.

Bushnell et al (2022) stated that full-thickness rotator cuff tears (FTRCTs) represent a common shoulder injury that, if untreated, could progress in size, become increasingly painful, and inhibit function. These lesions are often surgically repaired, with double-row arthroscopic repair often preferred for larger tears. Biological augmentation technologies have been developed to improve rates of post-operative radiographic re-tear and enhance patient-reported outcomes following surgical FTRCT repair. In a prospective, multi-center study, these researchers sought to confirm that augmented repair with a bioinductive bovine collagen implant would result in favorable re-tear rates and patient outcomes with follow-up to 2 years. Of 115 adult patients enrolled in this trial, 66 (57.4 %) had medium (1 to 3 cm) tears and 49 (42.6 %) had large (3 to 5 cm) tears. MRI and patient-reported outcomes (ASES Assessment Form and CMS) were performed and recorded at baseline, 3 months, 1 year, and 2 years. Mean duration of follow-up was 2.1 years (range of 1.5 to 2.9 years). Between baseline and 2-year follow-up, mean total thickness of the supraspinatus tendon increased by 12.5 % for medium tears and by 17.1 % for large tears. Radiographic re-tear was noted in 7 of 61 available patients (11.5 %) with medium tears, and in 14 of 40 patients (35.0 %) with large tears. In both groups, these tears primarily occurred before the 3-month follow-up visit (13 of 21 [61.9 %]). Radiographic re-tear with the supplemented double-row (DR) repair technique was 13.2 % overall (12 of 91 DR patients, 11.3 % for medium tears and 15.8 % for large tears). The minimal clinically important difference was achieved by more than 90 % of patients with both medium and large tears for both ASES and CMS. There were 2 serious adverse events (AEs) classified by the treating surgeon as being possibly related to the device and/or procedure (1 case of swelling/drainage and 1 case of intermittent pain); 9 patients (7.8 %; 4 medium tears and 5 large tears) required re-operation of the index rotator cuff surgery. The authors concluded that final 2-year data from this study confirmed that using this implant in augmentation of arthroscopic double-

row repair of FTRCTs provided favorable rates of radiographic re-tear and substantial functional recovery. The relative safety of the device was also further supported. Level of Evidence = IV.

McIntyre et al (2023) examined if the use of resorbable bio-inductive collagen implant (RBI) in addition to conventional rotator cuff repair (conventional RCR) is cost-effective when compared to conventional RCR alone, in the treatment of full-thickness (FT) rotator cuff tears. These researchers developed a decision analytic model to compare the expected incremental cost and clinical consequences for a cohort of patients with FT rotator cuff tears. The probabilities for healing or failure to heal (re-tear) were estimated from the published literature. Implant and healthcare costs were estimated from a payor's perspective in 2021 U.S. prices. An additional analysis included indirect cost estimations (e.g., productivity losses). Sensitivity analyses examined the effect of tear size, as well as the impact of risk factors. The base case analysis demonstrated that resorbable bio-inductive collagen implant + conventional RCR results in incremental costs of $232,468 and an additional 18 healed RCTs per 100 treated patients over 1 year. The estimated incremental cost-effectiveness ratio (ICER) was $13,061/healed RCT compared to conventional RCR alone. When return-to-work was included in the model, RBI + conventional RCR was found to be cost-saving. Cost-effectiveness improved with tear size with the largest benefit observed in massive tears compared to large tears, as well as patients at higher risk of re-tearing. The authors concluded that this economic analysis demonstrated that RBI + conventional RCR delivered improved healing rates at a marginal increase in costs when compared to conventional RCR alone and thus was cost-effective in this patient population. Considering indirect costs, RBI + conventional RCR resulted in lower costs compared to conventional RCR alone; therefore, it was deemed to be cost-saving.

The authors stated that this study had several drawbacks. First, these investigators made some simplifying assumptions, such as restricting the analysis to 12 months post-surgery. They acknowledged that in practice, re-tear may appear after 12 months. However, if this was the case, these researchers believed this analysis would be conservative, as it may bias the results against RBI, which was expected to lower the incidence of long-term failure. These investigators are awaiting long-term data to

validate this hypothesis. Second, these researchers assumed that risk factors that influence re-tear or failure rates are independent. Patients often present with multiple risk factors, such as diabetes and hypertension. Third, these investigators used data from non-comparative studies for RBI; however, they were aware of a RCT comparing RBI to conventional RCR, which has reported promising interim results showing 86 % reduction in non-healing rates in favor of RBI in 59 patients presented as a conference proceeding. The economic evaluation employed non-comparative single-arm studies to estimate the impact of RBI when compared to conventional RCR, showing a 66 % reduction in non-healing RCT which was less than what the interim results were showing, suggesting their analysis used conservative estimates of the potential impact of RBI. Fourth, the studies for RBI, which provided evidence on return-to-work did not report by type of work (sedentary or manual). This information should be collected and be reported separately in future studies. The authors suggested updating their study once further comparative evidence becomes available. Furthermore, additional comparative investigation is needed to address return-to-work, segmented by occupational type (e.g., sedentary or manual).

Ceramic-Based Products (e.g., AttraX Putty, Beta Tri-Calcium Phosphate (b-TCP) and Bioglass (45S5)) for the Enhancement of Bone Healing and/or Fusion

Parker and Malham (2017) stated that recombinant human bone morphogenetic protein-2 (rhBMP-2) generally provides high rates of clinical improvement and fusion; however, rhBMP-2 has been associated with adverse effects. Recently, a beta tricalcium phosphate (β-TCP) bone substitute has been developed. In a prospective study, these researchers examined the fusion rates and clinical outcomes of patients treated with β-TCP compared to rhBMP-2. A total of 135 consecutive patients who underwent lateral lumbar interbody fusion with β-TCP (n = 25) or rhBMP-2 (n = 110) in the interbody cage were included in the study. The 25 β-TCP patients were a group of consecutive patients from numbers 46 to 70. Clinical outcomes included back and leg pain, ODI, and SF-36 physical and mental component scores (PCS and MCS). CT scans were carried out at 6, 12, 18, and 24 months until confirmation of solid interbody fusion, with no further scans conducted once fusion was achieved. Targeted CT at the operative level(s) was carried out to reduce

radiation exposure. At 24 months there was no significant difference between clinical outcomes of the β-TCP or rhBMP-2 patients, with improvements in back pain (46 % and 49 %; p = 0.98), leg pain (31 % and 52 %; p = 0.14), ODI (38 % and 41 %; p = 0.81), SF-36 PCS (37 % and 38 %; p = 0.87), and SF-36 MCS (8 % and 8 %; p = 0.93). The fusion rate was significantly higher for rhBMP-2 with 96 % compared to 80 % for β-TCP (p = 0.01). Separating patients into those with a stand-alone cage and those with supplemental posterior instrumentation, there was no significant difference between instrumented fusion rates of the β-TCP and rhBMP-2 patients at 6 (p = 0.44), 12 (p = 0.49), 18 (p = 0.31) or 24 (p = 0.14) months. For stand-alone patients there was a significant difference at 6 (p = 0.01), 12 (p = 0.008) and 18 months (p = 0.004) with higher fusion rates in the rhBMP-2 group; however, by 24 months this was not significant (p = 0.18). The authors concluded that comparable clinical outcomes and complication rates suggested that β-TCP was a viable alternative to rhBMP-2. The difference in fusion rates for the stand-alone patients suggested that β-TCP may require supplemental posterior instrumentation to enhance fusion.

Lehr et al (2020a) noted that spinal fusion with autologous bone graft is a frequently performed surgical treatment. Several drawbacks of autografting have driven the development of numerous alternatives including synthetic ceramics. However, clinical evidence for the stand-alone use of these materials is limited. In a randomized, intra-patient controlled, multi-center study, these researchers determined non-inferiority of a biphasic calcium-phosphate (AttraX putty) as a bone graft substitute for autograft in instrumented posterolateral fusion (PLF). This trial included 100 non-traumatic adults who underwent a primary, single- or multi-level, thoracolumbar, instrumented PLF. After instrumentation and preparation for grafting, the randomized allocation side of AttraX Putty was disclosed. Autograft was applied to the contralateral side of the fusion trajectory, so each patient served as his/her own control. For the primary efficacy outcome, PLF was assessed at 1-year follow-up on CT scans. Each segment and side were scored as fused, doubtful fusion, or non-union. After correction for multi-level fusions, resulting in a single score per side, the fusion performance of AttraX Putty was tested with a non-inferiority margin of 15 % using a 90 % CI. There were 49 men and 51 women with a mean age of 55.4 ± 12.0 (range of 27 to 79) years; 2/3 of the patients underwent a single-level fusion and 62 % an additional

interbody fusion procedure. The primary analysis was based on 87 patients, including 146 instrumented segments. The fusion rate of AttraX Putty was 55 % versus 52 % at the autograft side, with an overall fusion rate of 71 %. The 90 % CI around the difference in fusion performance excluded the non-inferiority margin (difference = 2.3 %, 90 % CI: -9.1 % to +13.7 %). The authors concluded that the findings of this non-inferiority trial supported the use of AttraX Putty as a stand-alone bone graft substitute for autograft in instrumented thoracolumbar PLF. Level of Evidence = I.

Lehr et al (2020b) provided 2-year clinical and radiographic follow-up of a double-blind, multi-center, randomized, intra-patient controlled, non-inferiority study comparing AttraX Putty with autograft in instrumented PLF surgery. These researchers compared PLF rates between 1 and 2 years of follow-up and between graft types; and examined the role of bone grafting based on the location of the PLF mass. A total of 100 adult patients underwent a primary, single- or multi-level, thoracolumbar PLF. After instrumentation and preparation for grafting, the randomized allocation side of AttraX Putty was disclosed. The contralateral posterolateral gutters were grafted with autograft. At 1-year follow-up, and in case of no fusion at 2 years, the fusion status of both sides of each segment was blindly assessed on CT scans. Inter-transverse and facet fusion were scored separately. Difference in fusion rates after 1 and 2 years and between grafts were analyzed with a Generalized Estimating Equations (GEE) model ($p < 0.05$). The 2-year PLF rate (66 patients) was 70 % at the AttraX Putty and 68 % at the autograft side, compared to 55 % and 52 % after 1 year (87 patients). GEE analysis demonstrated a significant increase for both conditions (OR of 2.0; 95 % CI: 1.5 to 2.7, $p < 0.001$), but no difference between the grafts ($p = 0.595$). Ongoing bone formation was only observed between the facet joints. The authors concluded that this intra-patient, controlled trial demonstrated a significant increase in PLF rate between 1 and 2 years after instrumented thoracolumbar fusion, but no difference between AttraX Putty and autograft. Based on the location of the PLF mass, this increase was most likely the result of immobilization instead of grafting. Level of Evidence = I.

Plantz et al (2021) stated that historically, iliac crest bone autograft has been considered the gold standard bone graft substitute for spinal fusion; however, the significant morbidity associated with harvesting procedures has influenced decision-making and practice patterns. To minimize these side effects, many clinicians have pursued the use of bone graft extenders to minimize the amount of autograft needed for fusion in certain applications. Synthetic materials, including a variety of ceramic compounds, are a class that has been studied extensively as bone graft extenders. These have been used in combination with a wide array of other biomaterials and examined in a variety of different spine fusion procedures. These investigators noted that ceramics have historically been used as bone graft extenders, often in combination with autograft materials. In a recent systematic review, Nickoli and Hsu (2014) examined the effectiveness of ceramic-based bone grafts in lumbar spinal fusion. Most commonly, ceramics were combined with autograft materials -- local autograft, ICBG, and/or bone marrow aspirate (BMA). It was demonstrated that the overall fusion rate for ceramic-based grafts was 86 %, and that the specific type of ceramic used did not significantly influence fusion rate. In general, the highest fusion rates were achieved when ceramics were combined with local autograft relative to all other adjunct materials. This review evaluated 30 studies in total; 3 level I studies, 9 level II studies, 3 level III studies, and 14 level IV studies. More importantly, ceramics were used as bone graft extenders (e.g., in combination with another adjunct) in all but 2 of the included studies. The average fusion rate in studies using ceramic with local autograft was 89.8 % (range of 52.6 % to 100 %), and with ceramic and ICBG it was 88.5 % (range of 78.9 % to 100 %). The fusion rate for ceramics and BMA combinations was demonstrated to be lower, with an average fusion rate of 72.6 % (range of 4.5 % to 95.5 %). When ceramics were combined with both local autograft and BMA, the fusion rate was greater -- 91.3 % (range of 85 % to 95 %), relative to ceramics and BMA alone. Notably, in the 2 studies that examined ceramic alone as a bone graft substitute, the average fusion rate was 81.5 % (range of 77.6 % to 90 %). This systematic review also compared the fusion rates between ceramic-based grafts used in different surgical approaches, namely PLF and circumferential fusion combining interbody fusion with posterior instrumentation -- anterior lumbar interbody fusion (n = 4 studies), posterior lumbar interbody fusion (PLIF) (n = 6 studies), and transforaminal lumbar interbody fusion (n = 2 studies). On average, there

were no significant differences in fusion rate between PLF alone versus circumferential fusion. The authors concluded that despite the heterogeneity of the included studies, the results were suggestive that ceramic materials were effective as a bone graft extender in a variety of applications when combined with an osteo-inductive component, such as autograft.

Celling Biosciences Solum IV Allograft

Solum IV is a resorbable porous calcium phosphate (CaP) bone void filler. It is an osteo-conductive implant with a multi-dimensional porosity similar to human cancellous bone and acts as a scaffold for the in-growth of new bone. Using a naturally derived biopolymer, Solum IV, when combined with bone marrow aspirate, becomes a moldable, 3D structure capable of conforming to the full geometry of the recipient site creating an in-vivo bioreactor for bone growth. Solum IV is composed of porous hydroxyapatite (HAp) granules and a type A porcine gelatin-based carrier. The granules consist of macro-porous ceramic granules composed of greater than 95 % HAp. The CaP granules are presented as ground particles. The macro-porous structure of Solum IV provides a resorbable osteo-conductive scaffold. The gelatin-based carrier is presented as ground freeze-dried particles. The product forms a cohesive and adhesive dough with a puffy-like consistency upon hydration that allows the shape of the implant to conform to the defect maximizing direct contact with viable host bone. Solum IV is substantially equivalent to Bioactive graft (nano-structured hydroxyapatite material).

Healos Bone Graft Replacement / Healos Sponge

Ploumis et al (2010) examined the safety and effectiveness of Healos graft carrier with bone marrow aspirate (BMA) and local autograft compared with the results of allograft in patients with lumbar degenerative scoliosis undergoing posterolateral fusion (PLF). A total of 28 patients with degenerative scoliosis underwent posterolateral instrumented fusion and decompression. Patients were grouped according to the graft used. Group A consisted of 12 cases in which the authors used a Healos graft carrier, BMA, and local autograft, and Group B consisted of 16 cases in which the authors used cancellous allograft and local autograft. Patients were followed for a minimum of 2 years post-operatively in terms of pain

(VAS), function (ODI), curve magnitude (Cobb angle), and fusion status (plain and dynamic radiographs). The 2 groups did not differ statistically significantly ($p > 0.05$) in age, sex, smoking habits, magnitude of pre-operative VAS, ODI score, Cobb angle, or number of levels requiring decompression and fusion. The groups had similar ($p > 0.05$) results in terms of pain, function, curve progression, and fusion rates at the 2-year follow-up examination. Radiographic fusion was achieved in all but 2 cases, 1 in each group, in which the patients were asymptomatic. Patients in the allograft group (Group B) showed evidence of fusion earlier than in the Healos group ($p < 0.05$). No toxicity from Healos graft was recorded. The authors concluded that the combination of Healos hydroxyapatite sponge and BMA plus local allograft had significantly slower fusion rates; but equal clinical outcomes compared with cancellous allograft plus local autograft when used for PLF in patients with degenerative lumbar scoliosis.

Kunakornsawat et al (2013) stated that autologous bone harvested from the iliac crest is the gold standard for lumbar spinal fusion; however, post-operative donor site pain and morbidity have been reported. Local bone graft is insufficient and contains some soft-tissue attachment. In a retrospective, radiographic outcome study, these researchers examined the radiographic fusion rate of Healos with BMA versus autologous bone graft in the same patients PLF. They examined radiographic fusion rate from plain radiographs in 55 patients indicated for PLF in Lerdsin General Hospital between April 2005 and December 2006. Subjects were implanted with Healos collagen-hydroxyapatite sponge with BMA and local bone graft (LBG) on each side of PLF. A total of 27 patients were included in the present study according to the authors' inclusion criteria. Plain radiographs were collected; and radiographic fusion rate was assessed for at least 2 years follow-up. The 2-year post-operative radiographic fusion rate was 29.63 % (8/27) in Healos/BMA group and 62.96 % (17/27) in LBG group. At 3-year follow-up, radiographic fusion rate of 36.84 % (7/19) was achieved in the Healos/BMA group and 78.93 % (15/19) in the LBG group. The authors concluded that Healos collagen-hydroxyapatite sponge with BMA had lower radiographic fusion rate when compared to LBG in PLF at 2 years after the surgery. Healos/BMA increased fusion rate with time but remained lower than LBG group at 3- and 4-year follow-ups.

Plantz et al (2021) note that historically, iliac crest bone autograft has been considered the gold standard bone graft substitute for spinal fusion; however, the significant morbidity associated with harvesting procedures has influenced decision-making and practice patterns. To minimize these side effects, many clinicians have pursued the use of bone graft extenders to minimize the amount of autograft needed for fusion in certain applications. Synthetic materials are a class that has been studied extensively as bone graft extenders. These have been used in combination with a wide array of other biomaterials and examined in a variety of different spine fusion procedures. These investigators stated that although there are a wide range of polymer-based composites for use in spine fusion, the most robust evidence exists for HA + collagen composites (Healos) combined with autologous materials, such as BMA. Notably, combining Healos with BMA eliminated the need for harvesting ICBG in some cohorts. This has the potential to reduce donor-site morbidity and other complications associated with ICBG harvesting.

Subchondroplasty

In a prospective study, Pasqualotto et al (2019) examined the effectiveness in terms of pain relief and functional improvement of subchondroplasty in the treatment of OA-related bone marrow lesions (BMLs) of the knee. The trial included first 15 consecutive patients who underwent subchondroplasty for the treatment of chronic degenerative BMLs in which previous conservative treatment have failed. Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) scores, KOOS, and VAS pain scores were obtained pre-operatively and at 1, 6, and 12 months of follow-up. WOMAC scores significantly improved from 39.7 ± 20.2 before surgery to 26.8 ± 16.1 at the 1-month follow-up ($p = 0.045$). A further significant improvement to 15.5 ± 12.7 ($p = 0.02$) and to 8.6 ± 3.1 ($p < 0.01$) was obtained both at 6-month and at 1-year follow-up. KOOS scores improved significantly from 47.5 ± 16.6 before surgery to 65.4 ± 14.9 at 1 month ($p = 0.013$) and to 80.4 ± 15.1 at 6-month follow-up ($p = 0.01$). A further improvement to 85.6 ± 15.1 was recorded 1 year post-operatively, although not significantly. VAS score showed a significant improvement from 55.8 ± 20.5 pre-operatively to 36.2 ± 16.9 at 1 month ($p = 0.008$) and to 18.2 ± 17.3 at 6-month follow-up ($p = 0.005$). This further improved to 12.8 ± 17.9 at 1-year follow-up, although not significantly. The authors concluded that subchondroplasty

represented a safe and valid surgical option in the treatment of OA-related BMLs of the knee, providing an improvement in terms of pain relief and functional recovery. Moreover, these researchers stated that RCTs with broader cohort and longer follow-up are needed to examine how long these improvements may last. Level of Evidence = IV.

The authors stated that this study had several drawbacks. First, the follow-up period was limited to 1 year whereas a study with a longer follow-up would be able to better assess the durability of the improvement. Second, arthroscopic treatment of intra-articular pathology could represent an important limitation since the relative contribution of each procedure could not be evaluated. However, it should be noted that several studies in patients with knee OA showed that arthroscopic debridement was unable to provide a durable relief of OA symptoms. Third, these researchers acknowledged that these findings referred to a small, non-randomized cohort without a control group.

In a systematic review, Di Matteo and colleagues (2021) examined the available clinical evidence regarding the safety and effectiveness of knee intra-osseous injections for the treatment of bone marrow lesions in patients affected by knee OA. These researchers performed a literature search on PubMed, Embase, and Google Scholar databases in January 2020. The following inclusion criteria were adopted: studies of any level of evidence dealing with subchondral injection of bone substitute materials and/or biologic agents; studies with minimum 5 patients treated; and studies with at least 6 months' follow-up evaluation. All relevant data concerning clinical outcomes, AEs, and rate of conversion to arthroplasty were extracted. A total of 12 studies were identified: 7 dealt with calcium phosphate administration, 3 with PRP, and 2 with bone marrow concentrate (BMC) injection. Only 2 studies were RCTs, whereas 6 studies were prospective and the remaining 4 were retrospective. Studies included a total of 459 patients treated with intra-osseous injections. Overall, only a few patients experienced AEs and clinical improvement was documented in the majority of trial. The authors concluded that knee intra-osseous injections are a minimally invasive and safe procedure to address subchondral bone damage in osteoarthritic patients. They are able to provide beneficial effects at short-term evaluation. Moreover, these researchers stated that more high-quality evidence is needed to confirm their potential and to identify the best

product to adopt in clinical practice. These investigators stated that the lack of any comparative evaluation versus subchondral drilling alone was the main limitation of the available evidence; subchondroplasty and tricalcium phosphate were key words of this study.

Nairn and associates (2021) noted that subchondroplasty is a novel minimally invasive procedure for painful subchondral BMLs. In a systematic review, these researchers examined the clinical outcomes of the subchondroplasty for the treatment of BMLs. The hypothesis tested was that patients would experience improvements in pain and functional outcomes following subchondroplasty. Medline, Embase, Web of Science, and Clinicaltrials.gov were searched from database inception to June 10, 2020 for all clinical studies that discussed subchondroplasty. Two reviewers independently screened 45 unique results and 17 studies were included in the final analysis. Data were collected regarding patient demographics, indications, pain, functional scores, conversion to total knee arthroplasty (TKA), and complications of the procedure. All but 1 study were level IV evidence; the mean methodological index for non-randomized studies (MINORS) score was 9 ± 2. There were 756 patients included, 45.1 % were women, and the mean age was 54 years (range of 20 to 85); 13 studies examined the effect subchondroplasty to the knee, while 4 examined the impact on the foot and ankle. Median length of follow-up was 12 months. The most common indication for subchondroplasty was joint pain with corresponding BML. Major contraindications to subchondroplasty included severe OA, joint instability, and malalignment. Mean pain score on VAS before subchondroplasty was 7.8 ± 0.6; but decreased to 3.4 ± 0.7 post-operatively. All studies examining functional scores reported improvement following subchondroplasty (IKDC 31.7 ± 1.9-54.0 ± 4.2 and KOOS 38.1 ± 0.6-70.0 ± 4.1). There were consistently high levels of patient satisfaction; 87 ± 8 % of patients would be willing to undergo the procedure again. Seven cases of complications were reported, most seriously osteomyelitis and avascular necrosis. Conversion to TKA ranged from 12.5 to 30 % with length of follow-up ranging from 10 months to 7 years. The authors concluded that existing low-quality studies showed subchondroplasty to benefit patients with BMLs via reduction in pain and improvement in function, along with a high degree of satisfaction following the procedure. The low short-to-medium term conversion rate to TKA suggested that subchondroplasty may play a role in delaying more

invasive and expensive procedures in patients with BMLs. These researchers stated that subchondroplasty is a novel procedure that has promising initial findings; however, further high-quality, comparative studies with long-term follow-up are needed to better understand the outcomes of the procedure before the procedure should be incorporated into routine clinical practice. Level of Evidence = IV.

The authors stated that this systematic review had several drawbacks. First, the results were limited by the relatively low quality of evidence of included studies. All but 1 were level IV studies; therefore, there was significant risk of bias within the presented results. Two of these studies were single-case reports and 4 had 5 or fewer patients. The heterogeneity of included studies in design and reported outcomes limited the ability of this review to draw meaningful conclusions from the results. Furthermore, the median length of follow-up of included studies was 12 months, limiting the understanding of long-term outcomes of the procedure. Five of the included studies disclosed some conflict of interest, most commonly that 1 or more authors were consultants for Zimmer Knee Creations, the developers of the Subchondroplasty procedure. This is especially relevant as most of the studies with large sample sizes had some industry conflict, making up 37 % of the total included sample. Finally, the majority of the studies available were centralized to the U.S. and were limited to English. It was also concerning that only 2 of the 9 registered trials were comparative in nature, although 1 prospective study does plan to enroll 1,000 patients that would aid in further elucidating the safety of the procedure. Future studies should also include long-term follow-up to better characterize long-term pain relief, conversion to TKA, and complications to guide surgeons in selecting patients who are good candidates for the procedure, better inform patient expectations, and direct post-operative follow-up.

In a systematic review, Betzler and co-workers (2021) examined clinical outcomes following intra-osseous injection for the treatment of knee OA. These researchers carried out a systematic search of the PubMed, Embase, and CINAHL databases in November 2020. The search workflow was in adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The following inclusion criteria were adopted: clinical trials of any level of evidence,

reporting clinical outcomes following intra-osseous injections of bone substitutes or biologic agents, and MSCs or PRP into the knee as treatment modalities for OA. Duplicate data and articles not written in English were excluded from this review. A total of 6 studies were identified and included in this review, with a total of 167 patients – 2 studies used subchondroplasty CaP injections, while 4 studies used intra-osseous injections of PRP. Two studies provided level II evidence, 2 studies provided level III evidence, and a further 2 provided level IV evidence; 5 out of 6 studies reported data using the VAS, 4 studies used the KOOS, while 3 studies used the WOMAC. Clinical improvements in pain and functionality were documented in all trials, with only a few patients experiencing AEs. The authors concluded that intra-osseous injections for knee OA are safe and effective; however, multiple pertinent variables such as safety, cost of treatment, and performance against placebos and other treatment modalities require further evaluation before intra-osseous injections can be considered as standard treatment for patients presenting with knee OA.

The authors stated that the findings discussed in this review should be carefully considered alongside the drawbacks of the current literature. These included the small number of studies available (n = 6), the small number of patients included in the studies (n = 167), as well as the lack of post-injection imaging (including MRI) to examine the healing potential of the treatment. Furthermore, the lack of RCTs on the subject was apparent (only 1 included in the literature). It was also worth considering that clinically significant changes in pain were not uniform along the entire VAS, as the evaluation of pain was ultimately subjective to each patient. Another drawback was the need for research on the effect of intra-osseous injections on cartilage repair and regeneration; none of the included studies reported results in this area. These would give relevant insight into the correlations between subchondral bone pathology and the health of the articular cartilage. Besides, it is a challenge to gain an accurate assessment of the monetary costs of different treatment modalities across the studies originating from various research and medical centers. This would be important information in the final cost-benefit assessment of each treatment modality. A possible extension of the subject could be an evaluation into the cost-effectiveness of the

different intra-osseous injection treatment modalities. This would be pertinent, given the importance of quality, yet affordable, regimes in the treatment of knee pathologies, including OA.

Khokhar and Conaghan (2022) reviewed the recent literature on bone in OA, with a focus on imaging and intervention studies. Most studies focused on knee OA; hip and hand studies were uncommon. Bone shape studies demonstrated that shape changes precede radiographic OA, predict joint replacement, and have demonstrated high responsiveness. Novel quantitative 3D imaging markers (B-score) have better characterized OA severity, including pre-radiographic OA status. The addition of CT-derived 3D metrics has improved the prediction of hip joint replacement when compared to radiographs alone. Recent studies of bisphosphonates for knee OA have reported no benefits on pain or BML size. A meta-analysis on vitamin D supplementation in knee OA suggested minimal symptom improvement and no benefits on the structure. Cathepsin K inhibition demonstrated reduction in OA bone change progression, but with no symptom benefit. Studies of injections of bone substitutes into BMLs (subchondroplasty) have generally been small and potential benefits remain unclear. The authors concluded that subchondral bone features are associated with pain, incidence and progression of OA. Recent studies have validated quantitative bone shape as a biomarker for OA trials. Trials of bone-targeted OA therapies have been disappointing although cathepsin K inhibition may slow structural progression.

Xenograft Implantation into Articular Surfaces

It has been noted that transplantation of xenogenic porcine chondrocytes could represent a future approach for the treatment of human articular cartilage defects. Major barriers are humoral and cellular rejection mechanisms triggered by xenogenic epitopes that initiate hyperacute rejection which is primarily driven by activation of the hosts' classical complement pathway. Chondrocytes from multi-genetically modified pigs studied in vitro may offer a promising cell source for cartilage repair due to protection from humoral rejection (Tritschler et al., 2021).

There is insufficient evidence to support the use of xenograft implantation into the articular surface.

Ceracell Ortho Foam for Use in Cervical Instrumentation

Per its 510(k) clearance, the Ceracell Ortho Foam is intended to fill bony voids or gaps of the skeletal system (posterolateral spine). These osseous defects may be surgically created or from traumatic injury to the bone and are not intrinsic to the stability of the bony structure. In the posterolateral spine Ceracell Ortho Foam is to be mixed with autograft bone. The device resorbs and is replaced with bone during the healing process.

There is a lack of evidence regarding the effectiveness of the Ceracell Ortho Foam.

OptiMesh

OptiMesh (Spineology, St. Paul, MN) is a biocompatible, radiolucent, porous polyester mesh pouch knitted from polyethylene terephthalate thread. Polyethylene terephthalate material is widely used to make surgical sutures and meshes.

Chi et al (2021) stated that interbody fusion is a widely utilized and accepted procedure to treat advanced debilitating lumbar degenerative disc disease (DDD). Increasingly, surgeons are seeking interbody devices that are large for stability and grafting purposes but can be inserted with less invasive techniques. To achieve these contrary objectives, a novel, conformable mesh interbody fusion device was designed to be placed in the disc space via a small portal and filled with bone graft in-situ to a large size. This design can reduce the risk of trauma to surrounding structures while creating a large graft footprint that intimately contours to the patient's own anatomy. In an Investigational Device Exempt (IDE) trial, these researchers examined the peri-operative and long-term results of this novel conformable mesh interbody fusion device. This study was a prospective, multi-center, single-arm, Food and Drug Administration (FDA) and Institutional Review Board (IRB)-approved IDE, performance goal trial. A total of 102 adults presenting with DDD at a single level between L2 and S1 and unresponsive to 6 months conservative care had instrumented lumbar interbody fusion. Validated assessment tools include 100 mm visual analog scale (VAS) for pain, Oswestry Disability Index (ODI) for function, single question survey for

patient satisfaction, and computed tomography (CT) scan for fusion. Patients were enrolled across 10 geographically distributed sites. Pain/ODI surveys, physical evaluations, and imaging were performed serially through 24 months. Specifically, CT was performed at 12 and, if not fused, 24 months. Independent radiologists assessed CTs for fusion. An independent committee adjudicated adverse events (AEs). Patients with complete data at 24 months were included in the analysis. A total of 96 (94 % follow-up rate) patients (57.0 ± 12.0 years, 50.0 % women, BMI of 30.6 ± 4.9) reported average decreased LBP from baseline of 45.0 ± 26.6 at 6 weeks and 51.4 ± 26.2 at 24 months. Right/left leg pain reduced by 28.9 ± 36.7 / 37.8 ± 32.4 at 6 weeks and 30.5 ± 33.0 / 40.3 ± 34.6 at 24 months. Mean ODI improved 17.1 ± 18.7 from baseline to 6 weeks and 32.0 ± 18.5 by 24 months. At 24 months, 91.7 % of patients rated their procedure as excellent/good. Fusion rates were 97.9 % (94/96) at 12 months, and 99 % (95/96) at 24 months. Mean operative time, estimated blood loss (EBL), and length of stay (LOS) were 2.6 ± 0.9 hours, 137 ± 217 ml, and 2.3 ± 1.2 days, respectively; no device-related serious AEs were observed. The authors concluded that clinically significant outcomes for pain, function, fusion, and device safety were demonstrated in this population. Substantial clinical improvements occurred by 6 weeks post-operative and continued to improve to 24 months. The successful outcomes observed in this trial supported the use of this novel device in an instrumented lumbar interbody fusion. Level of Evidence = III. This report substantiated that the preliminary 1-year findings (published earlier for this investigation) were confirmed; and the fusion rates and that patient improvements reported were sustained through 2 years.

The authors stated that the drawbacks of this study were the relatively small number of subjects and that it was a single-arm trial that did not compare directly to other established methods for achieving interbody fusion. Despite these factors, demonstration of the positive outcomes including high rates of improved patient self-reported outcomes, excellent fusion rates, and strong safety profile for approximately 100 patients, identified OptiMesh as a viable option for interbody fusion. However, as with many advances in medicine, adoption of this technique must occur in the context of the abundant literature pertaining to interbody fusion.

It should be noted that this study only provided level III evidence; and some of the authors received financial support from the manufacturers. (John Chi discloses consulting fees from DePuy Spine and Stryker/K2M. Pierce Nunley discloses royalties or licenses from Stryker, Zimmer Biomet, and IMSE; consulting fees from Spineology, Stryker, IMSE, Integrity Implants, Kuros, NEO Spine, and Simplify Medical; payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing, or education events from Instrinsic Therapeutics and Organogenesis; and stock or stock options from Spinelogy and Surgalign. Stephane Lavoie discloses royalties or licenses from Spineology and consulting fees from Choice Spine and Spineology. Yi Lu discloses grants from the Stepping Strong Foundation (paid to institution); consulting fees from DePuy/Synthes; and stock or stock options in AxioMed LLC. Marcus Stone discloses support for the present article (e.g., funding, provision of study materials, medical writing, etc.) from Spineology; grants or contracts from Wright Medical (paid to institution); consulting fees from Spineology and Organogenesis (paid to Marcus Stone) and from Simplify Medical, Zimmer Biomet, and Spineology (paid to institution).

Signafuse Bioactive Bone Graft

Signafuse is a synthetic bone graft made up of bioglass and a biphasic mineral (60% hydroxyapatite, 40% β-tricalcium phosphate) available in putty and strip formats. Signafuse bioactive bone graft is designed to support spinal fusion and the formation of new bone. Signafuse has been available in putty form since 2014. The strip format offers more flexibility for surgical procedures, such as correction of spinal deformity.

SurGenTec ViMax Viable Fiber Allograft

The ViMax viable fiber allograft encompasses a vast number of cells that are of human origin. They include mesenchymal stem cells (MSCs).

There is a lack of evidence regarding the effectiveness of the ViMax viable fiber allograft.

Tactoset Injectable Bone Substitute

Tactoset Injectable Bone Substitute is a synthetic, biocompatible bone graft substitute material that hardens and converts to a poorly crystalline hydroxyapatite at body temperature. It is indicated for filling bone voids or defects of the skeletal system (i.e., extremities and pelvis) that are not intrinsic to the stability of bony structure. These defects may be surgically created osseous defects or defects created from traumatic injury to the bone. The device provides an injectable, self-setting, osteoconductive bone graft substitute that resorbs and is replaced by the growth of new bone during the healing process. Tactoset Injectable Bone Substitute can augment hardware and support bone fragments during the surgical procedure. The cured paste acts only as a temporary support media and is not intended to provide structural support during the healing process.

Stark et al (2020) noted that subchondral bone marrow edema (SBME) represents a pathologic alteration of subchondral bone. A strong correlation exists between its presence and the progression of osteoarthritis (OA). Very few treatment options exist between the spectrum of conservative management and the definitive treatment of total knee arthroplasty (TKA). Tactoset is an injectable synthetic, biocompatible hyaluronic acid-based bone graft substitute that allows for a minimally invasive treatment for painful SBME via percutaneous skeletal fixation (PSF). These researchers presented the technique of PSF using Tactoset.

There is insufficient evidence to support the use of Tactoset injectable bone substitute after digital amputation.

OSSURE Local Osteo-Enhancement Procedure (LOEP)

AgNovos Bioscience (Rockville, MD) developed the OSSURE Local Osteo-Enhancement Procedure (LOEP), a minimally-invasive treatment for osteoporotic bone loss in the proximal femur. The OSSURE implant is a triphasic, resorbable material consisting of calcium sulphate, brushite and ß-tricalcium phosphate (AGN1). The AGN1 implant material is injected into a site of osteoporotic bone loss. The OSSURE LOEP is

intended to provide immediate increases in strength of osteoporotic proximal femurs and ensure subsequent resorption and replacement by bone (AgNovos Bioscience, 2023; Kheder, et al., 2021).

Howe, et al. (2020) conducted a prospective cohort study to evaluate feasibility, safety, and in vivo response to treating proximal femurs of postmenopausal osteoporotic women with a minimally-invasive local osteo-enhancement procedure (LOEP). The LOEP involved the injection of a resorbable triphasic osteoconductive implant material (AGN1). This study enrolled 12 postmenopausal osteoporotic (femoral neck T-score ≤ -2.5) women aged 56 to 89 years. AGN1 LOEP was performed on left femurs and right femurs were untreated controls with follow up for 5-7 years. Study outcomes were adverse events, proximal femur areal bone mineral density (aBMD), AGN1 resorption, and replacement with bone by X-ray and CT, and finite element analysis (FEA) estimated hip strength. Baseline treated and control femoral neck aBMD was equivalent. Treated femoral neck aBMD increased by 68 ± 22%, 59 ± 24%, and 58 ± 27% over control at 12 and 24 weeks and 5-7 years, respectively ($p < 0.001$, all time points). Based on conservative assumptions, FEA-estimated femoral strength increased by 41%, 37%, and 22% at 12 and 24 weeks and 5-7 years, respectively ($p < 0.01$, all time points). Qualitative analysis of X-ray and CT scans demonstrated that AGN1 resorption and replacement with bone was nearly complete by 24 weeks. AGN1 appeared to be fully resorbed and replaced with bone integrated with surrounding trabecular and cortical bone at 5-7 years follow up. There was no occurrence of procedure- or device-related serious adverse events. The investigators concluded that treating femurs of postmenopausal osteoporotic women with AGN1 LOEP resulted in a rapid, durable increase in aBMD and femoral strength. The findings support the use and further clinical study of AGN1 LOEP in osteoporotic patients at high risk of hip fracture.

The systematic review on "Triphasic biomaterial for augmentation of the osteoporotic femoral neck" (Kheder, et al., 2021) included only one non-randomized comparative study which met the predefined inclusion criteria, comparing AGN1 LOEP with no treatment. Twelve postmenopausal women with osteoporosis received AGN1 LOEP in the left hip and no treatment in the right hip. More than 50% of the participants were taking anti-osteoporotic pharmacological agents at the

time of the trial. The results from this single prospective cohort study on 12 postmenopausal women showed significantly greater femoral neck aBMD and femoral strength favoring AGN1 LOEP over a long-term follow-up. However, the overall risk of bias for this study was serious. Moderate to serious biases were noted for the following domains: confounding, missing data , and possible selection of the outcome results. The study's small sample size would result in it being underpowered to detect a significant difference. This would reduce the likelihood that a statistically significant result is indicative of a true effect. The current evidence for AGN1 LOEP is derived from this single prospective cohort study on 12 postmenopausal women. The evaluation of AGN1 LOEP warrants further study via randomized control trials with large patient sample sizes and follow-up of at least 5 years. Currently, there is insufficient evidence to prove the effectiveness and safety of AGN1 LOEP in the prevention and treatment of osteoporosis-related hip fractures.

The clinical trials "RECONFIRM - Study of AGN1 LOEP in patients with osteoporosis" (NCT05202678) and "RESTORE - Study of AGN1 LOEP to prevent secondary hip fractures" (NCT04796350) are currently recruiting for participants.

Marlex Mesh

Marlex mesh (trademarked by Chevron Phillips Chemical Company and incorporated in C.R. Bard products) is a device made of polypropylene (synthetic plastic) that has been used in hernia repair procedures. Marlex mesh is being used as an option to treat disruption of the extensor mechanism after total knee arthroplasty (TKA),

Extensor mechanism reconstruction with Marlex mesh using a stepwise surgical approach is being performed in a similar manner for both quadriceps tendon and patellar tendon rupture. Prior to incision, a sheet of Marlex mesh is rolled onto itself and sewn together. "If the tibia is not being revised, a burr is utilized to create a trough in the tibia. Five centimeters of the tapered portion of the mesh are predipped in bone cement. The remaining cement is inserted into the trough. The tapered portion of the mesh is inserted into the tibial trough, ensuring that the mesh is fully seated. After the cement has cured, a lag screw is placed

across the mesh and cement and into host bone. If the tibia is being revised at the time of the Marlex mesh reconstruction, the 5 cm of predipped mesh is placed anteriorly in the medullary canal in line with the tibial crest. The remaining procedure is similar regardless of whether the components are revised. At the level of the joint, it is essential to ensure that the mesh is covered with host tissue. Next, the proximal reconstruction, which involves mobilizing the vastus lateralis and vastus medialis obliquus (VMO) distally by releasing all ventral and dorsal soft-tissue adhesions off the muscle bellies, is completed. Finally, the mesh is unitized to the vastus lateralis. With the limb maintained in full extension, the mesh is pulled directly proximally while another assistant pulls the vastus lateralis distally and medially. The vastus lateralis is deep, and the mesh is directly on top of it. Multiple nonabsorbable sutures are placed through the mesh and vastus lateralis. The VMO is then pulled distally and laterally over the mesh (which is now unitized to the vastus lateralis) by an assistant. Multiple nonabsorbable sutures (usually 8) are placed through the VMO, through the mesh, and through the vastus lateralis, unitizing the entire construct" (Abdel et al, 2019).

Morrey and colleagues (2016) state that quadriceps tendon rupture is an uncommon injury which has less than favorable surgical results. The authors state that a novel repair technique that uses Marlex mesh (C R Bard, Murray Hill, New Jersey) has been developed which may allow earlier range of motion and functional restoration of extension. Thus, in a retrospective study, the authors sought to evaluate the technical feasibility, durability, and postoperative function of this repair. Their study analyzed 8 knees (7 patients) operated on with this technique from 1990 to 2011. Mean follow-up was 9 years. Average age at the time of injury was 69 years. Mean operative time was 130 minutes. No perioperative complications occurred, no patients had clinical evidence of failure, and no patients required subsequent reoperation. Mean flexion at final follow-up was 100°. The authors found that of the 8 knees, 7 knees had no extensor lag at final follow-up. The authors concluded that Marlex mesh augmentation for quadriceps tendon ruptures has promising clinical results, despite significant comorbidities. The results showed that the technique was feasible, with low cost and reasonable operative time.

Hartline et al (2021) state that synthetic mesh has emerged as an option for reconstruction in the arthroplasty patient. The authors presented on two cases of chronic, re-ruptured quadricep tendon (QT) injuries in native knees treated with synthetic mesh reconstruction. Both patients had minimal ambulatory knee function prior to surgical intervention and were treated with a synthetic mesh reconstruction of their extensor mechanism. The authors technique had been modified from previously described techniques used in revision knee arthroplasty. At the one-year follow-up, both patients had improvement in their active range of motion and had returned to their previous activity. The authors concluded that synthetic mesh reconstruction of chronic extensor mechanism disruption is a viable technique that can be utilized as salvage for the persistently dysfunctional native knee. Longitudinal investigations should quantify and compare the efficacy of this novel technique.

Nucleus Pulposus Allograft

VIA Disc NP (VIVEX Biologics, Inc.) is a nucleus pulposus (NP) allograft intended to support the cushioning function of the intervertebral disc and supplement tissue loss. The VIA Disc NP procedure is considered minimally invasive and intended for patients who have experienced tissue loss within the lumbar intervertebral discs. The procedure involves fluoroscopy imaging to guide a spinal needle through Kambin's safe triangle into the nucleus pulposus of the intervertebral disc. The VIA Disc NP allograft is delivered into the target disc to rehydrate and support the cushioning function of the degenerated disc.

In a prospective clinical pilot study, Beall et al (2024) sought to evaluate the magnitude and durability of improvement in back function in patients with chronic lumbar discogenic pain followed for 6 months after a single intradiscal injection of minimally manipulated, off-the-shelf processed NP allograft (VIA Disc NP®, VIVEX Biologics, Inc.) at up to 2 vertebral levels. This single-arm, multicenter study included a total of 29 patients with symptomatic lumbar discogenic pain refractory to conservative care who had a back function score of 40-80 points on the Oswestry Disability Index (ODI), ≥ 6 on an 11-point back pain Numeric Rating Scale (NRS-11) and corresponding imaging evidence of disc degeneration were enrolled. A single dose, intradiscal injection of approximately 100 mg of NP allograft mixed with sterile saline was administered to the affected level or

levels. The average ODI and NRS-11 improvements between baseline and 6-months post-procedure were 54.8% (95% CI, 41.3-68.3) and 52.9% (95% CI, 34.7-71.1) respectively ($P < 0.001$). A minimal clinically important difference of ≥ 30% improvement over baseline was achieved in 79% (22 of 28) and 68% (19 of 28) of patients for ODI and NRS-11, respectively. At 6-months post-procedure, 64% (18 of 28) of patients had an NRS-11 score ≥ 3. The authors concluded that this pilot study demonstrate the feasibility of treating patients with symptomatic lumbar disc degeneration with a single intradiscal injection of allogeneic NP to provide significant and durable improvements in back function and pain. The authors acknowledged that this pilot study did not employ a concurrent control group and the clinical follow-up was limited to 6 months.

Autologous Platelet Concentrates in Maxillary Sinus Augmentation

In a meta-analysis, Xie et al (2025) compared the effectiveness of adding autologous platelet concentrates (APCs(+)) and sole bone grafting procedures (APCs(-)) in maxillary sinus augmentation (MSA). These investigators carried out a systematic search to identify RCTs from electronic database and clinical registers published up to June 1, 2024. A total of 6 evaluation indices comprising percentage of new bone formation (NBF%), percentage of residual bone substitute (RBS%), percentage of fibrous tissue (FT%), implant stability quotient (ISQ), bone height (BH) as well as bone density (BD) were identified. They examined 14 studies entailing 268 lifted sinuses and 119 implants. Compared with the APCs(-) group, the APCs(+) group showed a significant advantage regarding NBF% (weight mean difference (WMD) = 4.80, 95 % CI: 1.82 to 8.32), $p = 0.007$) and RBS% (MD = -4.48, 95 % CI: -6.01 to 2.96), $p < 0.01$), with no evident superiority in FT% (MD = -0.11, 95 % CI: -0.56 to 0.35, $p = 0.64$), ISQ (MD = -1.68, 95 % CI: -6.77 to 3.40, $p = 0.52$) and BH (standard MD (SMD) = -0.11, 95 % CI: -3.04 to 2.83), $p = 0.94$). While APCs(-) group demonstrated a significant benefit in BD compared to the APCs(+) group (MD = -28.34, 95 % CI: -48.98 to -7.71, $p = 0.01$). The authors concluded that compared with APCs(-), APCs(+) could be a better option for promoting NBF% and reducing RBS%, whereas these researchers observed no evident influence in FT% and ISQ under MSA. Concerning BD and BH results, there were no additional effects with the addition of APCs, which needs to be treated with caution due to the limited number

of studies. These researchers stated that APCs(+) procedures showed promising results, with a higher NBF% and lower RBS%, making them a potential routine MSA strategy.

Beta-Tricalcium Phosphate Combined with Native Bone Proteins for Ankle and Hindfoot Arthrodesis

In a prospective, multi-center study, Leppilahti et al (2025) examined the performance and safety of a combination of osteo-conductive beta-tricalcium phosphate and osteo-inductive native bone proteins (β-TCP - NBP) used as alternative for autograft in ankle and hindfoot arthrodesis. A total of 34 patients enrolled underwent ankle or hindfoot arthrodesis with β-TCP - NBP and were evaluated radiographically, clinically, and functionally up to 52 weeks. The primary performance endpoint was fusion rate evaluated with CT at 6 months. Safety was evaluated based on the severity and incidence of AEs. Functional evaluation was carried out by means of the AOFAS ankle-hindfoot score and pain was recorded using the VAS. CT at 6 months showed that 85.3 % had osseous bridging of the joint of ≥ 25 %, 52.9 % ≥ 50 %, 8.8 % < 25 % and 5.9 % showed no bridging. The AOFAS score increased significantly from 60.4 ± 17.6 points at operation to 68.6 ± 17.2 points at 6 months and to 73.5 ± 17.7 points at 12 months. The group with fusion rate ≥ 25 % showed significantly higher AOFAS score than that with fusion rate < 25 % at 12 months. The mean VAS pain score at rest and during weight bearing decreased significantly ($p < 0.0001$) from operation to 6 and 12 months. The authors concluded that the findings of this study showed that β-TCP - NBP was a valuable bone graft substitute for fusion of ankle and hindfoot due to debilitating OA and offered an alternative for autograft. Level of Evidence = II. The main drawback of this trial was the lack of a control group. Other drawbacks included small sample size (n =34) and a relatively short follow-up duration (up to 12 months).

Low-Crystallinity Apatite for Bone Reconstruction

Park et al (2025) noted that to overcome the shortcomings of sintered bone graft materials, low-crystallinity apatite (LCA) was developed using a non-heated approach to enhance resorption and integration during bone regeneration. These researchers examined the effectiveness of LCA as a synthetic bone graft material for bone reconstruction. LCA was

compared to 3 conventional synthetic bone graft materials: biphasic calcium phosphate (BCP) 37, BCP 64, and octa-calcium phosphate (OCP). Crystalline structure and surface morphology were examined using X-ray diffraction (XRD) and scanning electron microscopy (SEM). In-vivo testing was carried out using a rabbit calvarial augmentation model, in which the grafts were placed into standardized defects. Bone formation and graft resorption were analyzed using micro-computed tomography (micro-CT) and histomorphometric analyses at 3 and 6 post-implantation. LCA exhibited structural similarities to the allograft material and enhanced surface properties. Micro-CT and histomorphometric evaluations at 3 and 6 weeks post-implantation showed higher rates of bone formation and substantial volumetric changes with LCA, indicating efficient graft resorption and bone regeneration. The authors concluded that LCA exhibited superior integration, osteo-conductivity, and biodegradability compared to other synthetic grafts, suggesting the potential for improved clinical outcomes with its use. Moreover, these researchers stated that although the effectiveness of LCA has been validated, further studies in diverse biological environments are needed to confirm its safety and effectiveness for broader clinical use.

PerOssal for the Treatment of Osteomyelitis and Nonunions

Armbruster et al (2024) stated that traditional autologous bone grafts as a treatment for bone defects have limitations such as donor-site morbidity and limited supply. PerOssal, a ceramic bone substitute, may overcome those limitations and could offer additional benefits such as prolonged, local antibiotic release. In a retrospective, single-center study, these investigators examined the clinical and radiological outcomes, including PROs, of using PerOssal in nonunions (NU) and high-grade chronic osteomyelitis (COM). This trial included patients who were treated with PerOssal between January 2020 and December 2023. Collected data included patient characteristics as well as various surgical and outcome parameters including the Lower Extremity Functional Scale (LEFS). A total of 82 patients were analyzed. Re-infection occurred in 19.5 % of cases. Osseous integration of PerOssal was achieved in 89 % of cases, higher in cavitary defects (91.5 %) than segmental defects (72.7 %). The revision rate was 32.9 %, mainly due to wound healing disorders and re-infections. Mean LEFS score was 53.4, which was heavily influenced by sex (male: 50.7 versus female: 63.4), revision surgery (no: 55.7 versus

yes: 49.1), re-infection (no: 56.6 versus yes: 39.4), and osseous integration of PerOssal (yes: 55.8 versus no: 38.4). The authors concluded that PerOssal showed promising outcomes in treating NUs and high-grade COM, especially in cavitary defects, with high osseous integration rates and acceptable functional results. However, re-infection remained a concern, especially with difficult-to-treat pathogens and extensive surgical histories. These researchers stated that early, comprehensive surgical intervention and tailored antibiotic strategies were crucial; patient selection, defect characteristics, as well as co-morbidities significantly influenced success. Moreover, they stated that further investigations are needed to optimize treatment protocols, refine treatment strategies, and enhance functional recovery.

The authors stated that this study had several drawbacks. first, the parameters of revision surgery, re-infection, and osseous integration are interrelated factors, making it challenging to determine the independent effect of each. The subjects in this study primarily included patients with cavitary defects; thus, these findings were most applicable to this specific group. Second, with a minimum follow-up time of only 6 months, re-infection rates could potentially be under-estimated. However, the mean follow-up duration in this trial of 2.2 years significantly exceeded the latest observed re-infection, which occurred at approximately 1.5 years post-surgery. In addition, the median time to re-infection was 69 days or 2.3 months, suggesting that the majority of re-infections were likely accounted for within the follow-up period. Third, although a variety of pre-existing conditions were monitored, obesity was not included as a potential confounder, which limited the conclusions that could be drawn regarding its impact in this patient cohort. However, high blood pressure and diabetes mellitus, key components of metabolic syndrome, were analyzed, providing indirect insights into the potential influence that obesity could have on outcomes. Fourth, osseous integration of PerOssal was evaluated only via biplanar post-operative X-rays rather than CT scans. In addition, no specific scoring system, such as the Goldberg Score, was used, which may have limited the precision of these assessments. Fifth, as a retrospective study, there are inherent limitations, including potential selection and information bias associated with the patient cohort examined.

Platelet-Rich Fibrin for Lateral Window Sinus Lifts

Estrin et al (2025) stated that platelet-rich fibrin (PRF) is being increasingly used in surgical procedures due to various improvements in clinical outcomes. More recently, a heating process to denature albumin in the platelet-poor plasma (PPP) layer has been reported to extend the resorption time of PRF from a typical 2-week period to 4 to 6 months. Because of its greater than 4-month resorption properties, extended PRF (e-PRF) membranes have been employed in dentistry as an alternative to collagen membranes in alveolar ridge preservations, ridge augmentations, soft tissue grafting, and as a barrier membrane in lateral sinus grafting procedures. In a multi-center, case-series study, these researchers examined the use of e-PRF as a membrane for lateral window sinus lifts. A total of 22 patients from 3 different clinical facilities participated in this trial between June 2022 and December 2023. All clinicians completed a standard lateral window sinus technique with sticky bone using allograft, xenograft, or autologous dentin with an e-PRF membrane used for sole coverage of the lateral window. The survival rate at 2nd stage implant surgery (approximately 6 months) and radiographic analysis determined on consecutive radiographs: the mean residual bone height (RBH) under the maxillary sinus at implant placement; and the change in endo-sinus bone level, were calculated to evaluate the endo-sinus bone changes at 6 months. Participants included 12 women and 10 men with a mean age of 58.8 years, ranging from 32 to 78 years. In 22 lateral window sinuses completed, a total of 31 implants were placed. All subjects healed uneventfully without any post-operative complications. Post-operative cone beam computed tomograms (CBCTs) were taken at 6 months to show successful sinus floor augmentation and to calculate endo-sinus gain. No soft tissue invagination was encountered in any of the study participants nor displacement of the graft material into the buccal mucosa. In 1 case highlighted in this study, the e-PRF membrane was employed to repair a sinus perforation. The average 6-month post-operative sinus height was 12.3 mm with a mean change in endo-sinus bone gain of 8.0 mm. The authors concluded that using e-PRF membranes in place of collagen membranes for coverage of the lateral window during sinus floor elevation was shown to be a successful treatment modality in this case-series study. These findings provided new opportunities using denatured albumin membranes in implant dentistry and paved the way for additional

applications. The material is 100 % autologous, low cost. Moreover, these researchers stated that this was the 1st case-series study highlighting its novel use for coverage of the lateral window during sinus grafting, larger, longitudinal studies are needed with comparison to standard resorbable membranes as well as long-term follow-up of implants placed into sites grafted with e-PRF membranes.

Appendix

Bone and Tendon Graft Substitutes considered medically necessary when criteria are met (not an all-inclusive list):

- 3D Profuse (ATEC Spine)
- Accell Connexus (SeaSpine)
- Accell Evo 3 (SeaSpine)
- Accell TBM (SeaSpine)
- ACF (DePuy Synthes)
- Allocraft CL (Stryker)
- AlloFlex (Allosource)
- AlloFuse (Allosource)
- AlloGro DBM (Allosource)
- AlloPac (Allosource)
- AlloSource Femoral Head Bone Graft (Allosource)
- AlphaGraft DBM (ATEC Spine)
- AlloQuent (Orthofix)
- Atrix-C (Xtant Medical)
- Autograft (patient)
- Berkeley Advanced Biomaterials allograft (excludes the TCP versions)
- Bio DBM (Stryker)
- BioAdapt Bridge (Surgalign)
- BioMet Boost DBM
- BioMet DBM putty
- BioMet EBI DBM
- BioReady DBM Putty (Surgalign)
- BioReady DBM Putty with Chips (Surgalign)

- BioSet Allograft (Surgalign)
- BioSet DBM (Surgalign)
- BonePlast Quick Set
- Cadaveric Allograft
- Capistrano (SeaSpine)
- CeSpace Bone (Aesulap)
- Conform Sheet (DePuy Synthes)
- Cornerstone ASR (Medtronic)
- Cornerstone L-ASR (Medtronic)
- Cornerstone-SR Allograft Tissue (Medtronic) (not allowed in the cervical spine if INFUSE is used as part of the graft)
- DBM Pure Macro allograft (LES Spine/SpinalFrontier)
- DBM Pure Micro allograft (LES Spine/SpinalFrontier)
- Demineralized Bone Matrix
- Dense cancellous bone allograft
- DynaGraft-D (Cigagenix)
- Dynagraft II DBM (SeaSpine)
- Elemax cortical spacer allograft (Surgalign)
- Evologics DBM
- Exponent DBM (Bioventus)
- FiberFUSE demineralized bone matrix (DBM) allograft
- Fortitude Duo (Zimmer Biomet)
- Fortitude Osso (Zimmer Biomet)
- FusionFlex (Wright Medical)
- Grafton Crunch, Flex, Gel, Matrix, Orthoblend (Large or Small Defect), Putty, Strips (Medtronic)
- IC Graft Chamber (LifeNet Health)
- Impacted cortical bone
- INFUSE Bone Graft (Bone Morphogenic Protein-2) (Medtronic) (See Section I in the policy section for exclusions)
- Integra DBM
- InterGro DBM (Zimmer Biomet)
- Kore Fiber (MTF Biologics; offered/marketed by KolosisBIO)
- Magnifuse (Medtronic)
- MTF cube (Musculoskeletal Transplant Foundation)
- MTF Cortico/Cancellous ACF spacer (Musculoskeletal Transplant Foundation)
- MTF DBX (Musculoskeletal Transplant Foundation demineralized bone matrix)

- Optecure DBM (Exactech)
- Optefil (Rti Surgical)
- Opteform DBM (Exactech)
- Optium (LifeNet)
- Oragraft (LifeNet)
- Osprey Condor, Eagle, Talon (Skye Biologics)
- OsteoAMP (Bioventus)
- OsteoAMP Select (Bioventus)
- Osteofil DBM
- Osteogenic Protein-1 (OP-1)
- Osteo-link DBM (Life Spine)
- OsteoSelect (Xtant Medical)
- OsteoSparx DBM (SeaSpine)
- OsteoSponge DBM (Xtant Medical)
- Osteostrip (Alliance Spine)
- Osteosurge 100 DBM (SeaSpine)
- Osteosurge 300 (SeaSpine)
- Pinnacle DBM sponge
- Polymethylmethacrylate (PMMA) Antibiotic Beads
- Prime DBM HD (Musculoskeletal Transplant Foundation; offered/marketed by KolosisBIO)
- Progenix Plus (UMG Uysal Medikal)
- Progenix Putty (UMG Uysal Medikal)
- Promote Osteostrip (Alliance Spine)
- Promote Osteopro (Alliance Spine)
- Propel DBM (NuVasive)
- PureBone (Biventus)
- Puros DBM (Zimmer Biomet)
- RTI Biologics BioSet Allograft Paste
- RTI DBM Powder
- Skye Biologics DBM cancellous bone allograft (chips, crunch, cubes, paste, putty)
- SteriSponge (offered/marketed by Bio-Implant Solutions)
- Staygraft DBM
- SteriFuse (Aurora Spine)
- Triad Allograft (NuVasive)
- Tricortical bone allograft
- Trinnect (Zimmer Biomet)
- VegaGraft (GS Medical)

- Vertigraft (LifeNet)
- Vertigraft II (LifeNet)
- Vesuvius fibers DBM (Stryker)
- VG2 cervical allograft (LifeNet)
- Xemplifi DBM (Globus Medical)
- Wright Allomatrix DBM C, DR
- Wright Allopure.

Note: There are insufficient published data demonstrating that Vivigen has superior clinical outcomes compared to standard DBM products such as DBX putty, despite its generally higher cost. Therefore, use of Vivigen is considered not medically necessary.

Bone and Tendon Graft Substitutes considered experimental, investigational, or unproven (not an all-inclusive list):

- Accufill
- Acell Biologic Graft
- Acuity Allostem
- Actifuse (silicated calcium sulfate)
- Activize Nucel
- AlloGro Stem Cell Bone Growth Substitute
- Allosource AlloStem
- Alphatec NeoCore
- Alphatec VELOSSITY Moldable Synthetic Bone Graft
- Amniofix
- AmnioShield
- Arteriocyte \ Magellan Platelet Separator Systems
- Arthrex Biopaste (BioCartilage)
- Arthrex Quickset (calcium phosphate cement)
- AttraX Putty
- Augment
- Beta-BSM
- Bioactive graft (nano-structured hydroxyapatite material)
- BioD factor
- BioD Restore (placental tissue graft)
- Biogennix RPC
- Bioglass 45S5
- BIO MatrX

- BioMatrix Generate
- BioSphere Putty Bioactive Bone Graft
- BiOstetic
- BioViable Bone Matrix
- BonAlive
- BoneSave
- BoneSource BVF
- Bone X'TRUDABLE
- BonePlast
- Calceon 6
- Callos Bone Void Filler
- CarriGen
- Cartimax
- Cellentra Allograft
- Celling Biosciences Solum IV allograft
- Cerament
- Cem-Ostetic
- CLM Bioactive Scaffold
- CONDUIT TCP Granules
- CopiOs Bone Void Filler (Zimmer Biomet)
- Cortiva acellular dermal matrix
- Cyclone Bone Marrow Concentrate system
- Equivabone Graft (Zimmer Biomet)
- Fibergraft Bioactive Glass Allograft
- FM-02
- Formagraft
- Cymbicyte
- Gamma-BSM
- GenerOs
- GranOS
- Globus Matrix ceramic
- Globus Medical conduct
- Globus NuBone
- Graston
- H-Genin
- Healos Bone Graft Replacement
- Healos Sponge
- HydroSet
- i-Factor Peptide Enhanced Bone Graft

- Inductaputty (hydroxapatite)
- InQu
- Integra Mozaik beta tricalcium phosphate (experimental for indications other than spinal fusion)
- Interface allograft
- InterGro DBM Plus
- Kainos+Nano
- Kinex Bioactive Putty
- Life Net AVS
- MAP3
- Marlex mesh
- Mastergraft allograft
- Mastergraft Granules
- Mastergraft Putty
- Mastergraft Strip
- Matrix Ovation Stem Cell
- MCS Bone Graft
- Mesenchymal Stem Cell Therapy
- Mozaik allograft
- nanOss Bioactive
- nanOss Bioactive 3D
- NextGraft
- NEXoss
- Norian SRS
- Norian SRS Fast Set Putty
- Novabone
- NovaBone bioactive strip
- NovaBone-C/M
- Nucell
- OpteMx
- Optimesh
- Orthospine Allostem or Osteostem
- Osiris Stryker Bio4
- OsSatura TCP
- Ossifuse (boats, strips, and putty)
- OssiMend Putty
- OsteoBlast II
- Osteocel
- Osteocel Plus

- Osteofuse
- Osteomatrix
- OsteoVation QWIK
- OsteoVive
- Osiris Therapeutics Ovation
- Ovation OS
- PalinGen
- PerioGlas
- Pioneer FortrOss
- PlatForm CM
- Plexus M
- Plexus P
- PolyGraft
- Porcine Intestinal Submucosa Surgical Mesh
- Pro Osteon Bone Graft Substitute
- Pro Osteon Porous Hydroxyapatite Bone Graft Substitute
- PurBone
- ProDense (calcium sulphate/calcium phosphate composite)
- PurGEN
- Regeneration BioSet
- Regenexx PL-Disc
- Rybone
- Signafuse Bioactive Bone Graft
- Skaffold NMX
- Skaffold ReNu Flow
- SKYE Liquid Gel
- SureFuse (DBM containing growth factors)
- SurGenTec ViMax viable fiber allograft
- Spineology Optimesh
- Stryker Biotech OP-1
- SxBarrier (Surgilogix) (placental tissue and growth factors)Tactoset injectable bone substitute
- TriCore
- Trinity Elite
- Trinity Evolution
- Tripore
- TRS PCL Cranial Bone Void Filler
- TruFit bone plugs
- TruFUSE allograft bone dowel (miniSURG Corp.)

- TruRepair
- Venado Granules
- Viagraf
- Vitoss BA2X
- Vivex Via Graft (Amendia) (contains growth factors)
- Wright Cancello-Pure
- Wright Cellplex
- Wright Ignite
- Wright MIIGX3
- Wright Pro-Dense
- Wright Pro-Stim.

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